Experiments in Astroparticle Physics (Ground-based experiments)

(Lecture 12)

Satyendra Thoudam

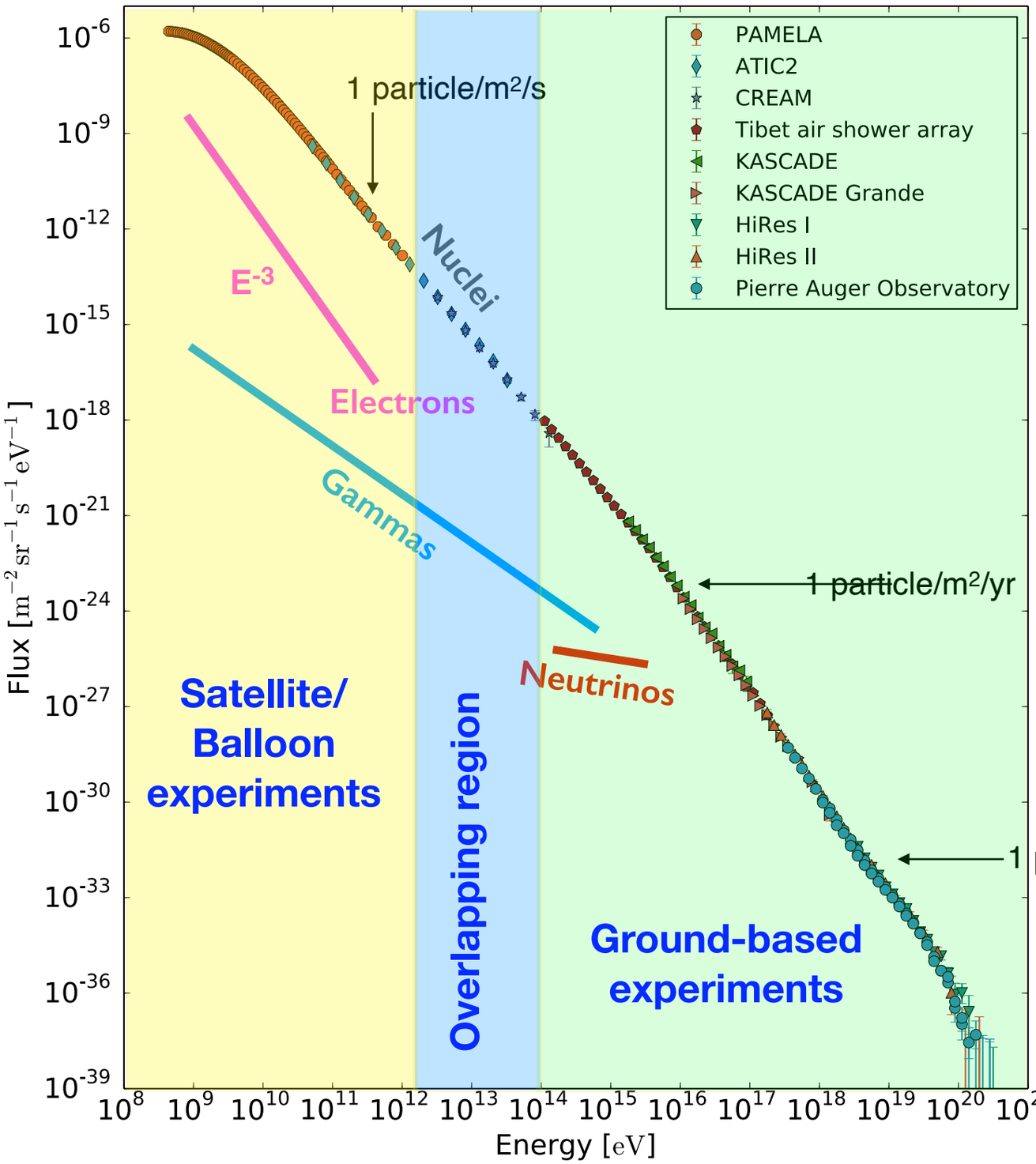
Khalifa University

Outline of this chapter

- Introduction: Ground-based experiments
- Cosmic-ray experiments:
    - Particle detector arrays
    - Radio antenna arrays
    - Fluorescence telescope
- Gamma-ray experiments:
    - Cherenkov telescopes
    - Particle detector arrays
- Neutrino experiments:
    - IceCube experiment in Antarctica
    - KM3NeT experiment in the Mediterranean Sea.

Introduction: Ground-based experiments

Spectra of cosmic rays, gamma rays and neutrinos



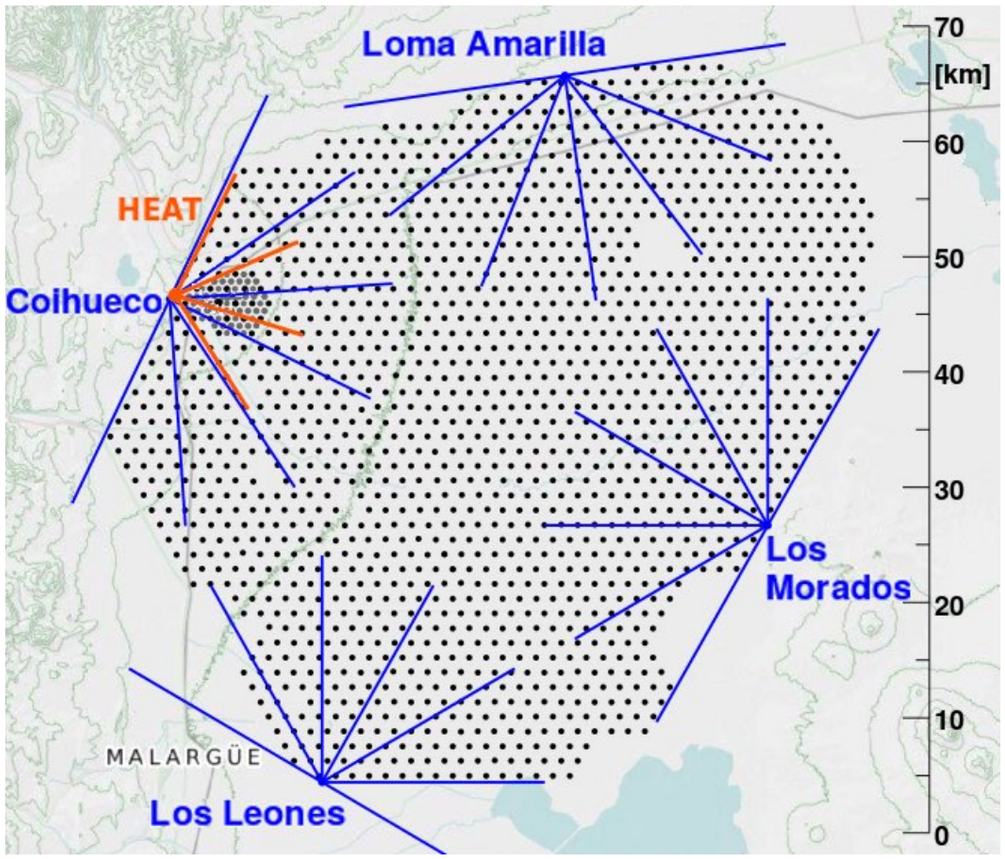
*Basic aim of every experiment is to measure the **Energy, Charge (or type) and Direction (or momentum)** of the incoming particle.

Introduction: Ground-based experiments

KASCADE-Grande (CR)

https://www.iap.kit.edu/kascade/english/

Pierre Auger Observatory (CR)



https://www.preprints.org/manuscript/201810.0656/v1

H.E.S.S telescope array

https://www.mpi-hd.mpg.de/hfm/HESS/

HAWC gamma-ray experiment

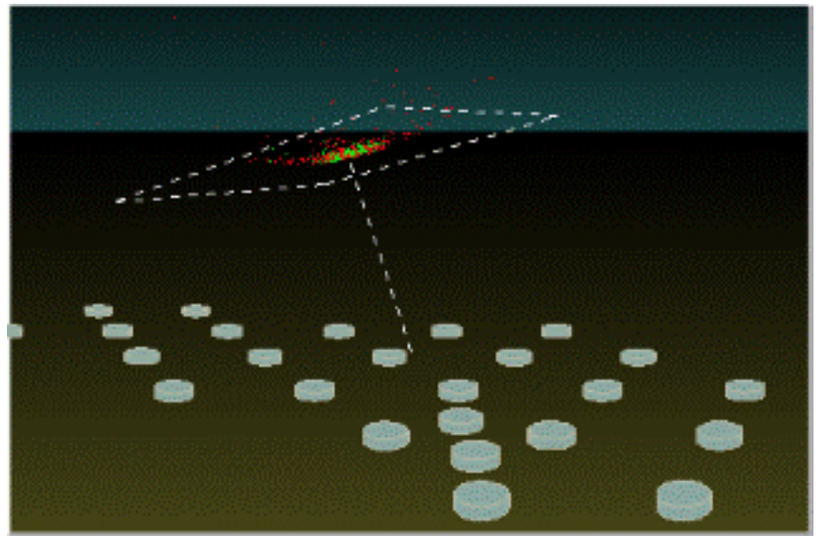
https://www.hawc-observatory.org/

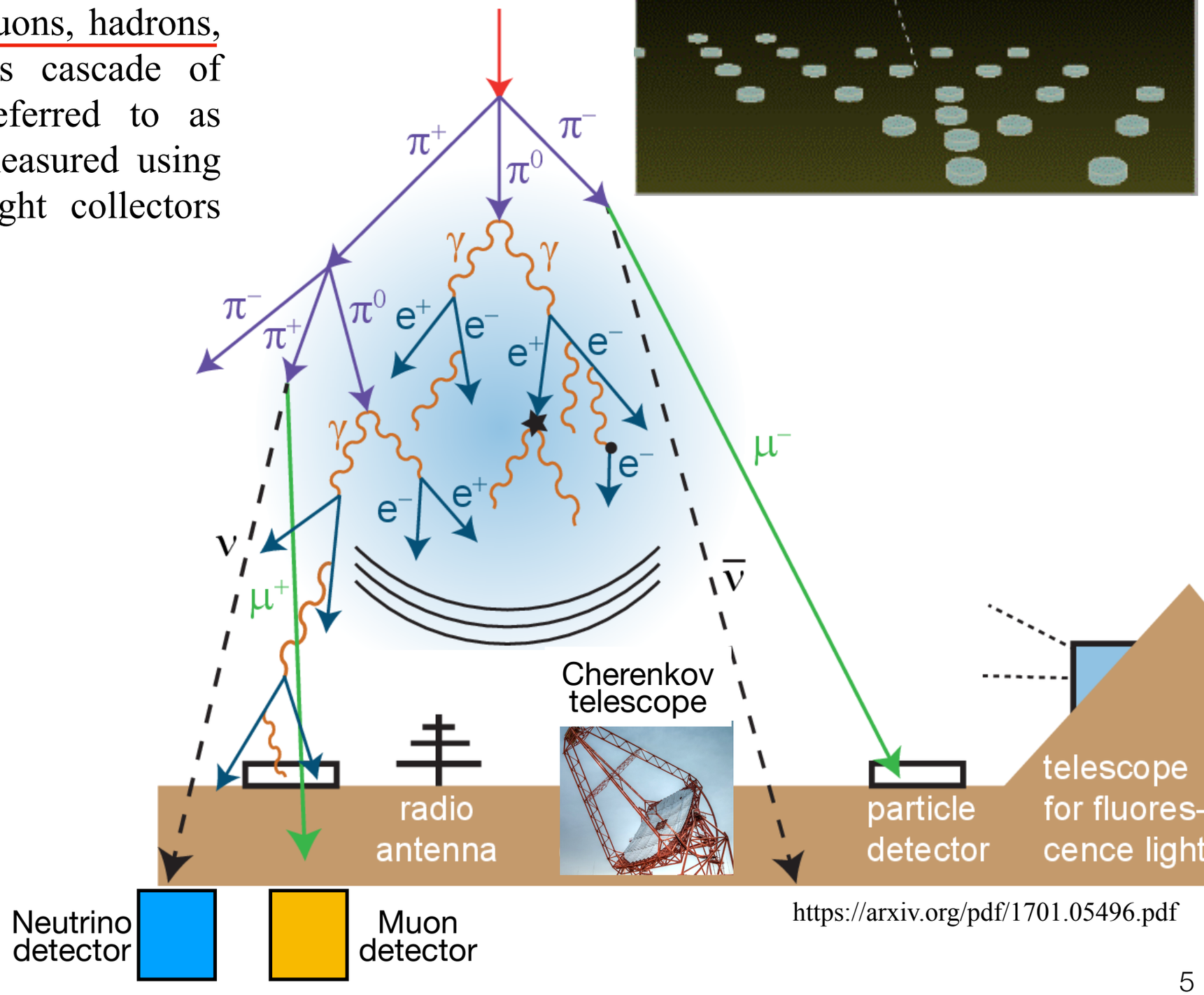
All ground-based experiments use the Earth's atmosphere as a calorimeter.

Introduction: Ground-based experiments

Extensive air shower

When high-energy cosmic rays or gamma rays undergo collisions with the atoms of the atmosphere, a cascade of secondary particles comprised mostly of electrons, muons, hadrons, and photons are produced. This cascade of particles, which is generally referred to as **Extensive air shower**, can be measured using array of particle detectors or light collectors placed on the ground.





https://arxiv.org/pdf/1701.05496.pdf

Introduction: Ground-based experiments

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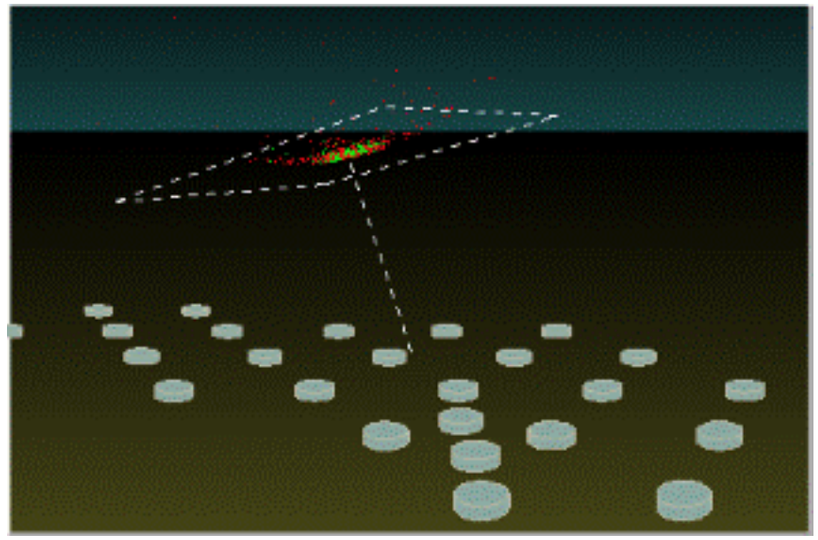
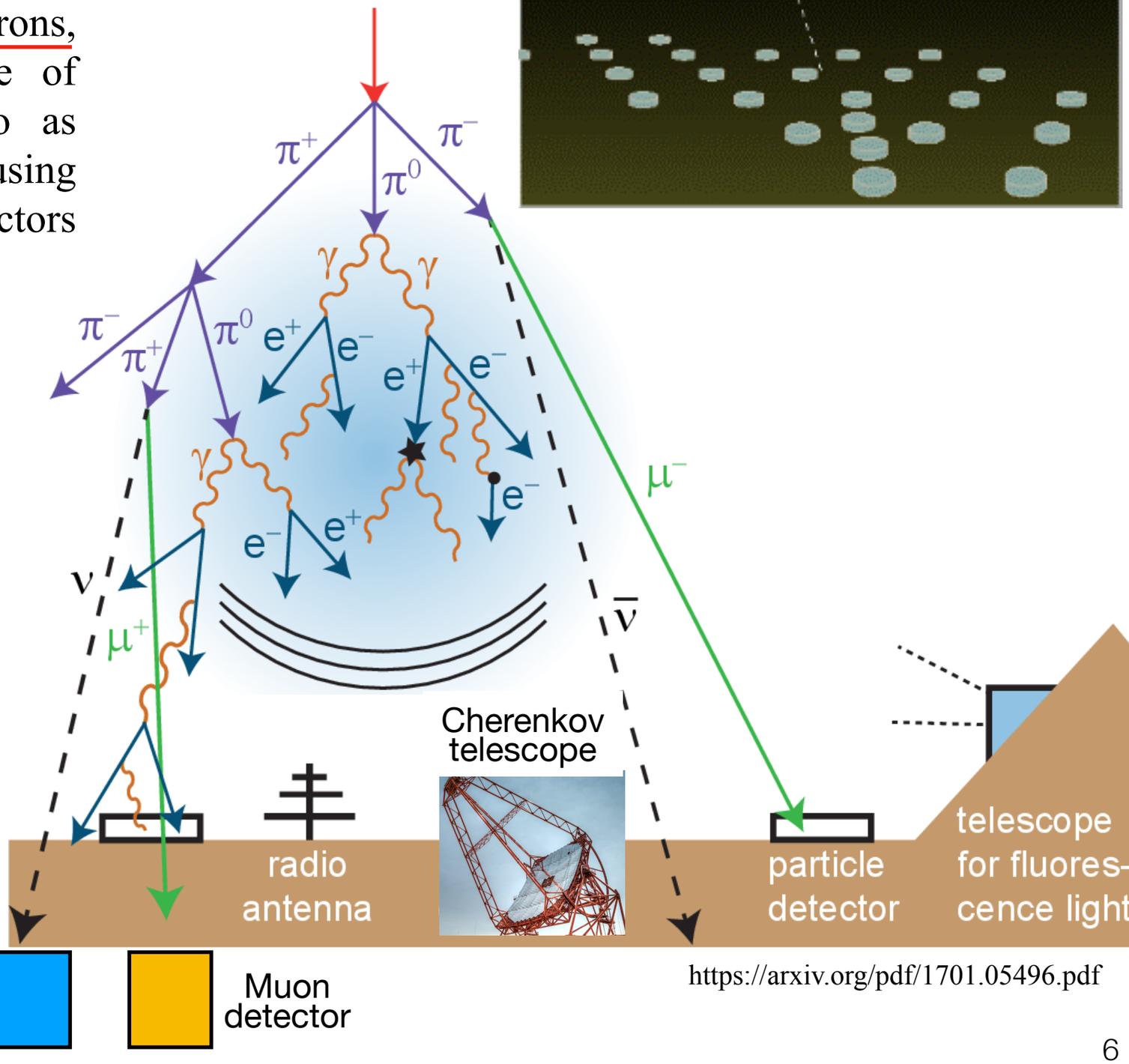
Air showers can produce Cherenkov light (visible region), radio emission and fluorescence light.





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Introduction: Ground-based experiments

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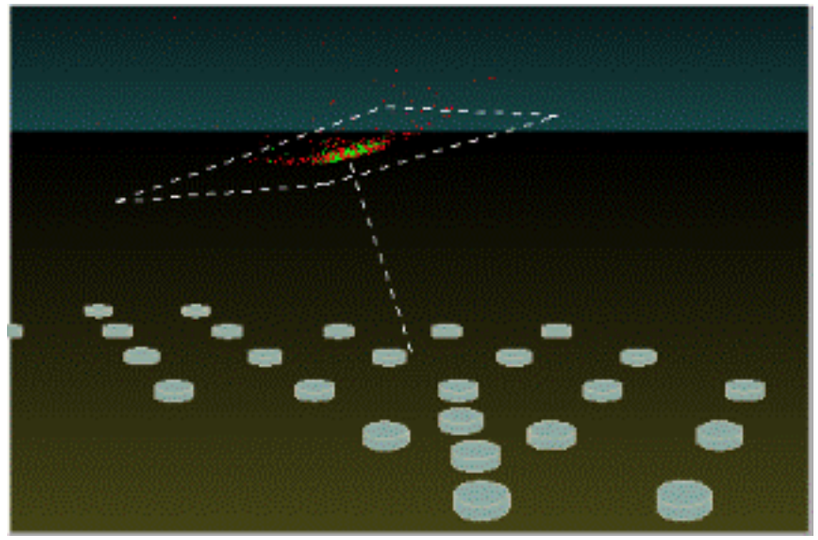
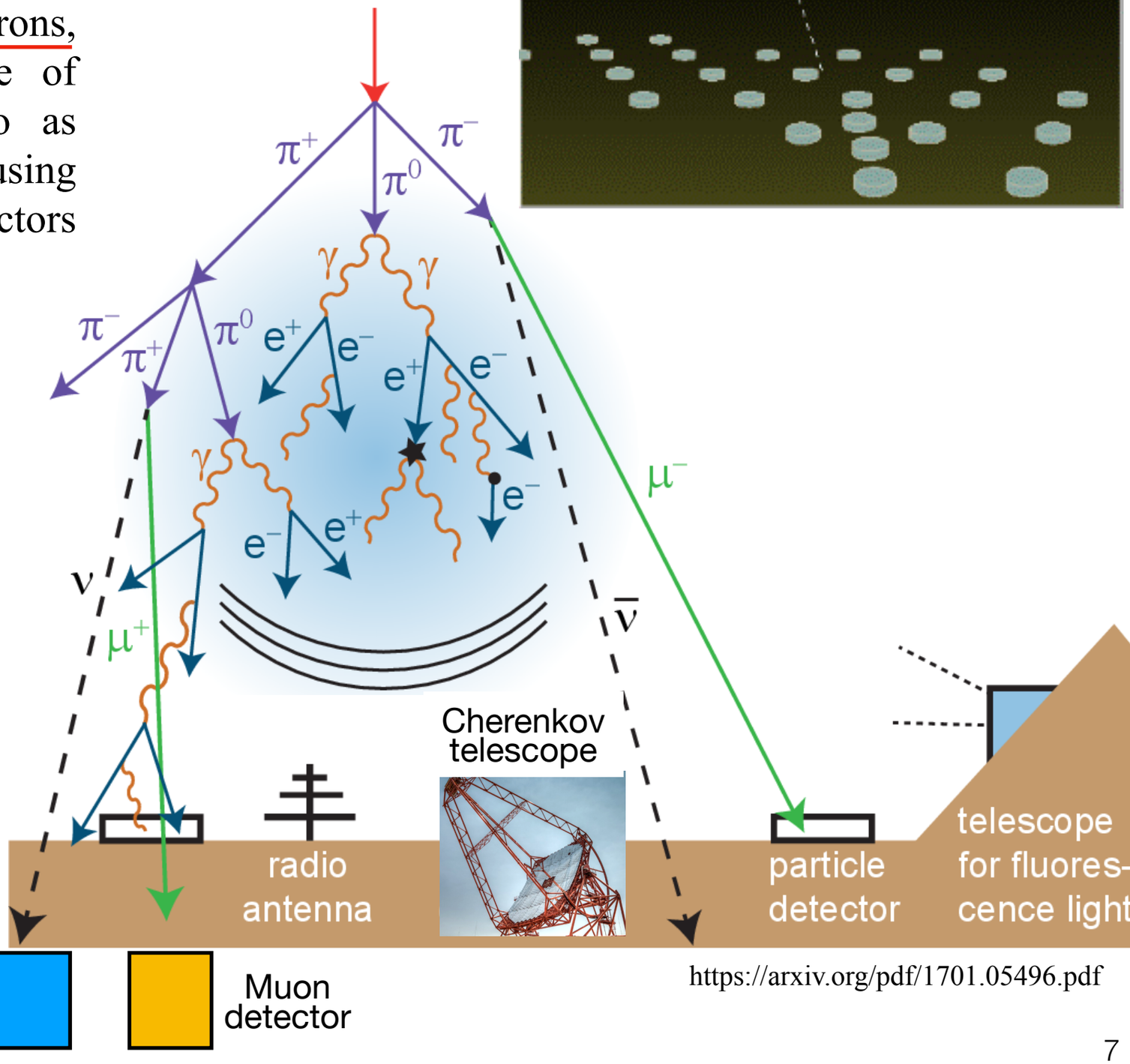
Detectors can be designed based on the type of the signal you want to detect:
- Particle detectors (CRs + gammas)
- Cherenkov telescopes (gammas)
- Radio antennas (CRs)
- Fluorescence telescopes (CRs)
- Neutrino detectors

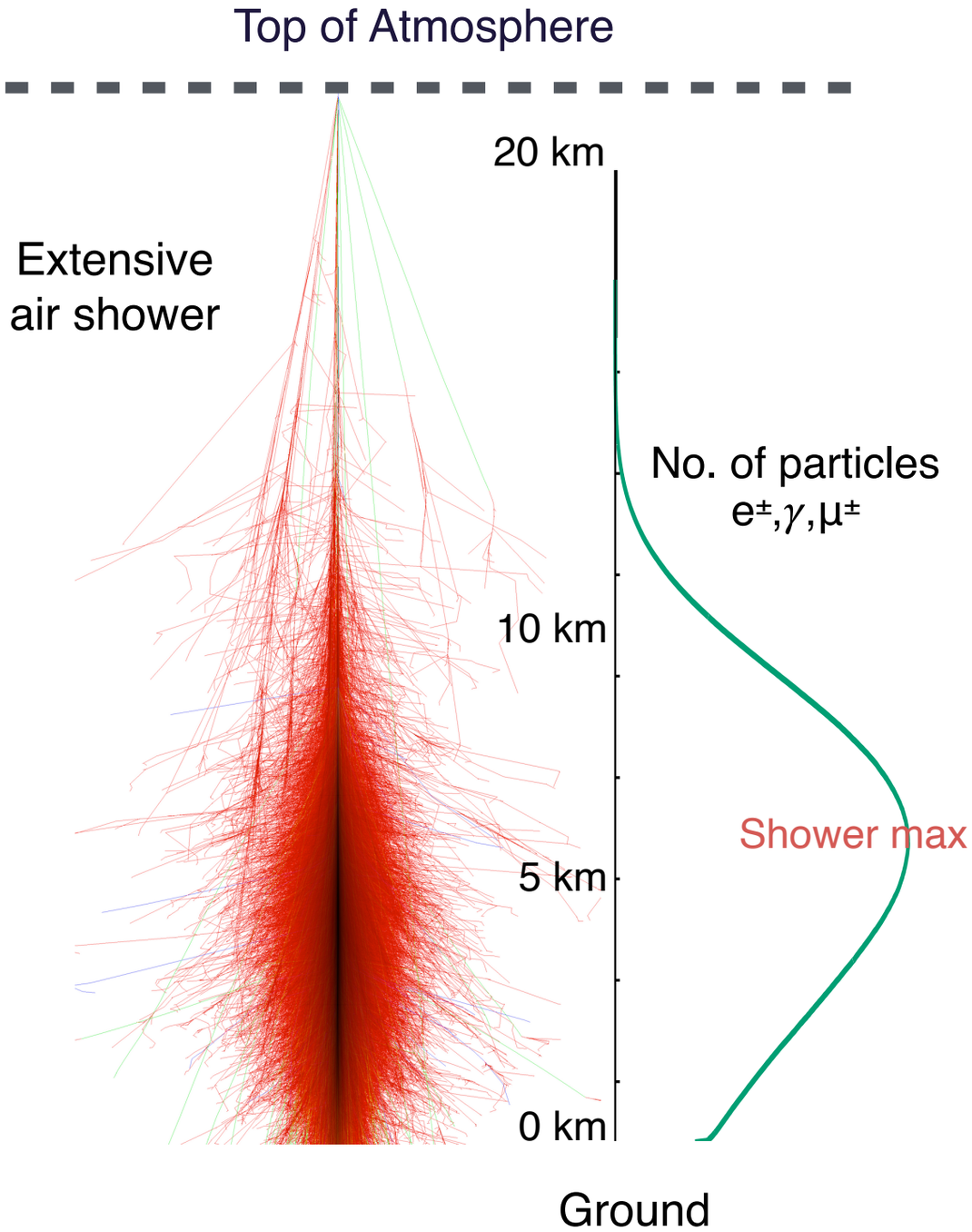




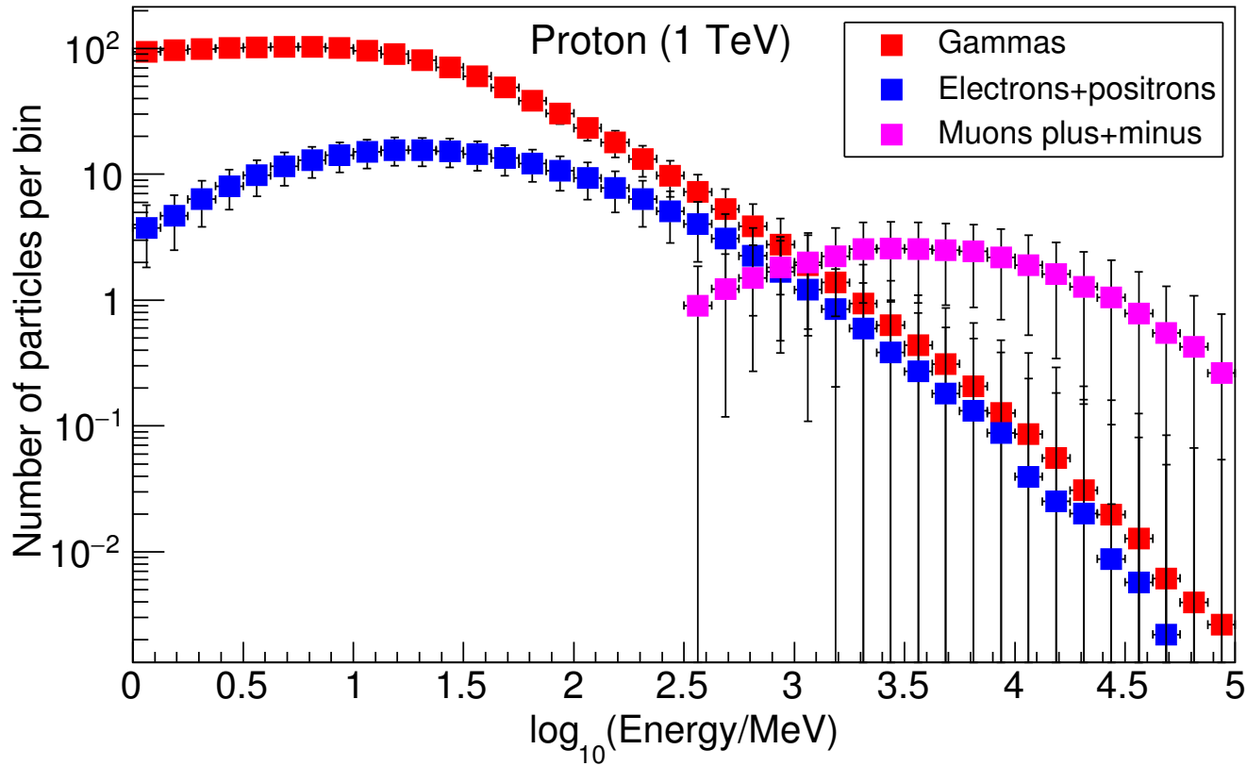
https://arxiv.org/pdf/1701.05496.pdf

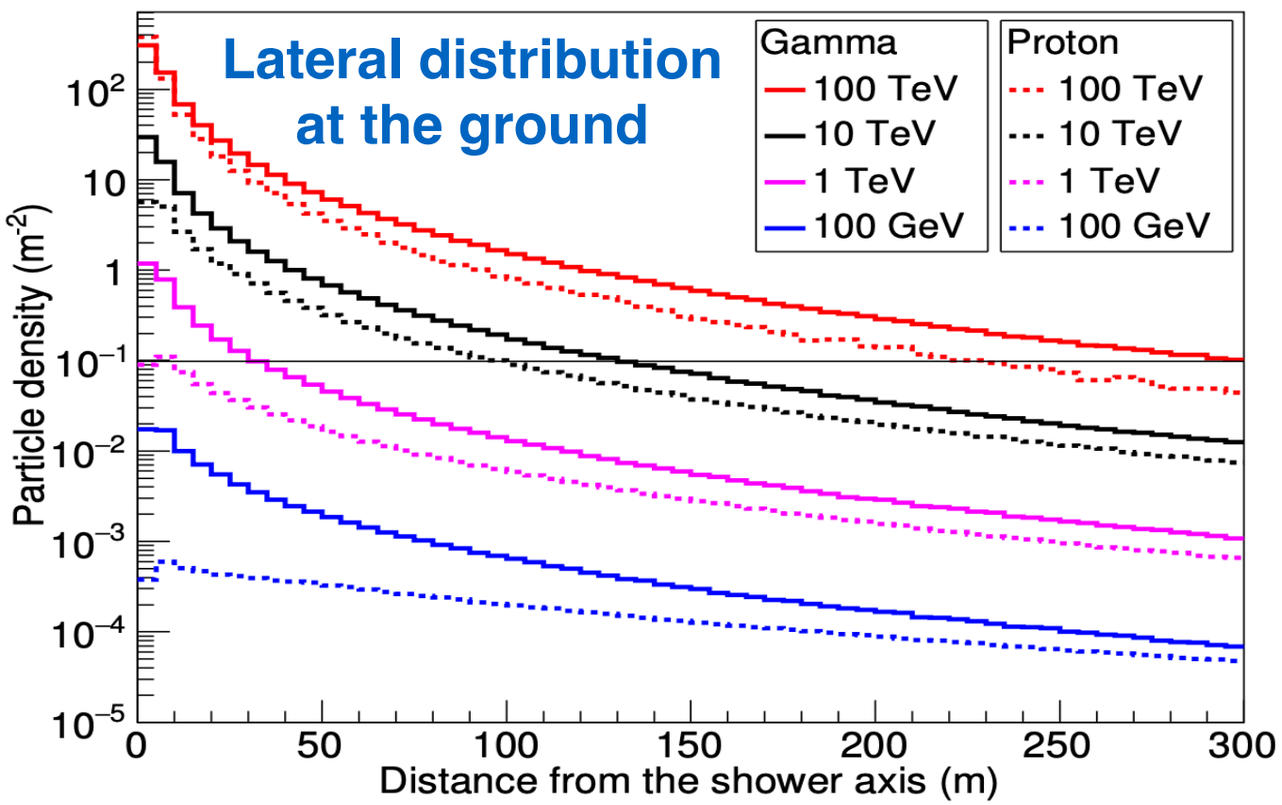
Some general properties of air showers

Longitudinal distribution of air shower



Energy distribution of particles at the ground





Thoudam, internal note, COMET collaboration

Cosmic-ray experiments

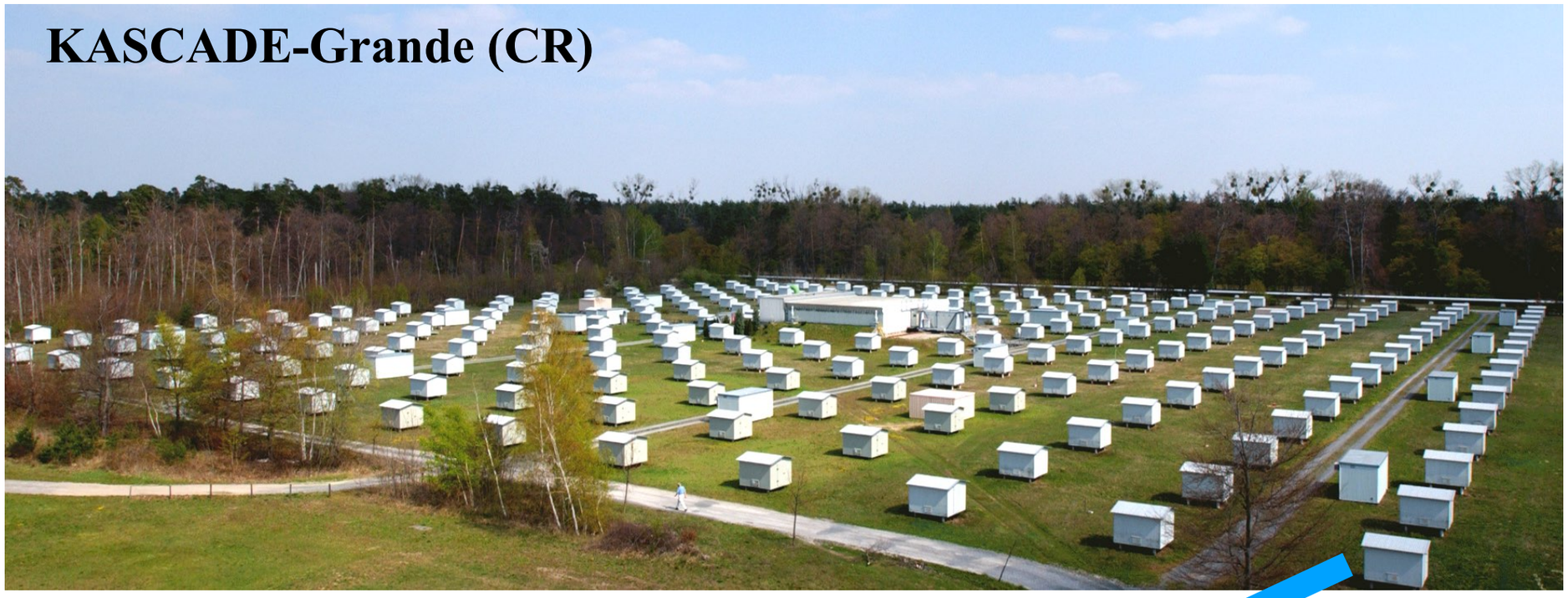
(1) Particle Detector Array (Scintillators and Water Cherenkov detectors)

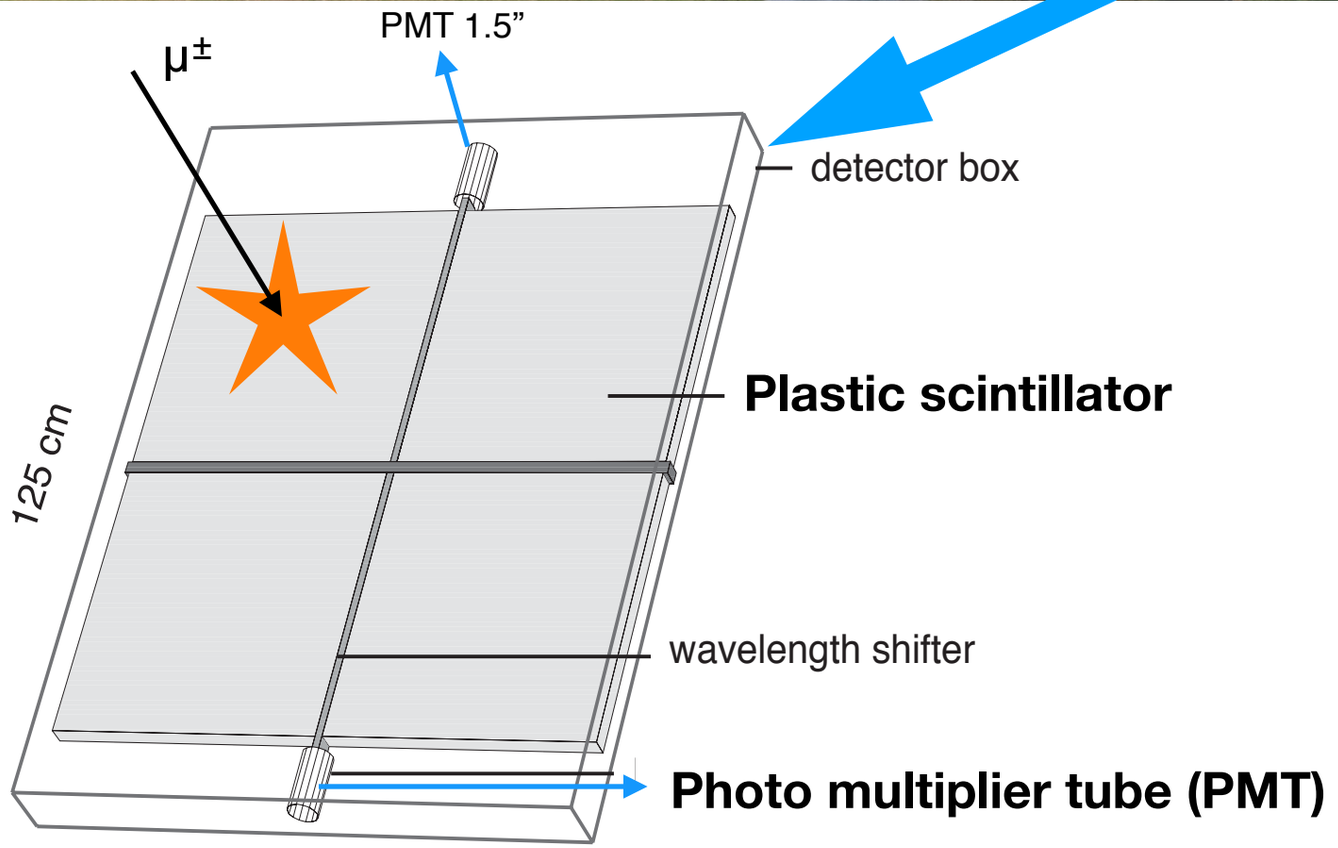
(2) LOFAR radio telescope in the Netherlands

(3) Fluorescence telescopes

Cosmic-ray experiments: (1) Particle detector array

Scintillation detector (plastic or liquid)

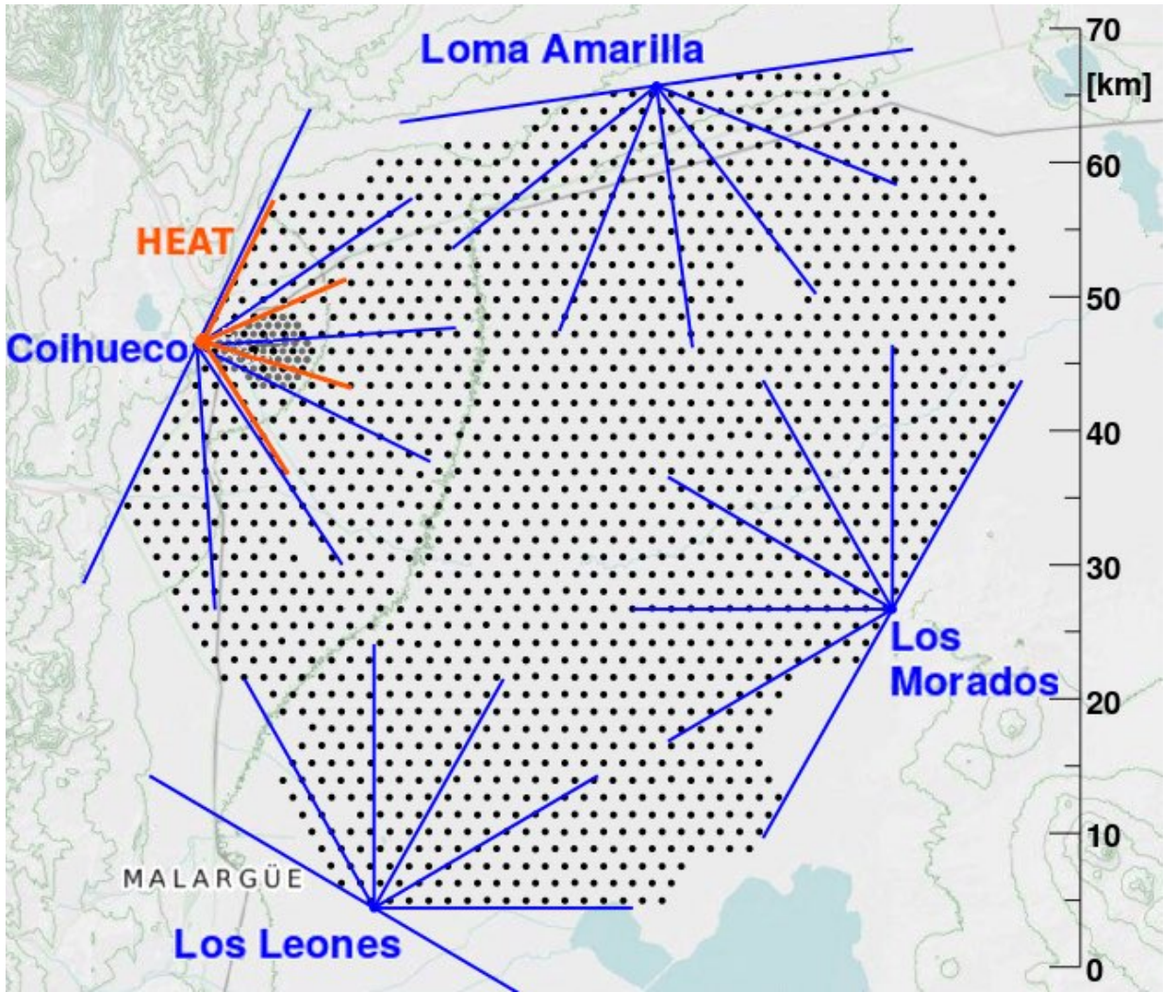




Cosmic-ray experiments: (1) Particle detector array

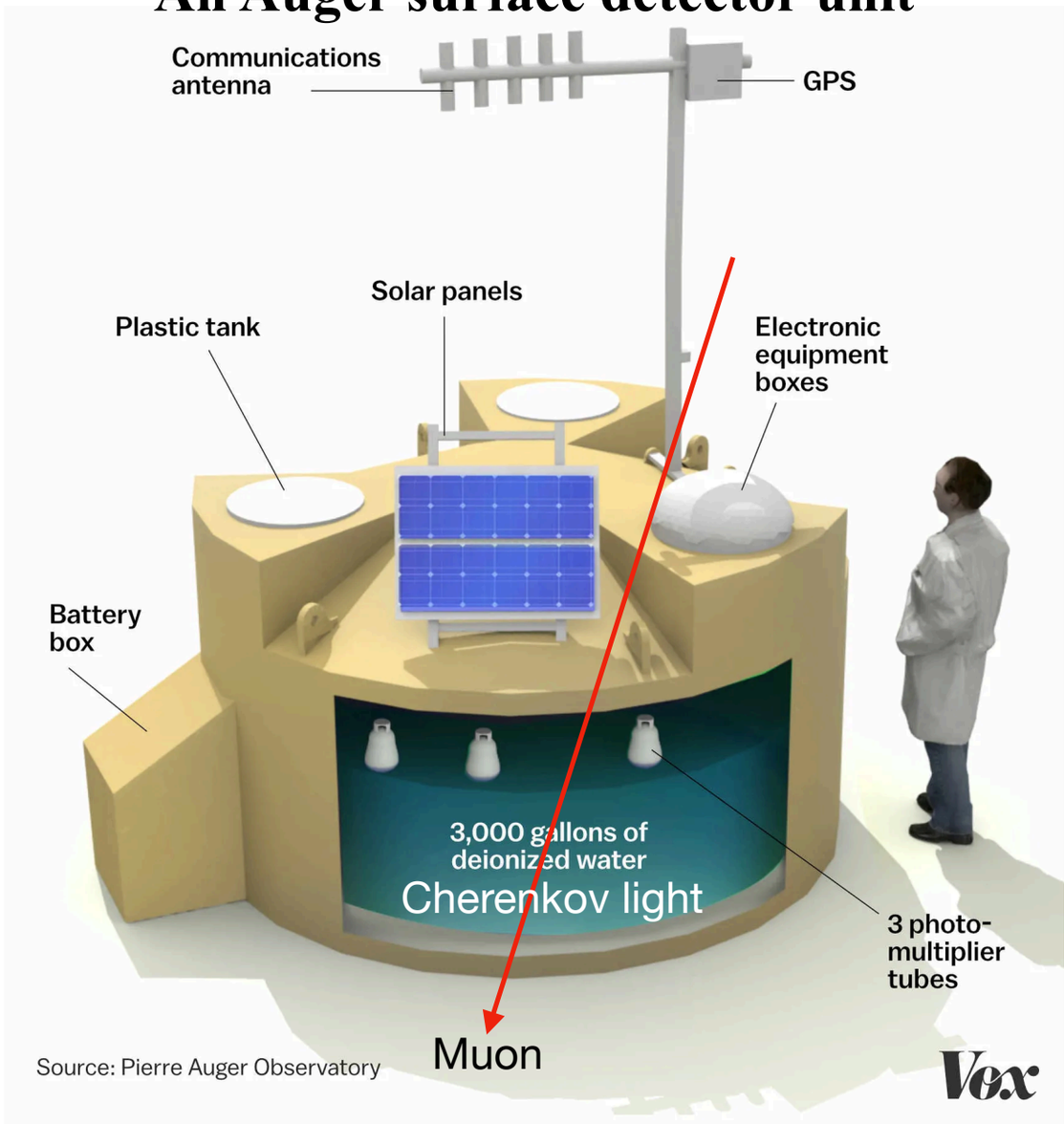
Water Cherenkov detector

Pierre Auger Observatory



https://www.preprints.org/manuscript/201810.0656/v1

An Auger surface detector unit

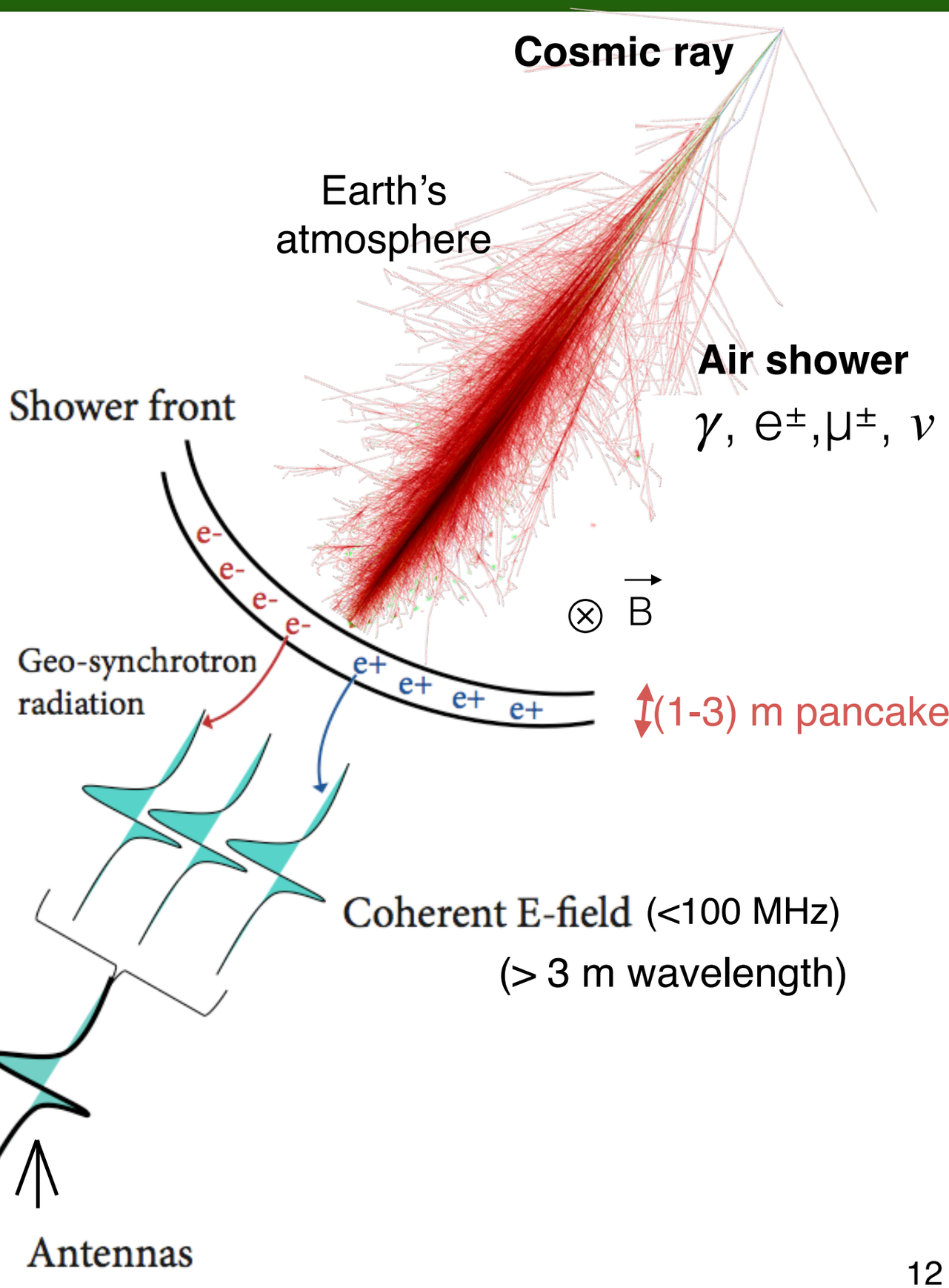


Introduction to the radio detection technique of Cosmic Rays

Radio emission from Extensive air showers

- Mainly geo-synchrotron radiation: Up to ~80%.
- Charge excess (Askaryan effect): Below ~20%.
- Subdominant effects:
  - Cherenkov radiation
  - Transition radiation
  - ….

- Coherence effect mainly below ~100 MHz.
- Corresponds to wavelength above 3 m.
- Shower disc thickness ~ (1-3) m.

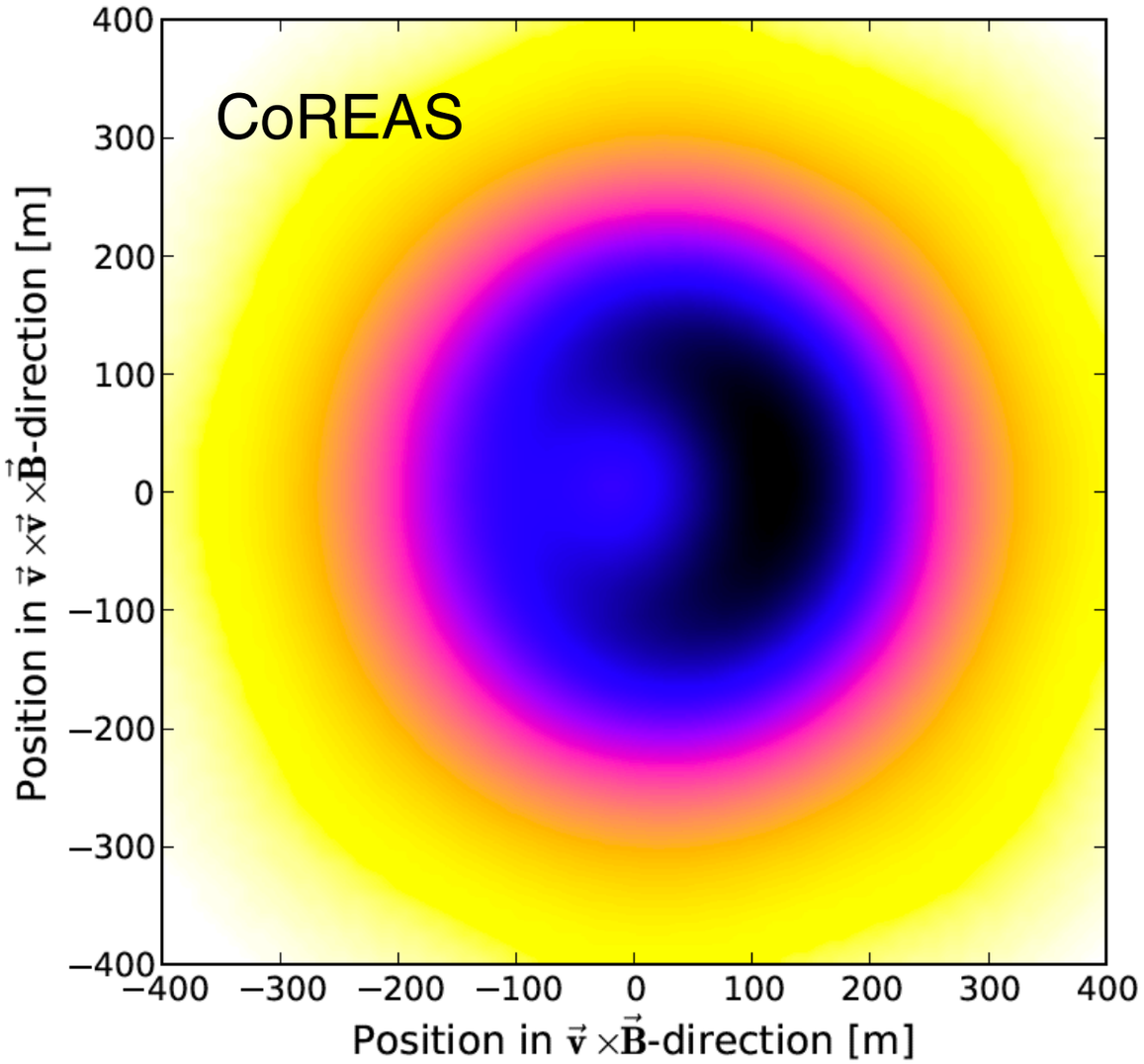


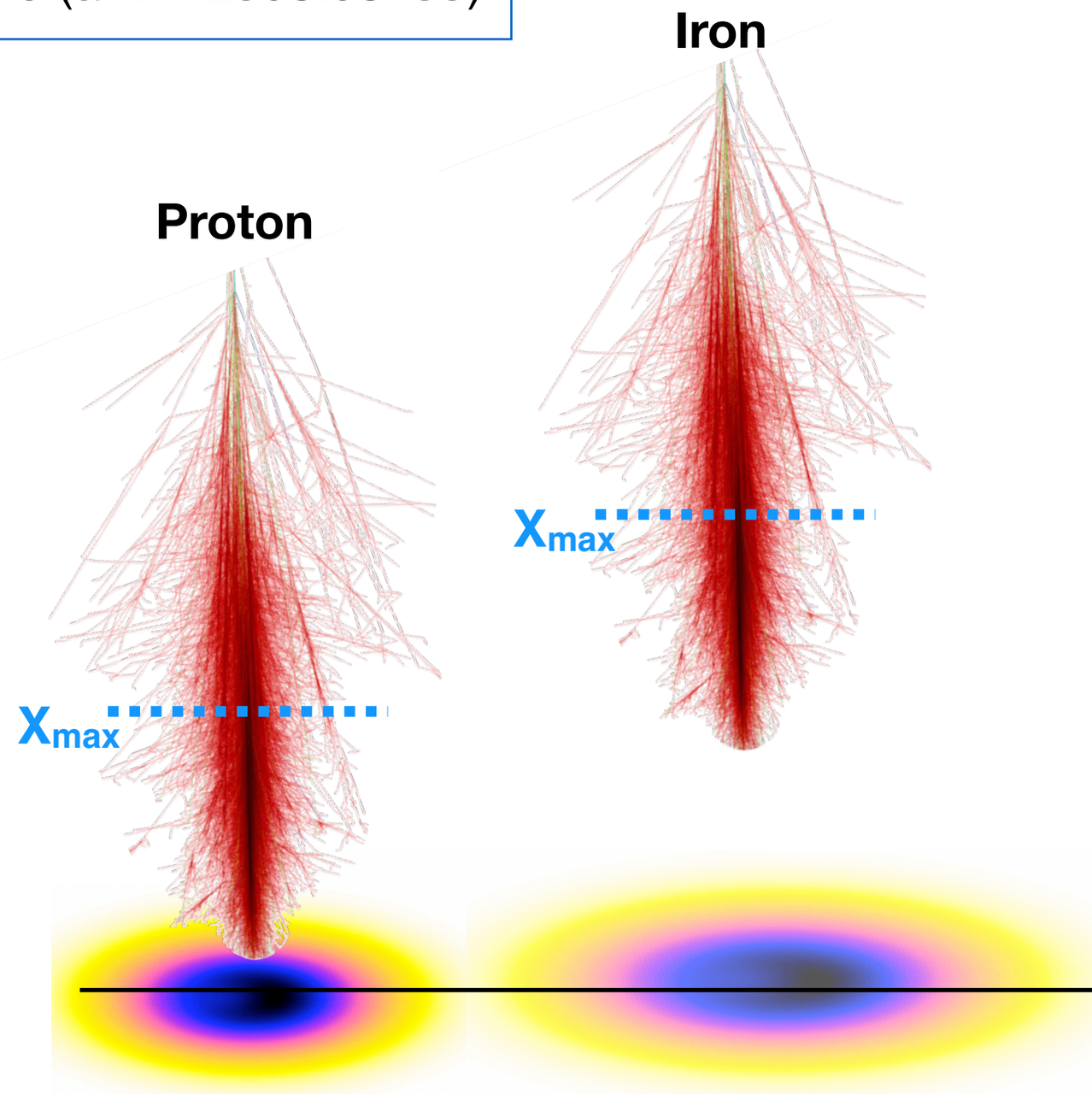
Introduction to the radio detection technique of Cosmic Rays

Simulation of Radio emission from air showers

- CORSIKA 7.0 + CoREAS simulation package OR ZHAireS.
- Radio packages now inbuilt in CORSIKA 8.0 (arxiv: 2508.08755).

Radio footprint in the shower plane





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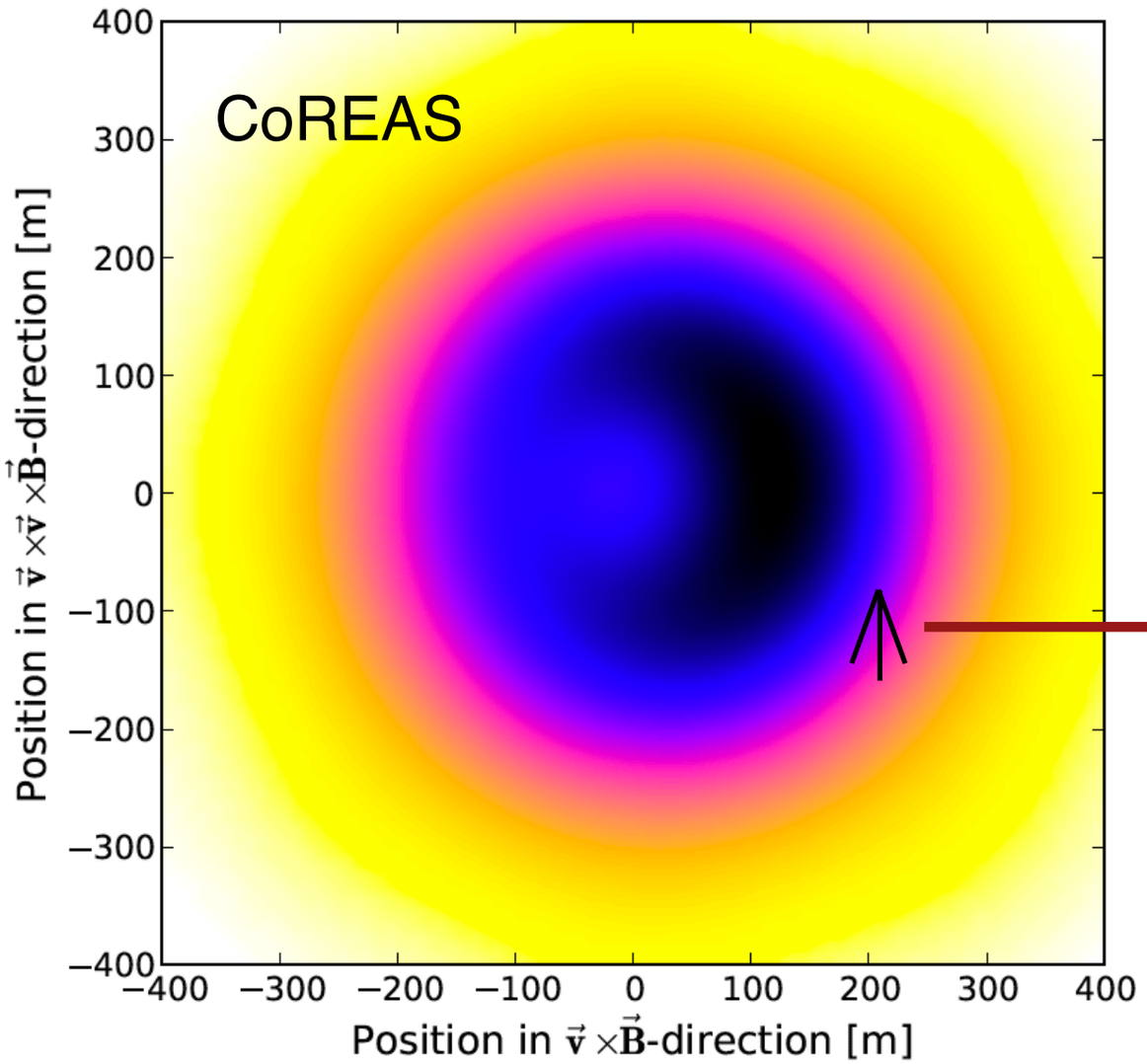
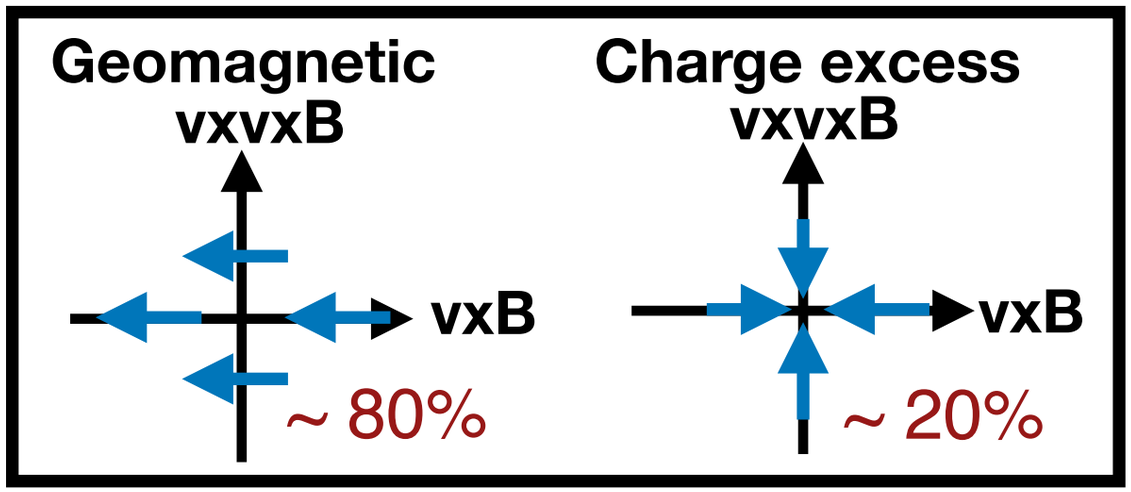
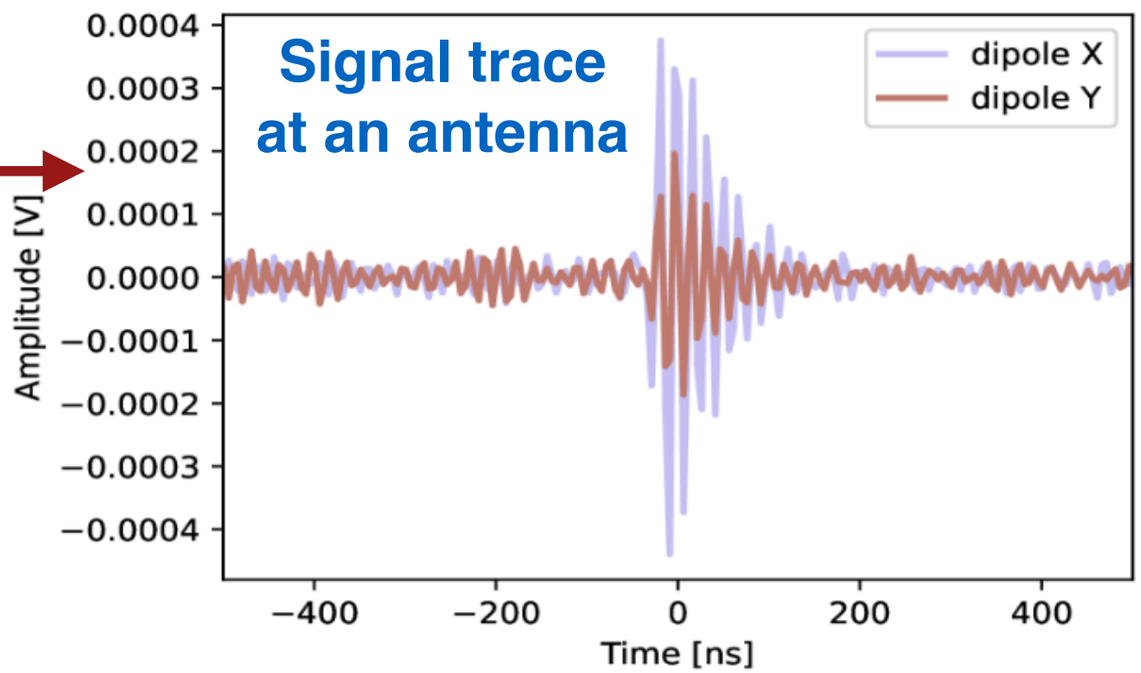


Polarization of the radio signal





Introduction to the radio detection technique of Cosmic Rays

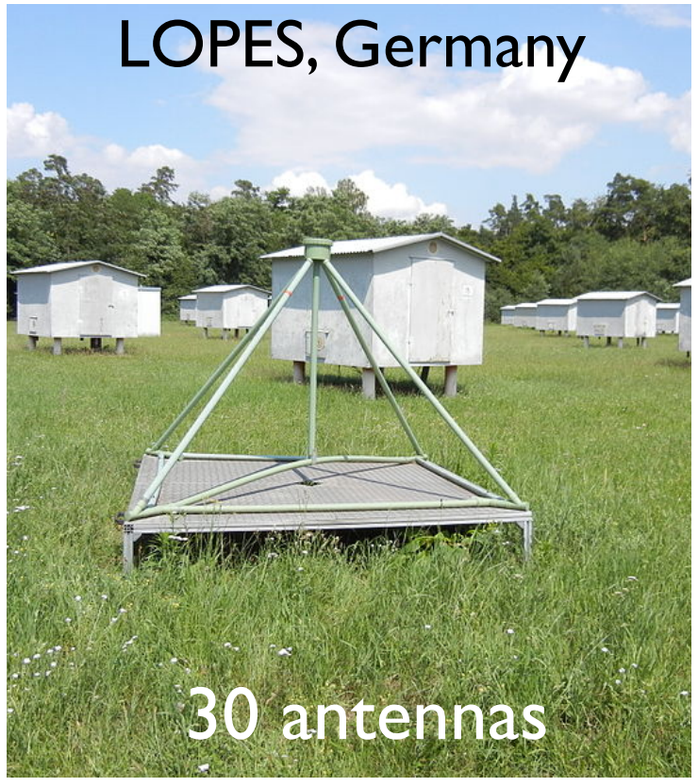
A brief history

- First radio signal detection: Jodrell Bank Observatory (Jelly+ 1965).
- Theoretical works: Kahn & Lerche (1968); Colgate (1967)
- Problems with noise, and other technical limitations.
- Not as successful as other techniques.
- Much later, in 2000s, new experiments came up.
- Modern computers, advanced electronics, …
- Radio detection technique revived.

Jelley et al. (1965)

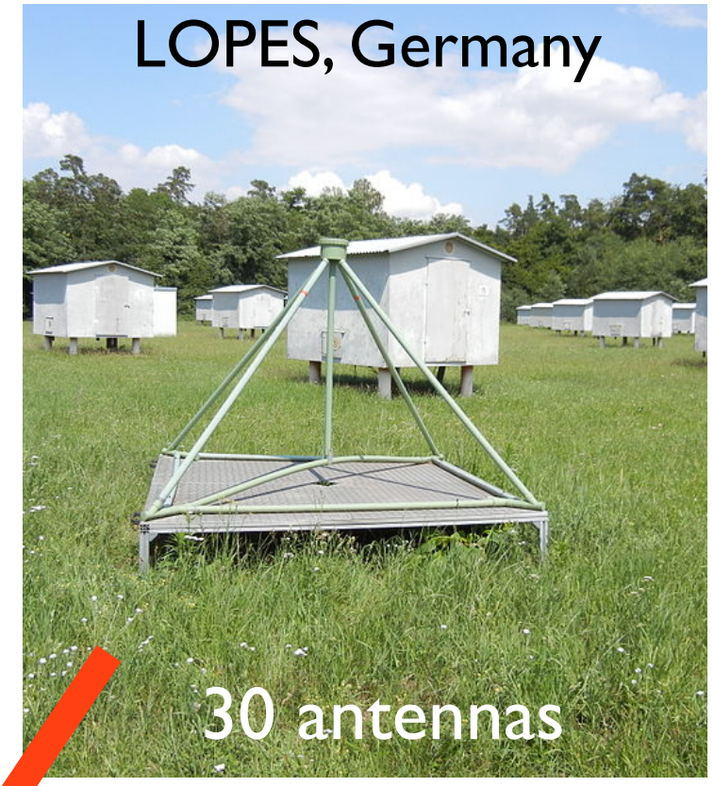
Introduction to the radio detection technique of Cosmic Rays

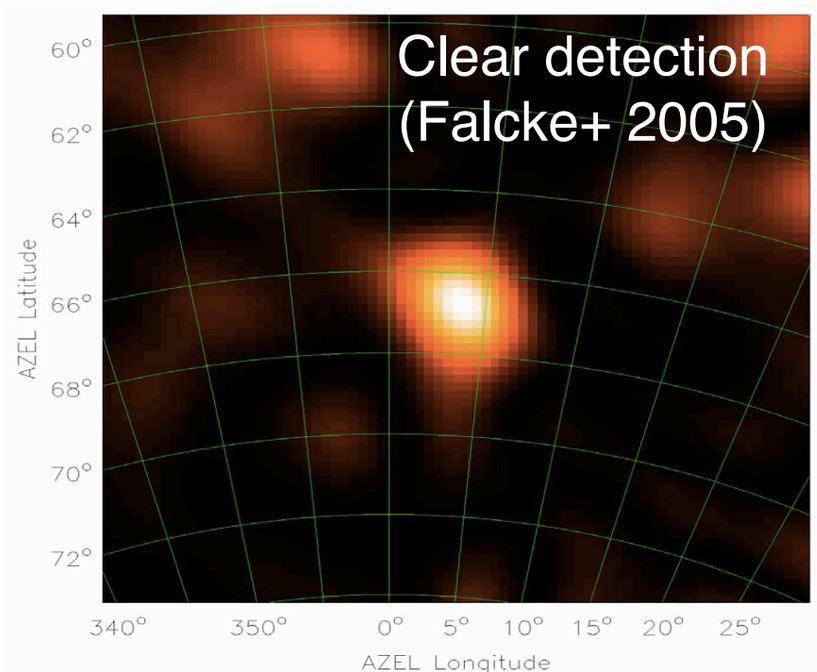
Past and Present experiments

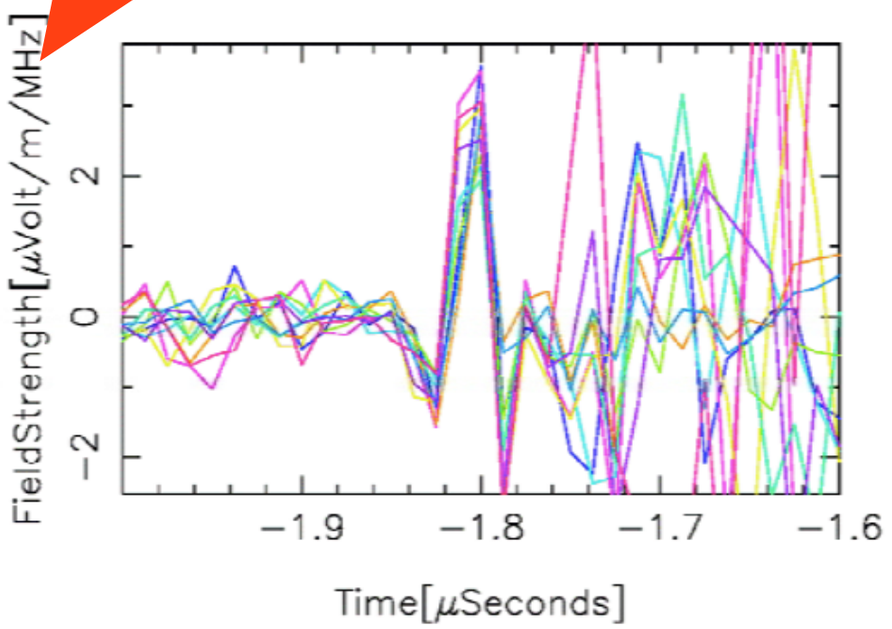


Introduction to the radio detection technique of Cosmic Rays

Past and Present experiments







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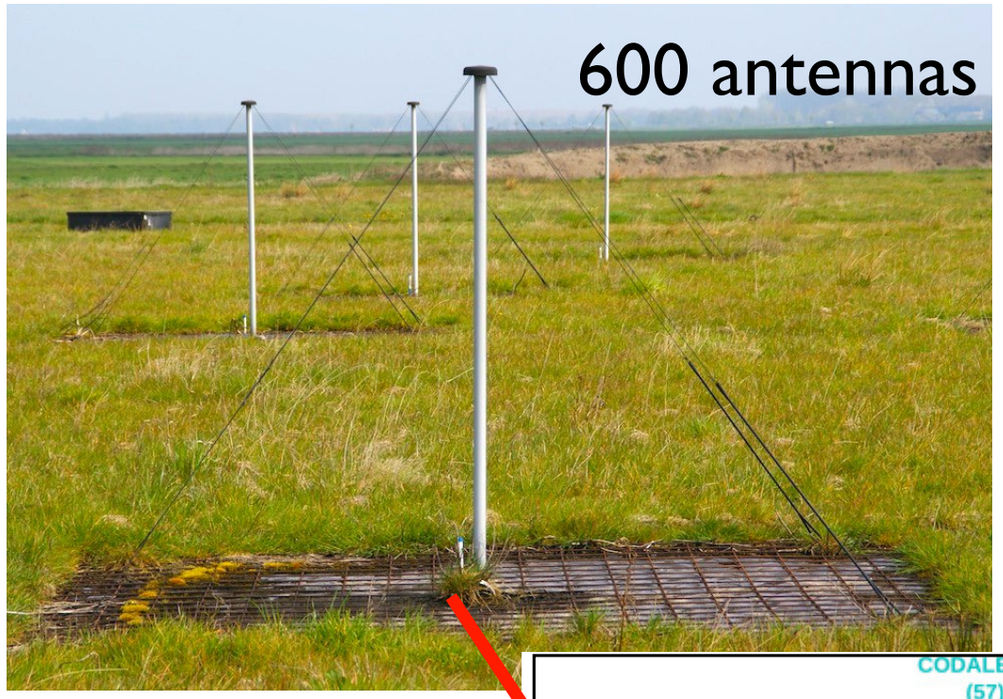
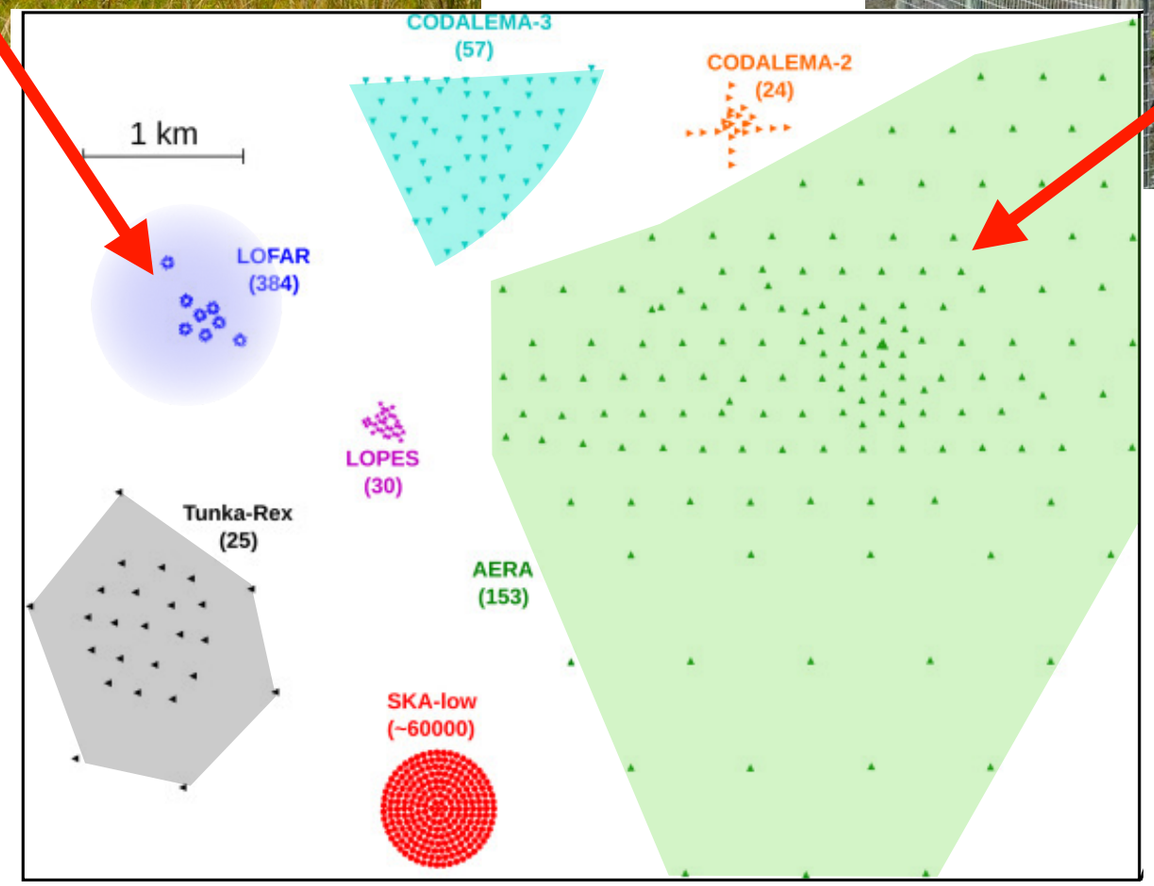
Introduction to the radio detection technique of Cosmic Rays

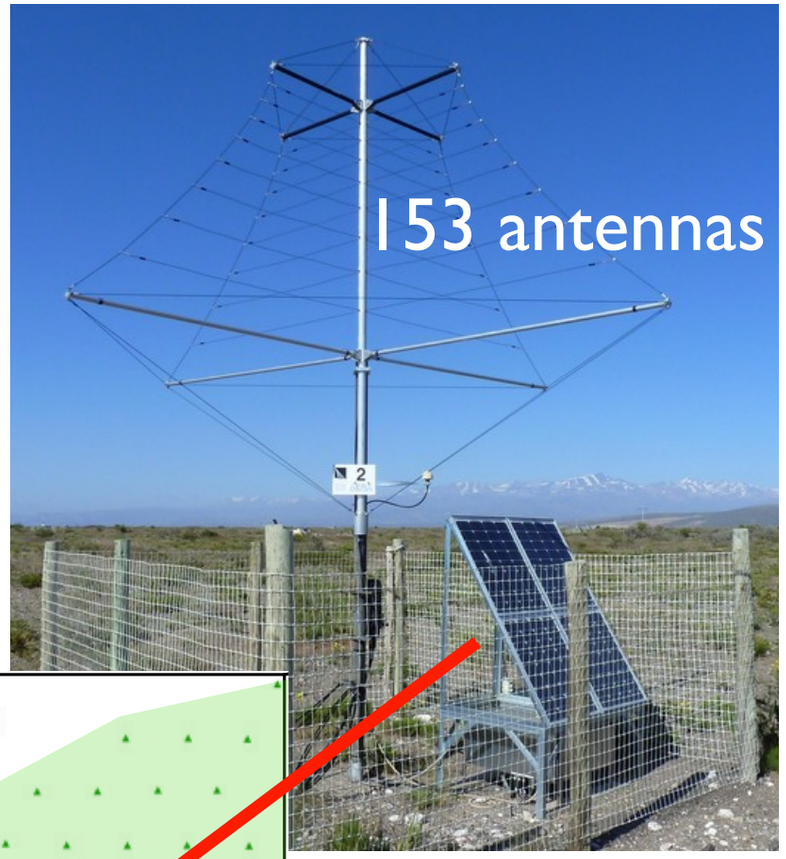
LOFAR, Netherlands+



Densest array at present



AERA, Argentina



Largest array at present

T. Huege, Physics Reports, 620 (2016)

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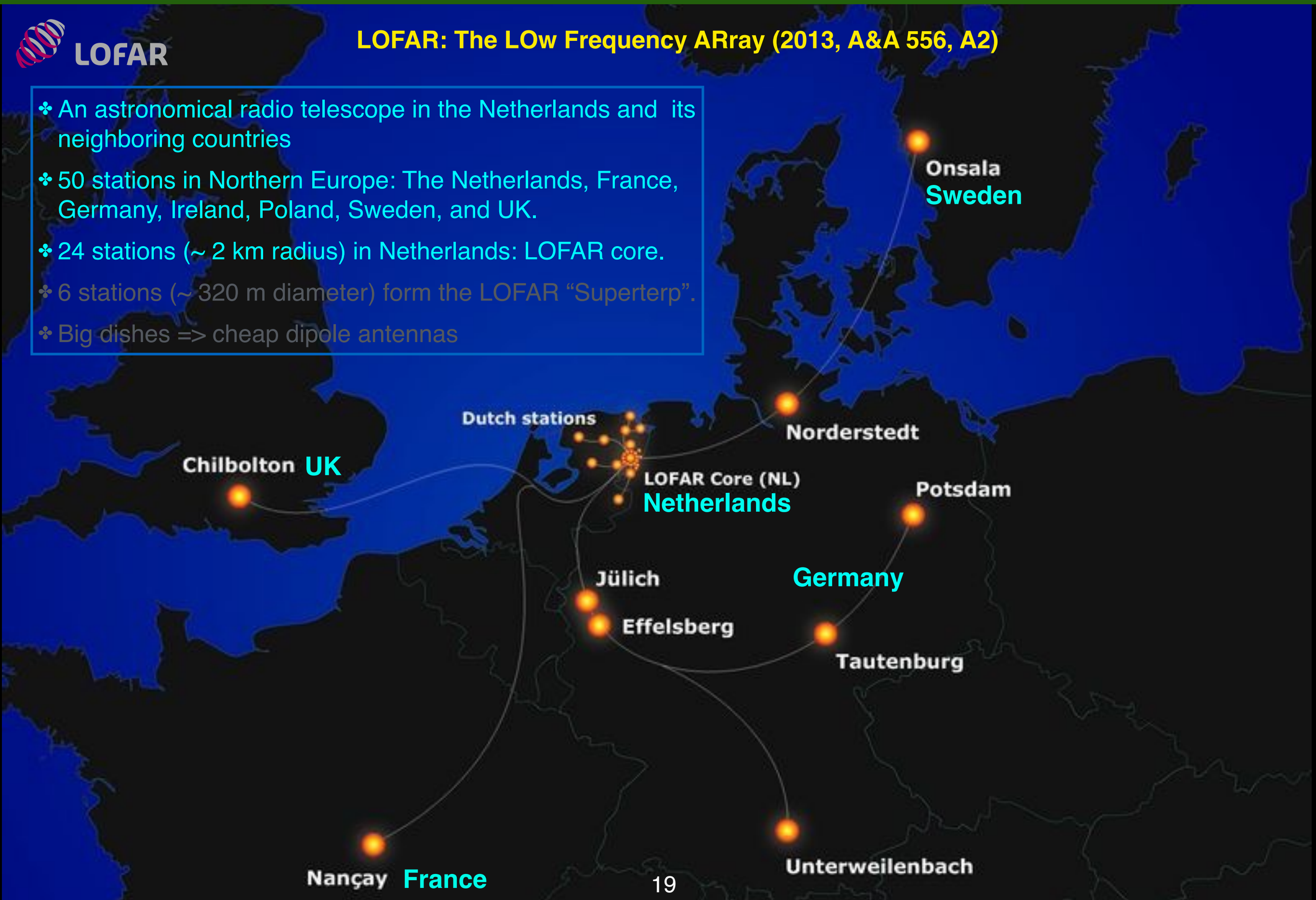
Cosmic-ray experiments: (2) Radio Antenna array

LOFAR

LOFAR: The LOw Frequency ARray (2013, A&A 556, A2)

- An astronomical radio telescope in the Netherlands and its neighboring countries
- 50 stations in Northern Europe: The Netherlands, France, Germany, Ireland, Poland, Sweden, and UK.
- 24 stations (~ 2 km radius) in Netherlands: LOFAR core.
- 6 stations (~ 320 m diameter) form the LOFAR "Superterp".
- Big dishes => cheap dipole antennas



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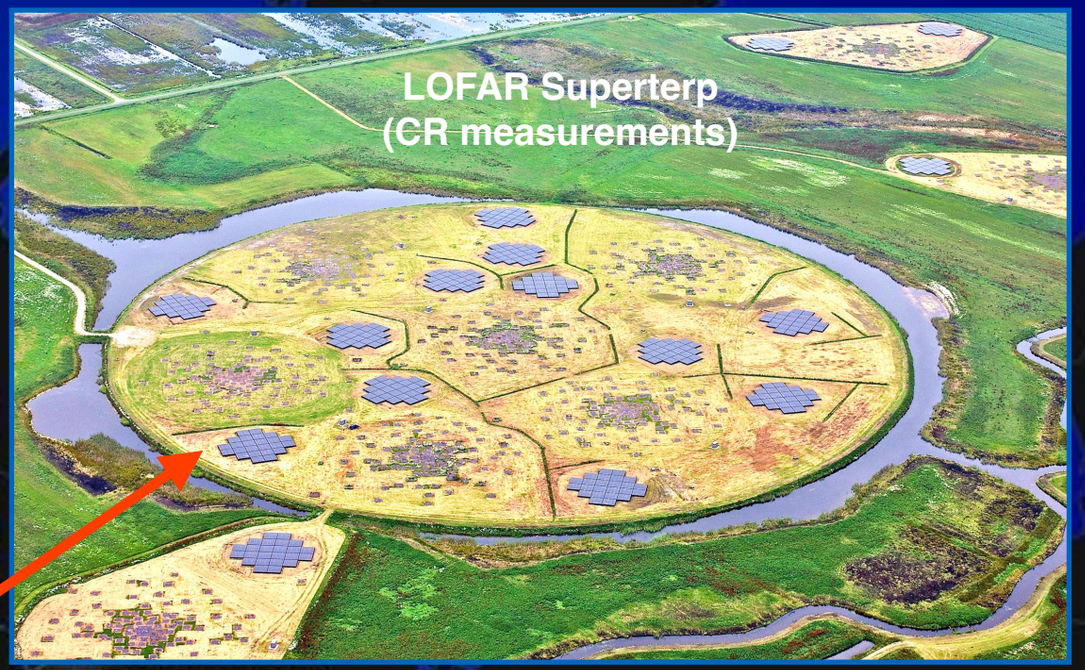

Dutch stations

Norderstedt

Chilbolton UK

LOFAR Core (NL)
Netherlands

Potsdam

Jülich

Germany

Effelsberg

Tautenburg

Nançay France

Unterweilenbach

Cosmic-ray experiments: (2) Radio Antenna array

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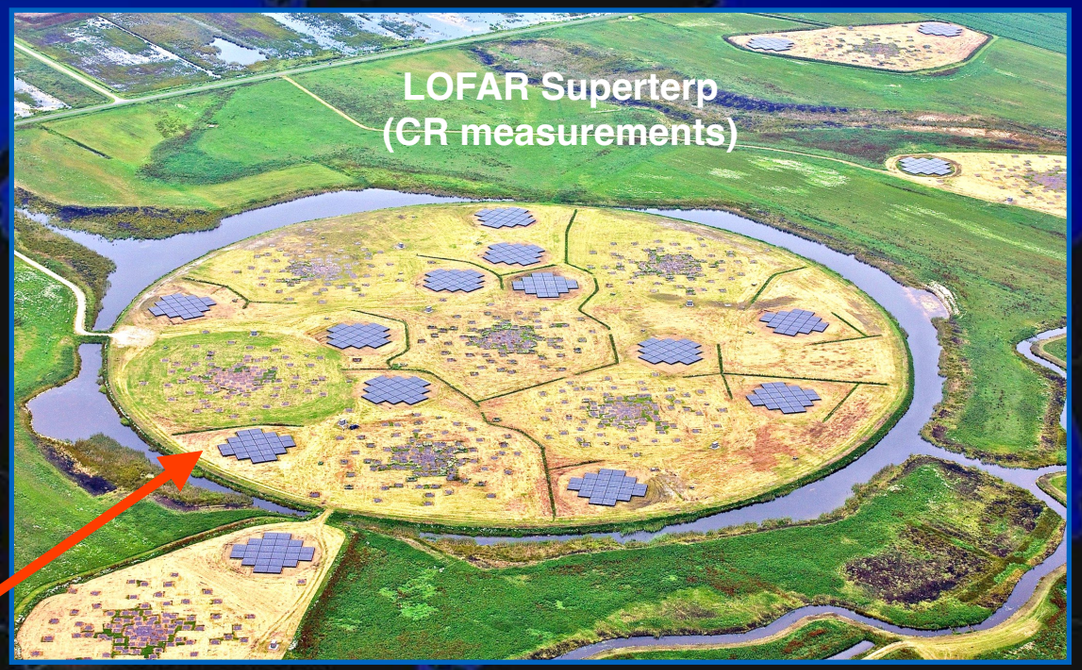


Norderstedt

Dutch stations

Chilbolton UK

LOFAR Core
Netherlar

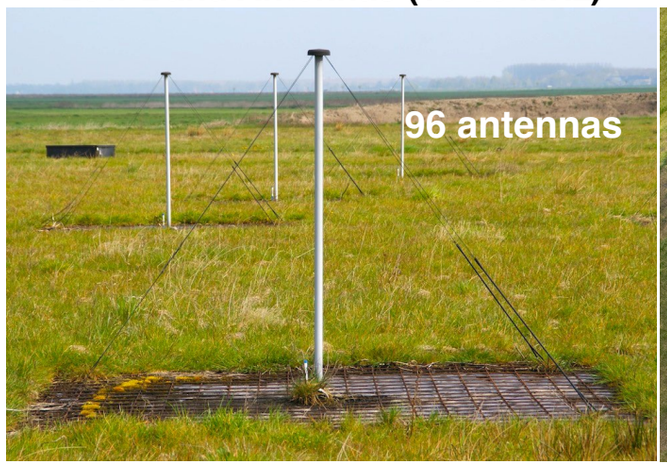
erg

Nançay France

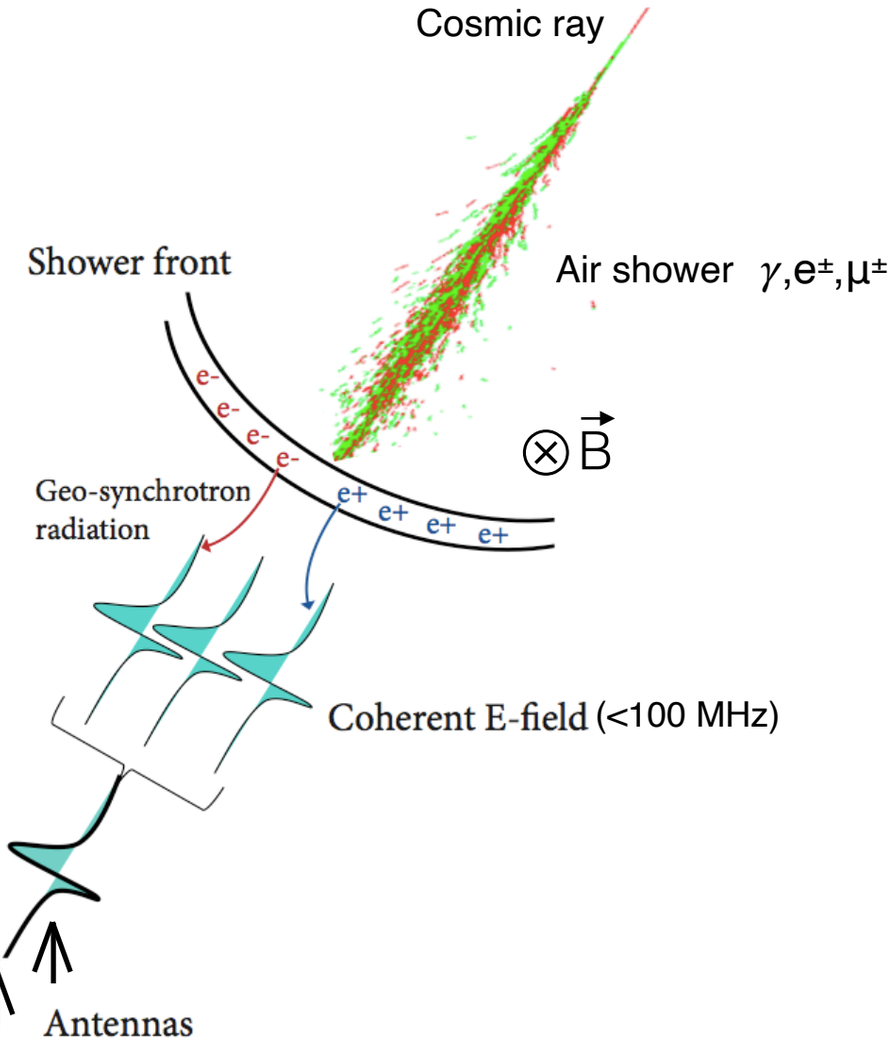
Unterweilenbach

Cosmic-ray experiments: (2) Radio Antenna array

Low Band Antennas (10-80 MHz)



High Band Antennas (100-240 MHz)



Radio emission mechanisms from air showers

- Mainly geo-synchrotron radiation: Up to ~80%.
- Charge excess (Askaryan effect): Less than ~20%.
- Subdominant effects:
  - Cherenkov radiation
  - Transition radiation
  - ....
- Coherence effect mainly below ~100 MHz.

Cosmic-ray experiments: (2) Radio Antenna array

Low Band Antennas (10-80 MHz)

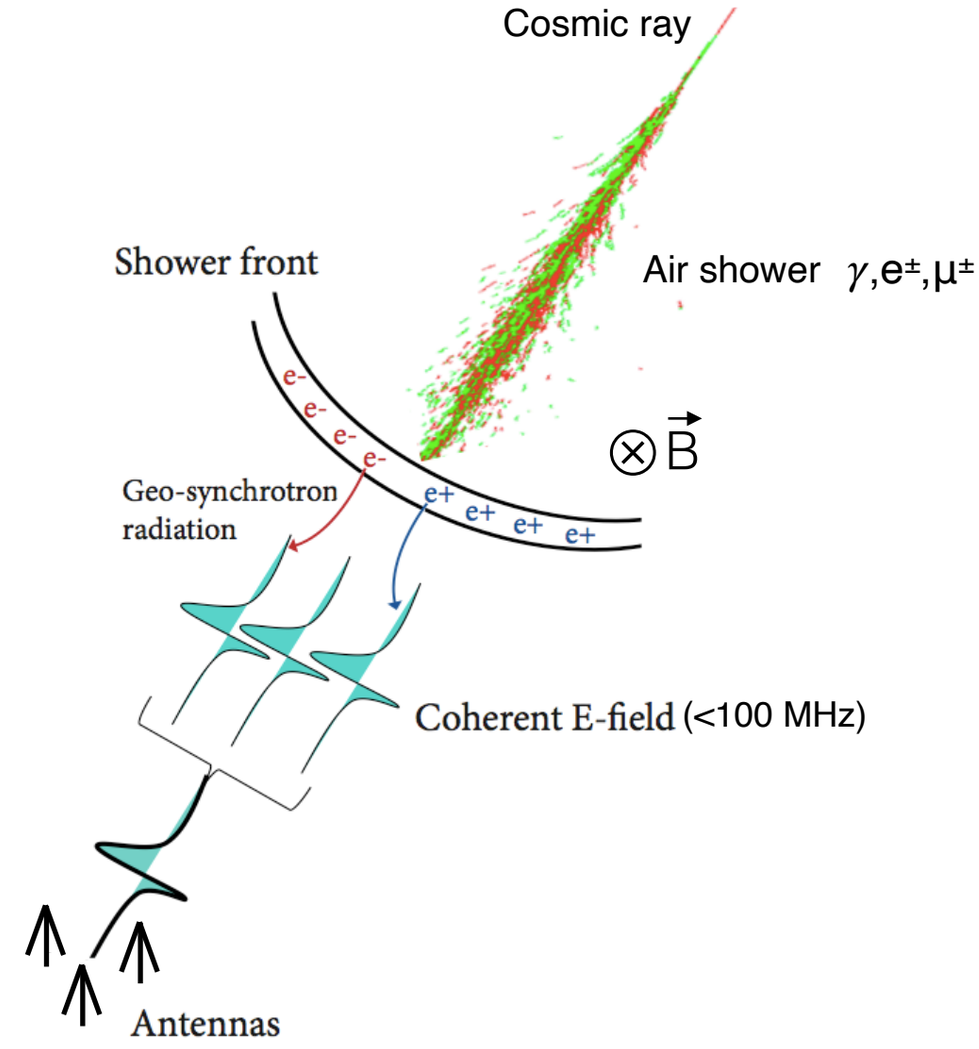
High Band Antennas (100-240 MHz)

Main advantages of radio detection

- Cheap detectors.
- Almost 100% duty cycle.
- CR mass (X_{max}) sensitivity less than 20 g/cm^2.

Major science goals of LOFAR

- CR composition measurement ~$10^{16\text{-}18}$ eV.
- Understanding radio emission of air showers.



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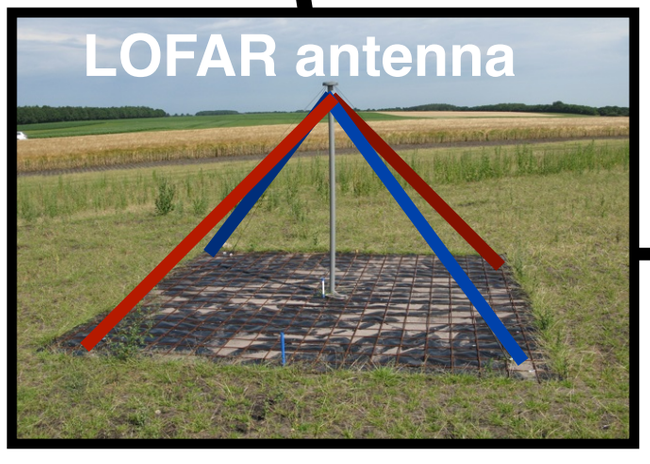
Cosmic-ray detection strategy with LOFAR

LORA particle detector array

- LOFAR Radboud Array (LORA) scintillators
- Provides trigger for antenna readout

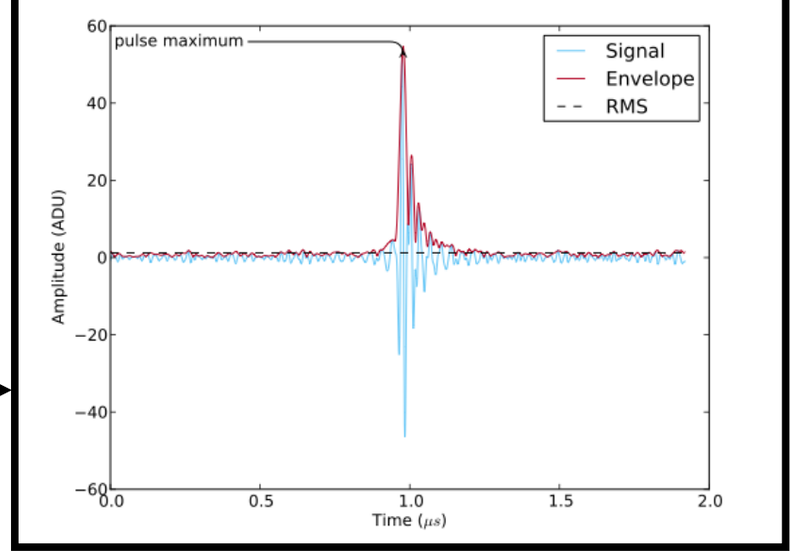
trigger



buffer

2 ms read-out

offline analysis



- RFI cleaning
- direction reconstruction
- antenna pattern unfolding
- calibration

P. Schellart et al., A&A 560, 98 (2013)

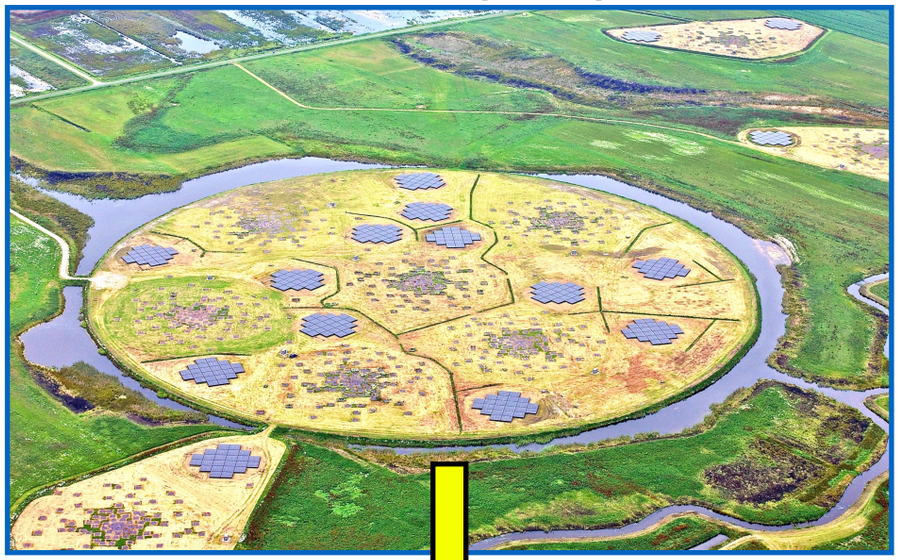
LOFAR Radboud Air Shower Array (LORA)

Thoudam+ 2014, NIMA, 767, 339
Thoudam+ 2016, APh, 73, 34

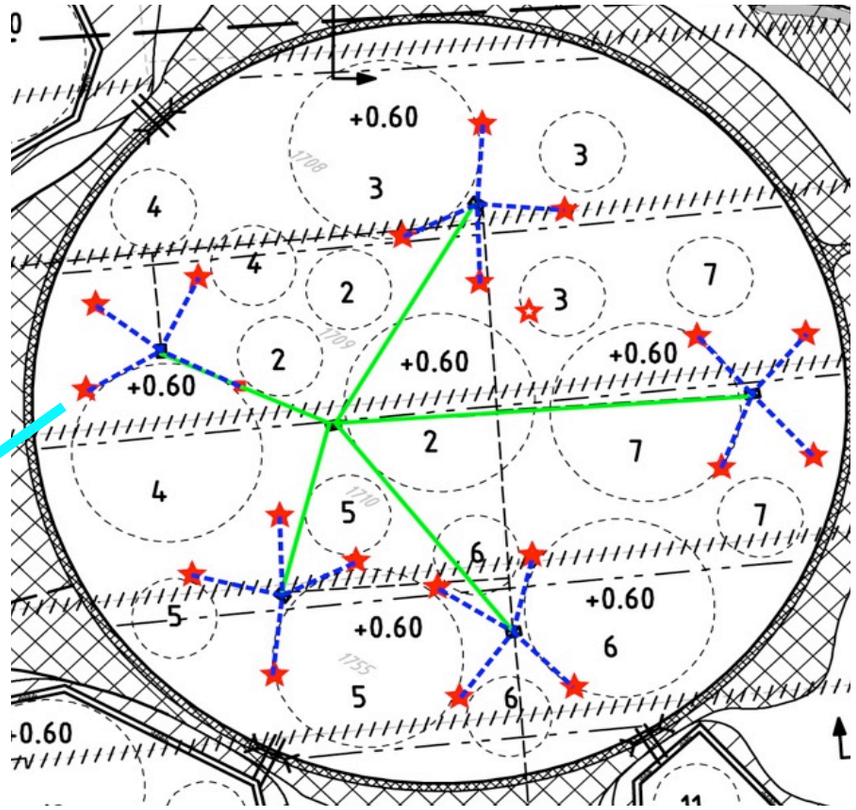
- 20 plastic scintillation detectors in the LOFAR Superterp.
- Detector size ~ 1m x 1m, distributed over ~300 m ring.
- Sensitive to cosmic rays above ~10^{16} eV.

- **Primary purpose:**
  - To trigger LOFAR antennas for cosmic-ray measurements.
  - Provide basic air-shower parameters: Shower core position, arrival direction and energy.

LOFAR Superterp



LORA array configuration



Air Shower Array (LORA)

+ 2014, N
+ 2016, A

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- Sens
- Prim
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  - P
    p

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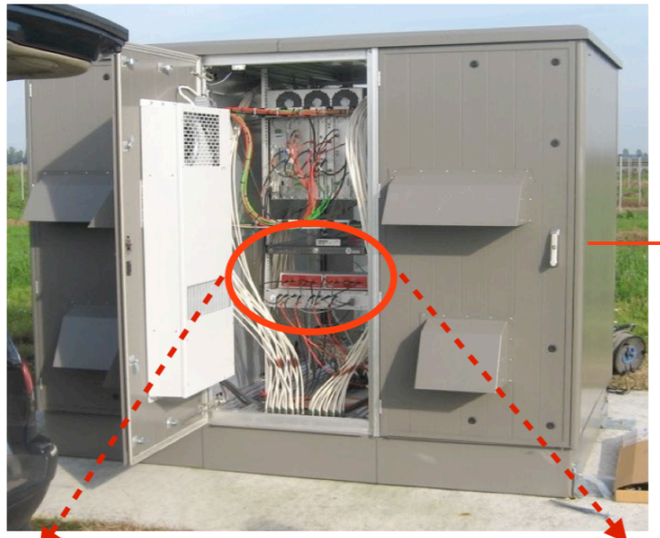
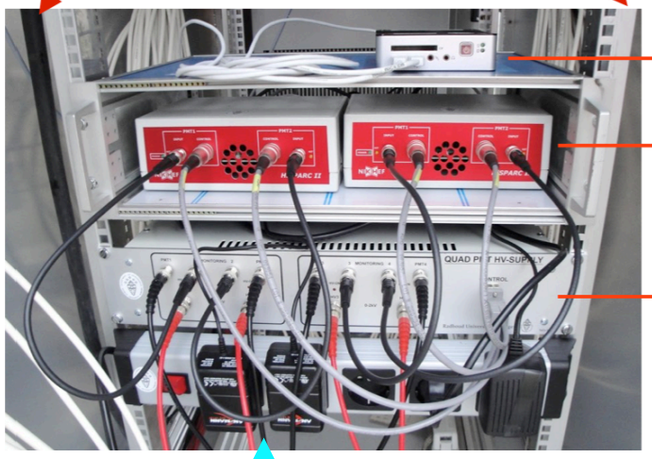
easureme
r core

Electronics

Developed for the HISPARC experiment
(www.hisparc.nl)



LOFAR Station cabinet



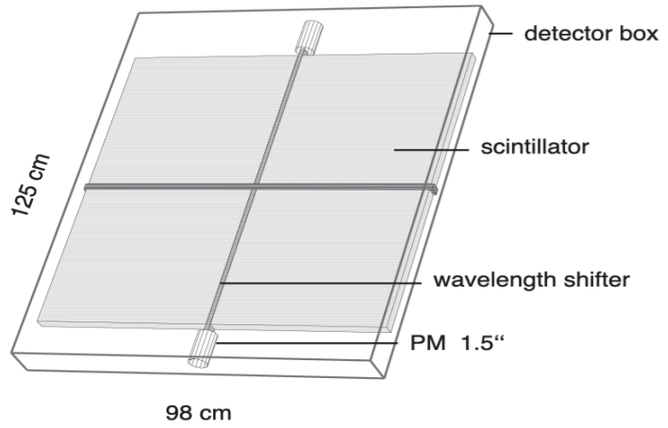
Station-PC

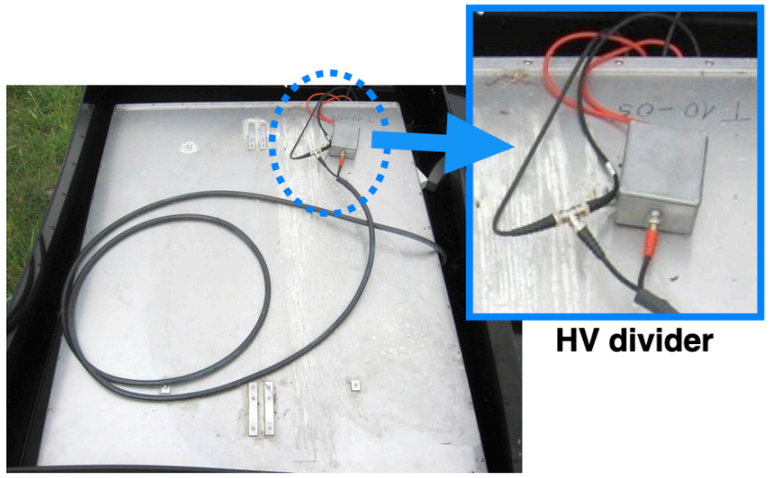
Digitizers

HV supply

Detector

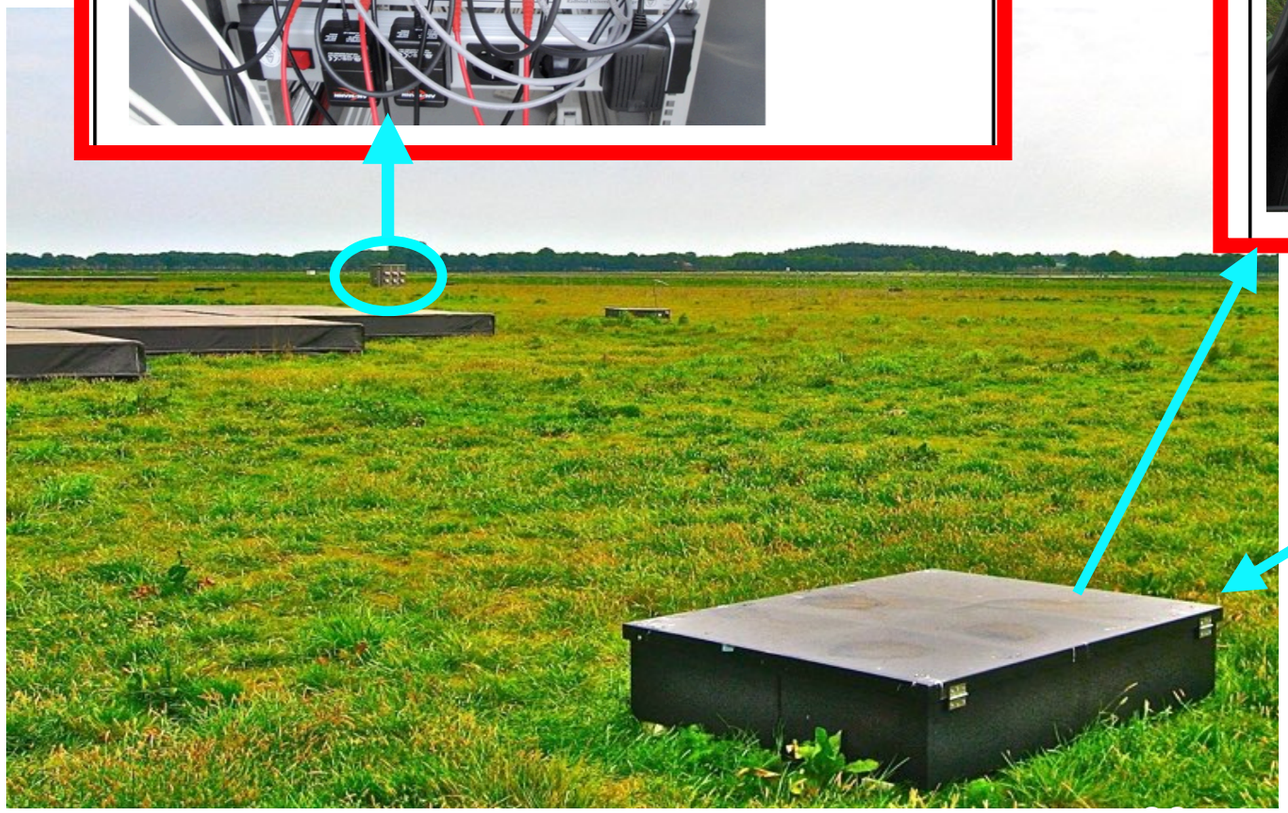
Taken over from the KASCADE experiment (Antoni et al. 2003)

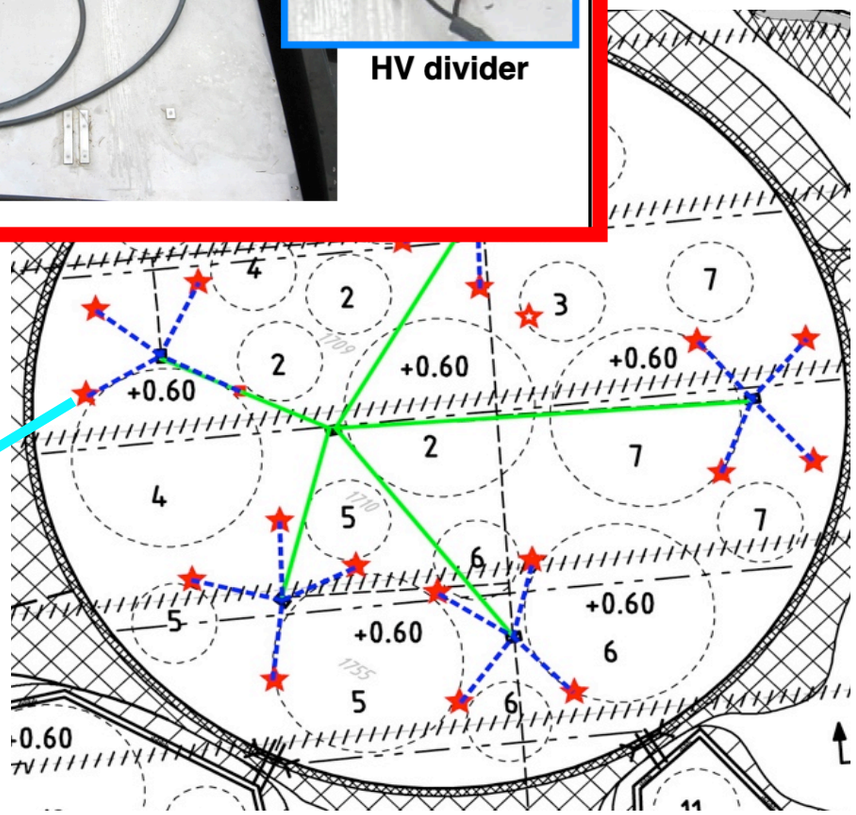




HV divider

ration





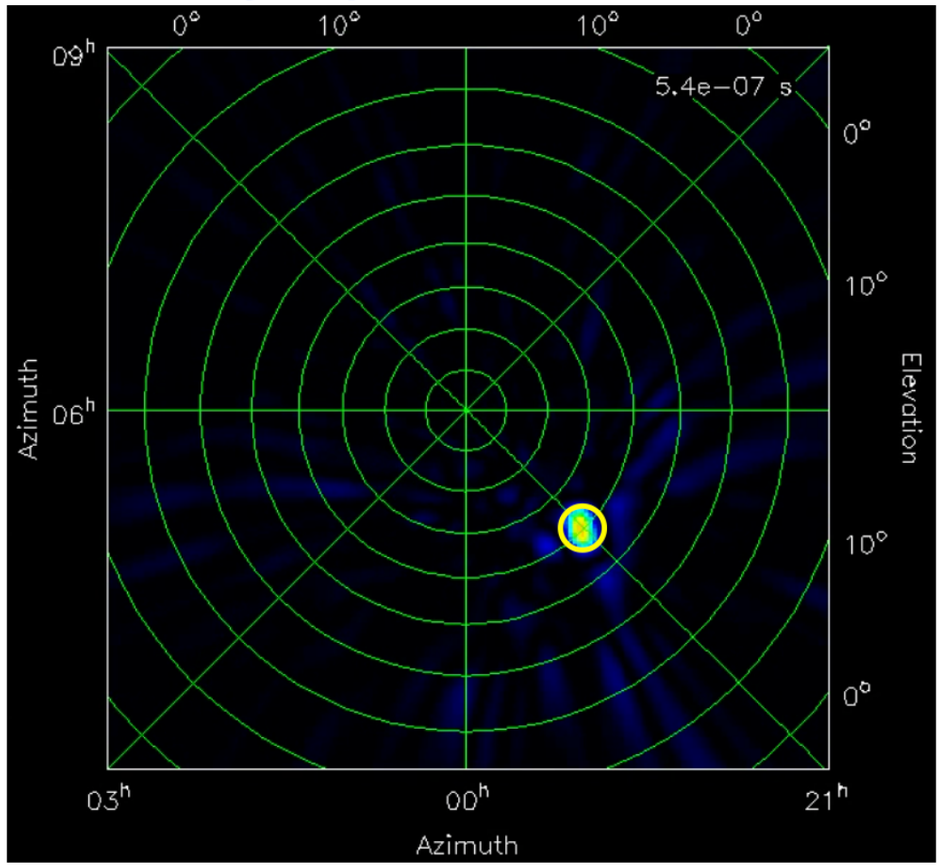
Some highlights from LOFAR cosmic-ray measurements:

A measured cosmic-ray event

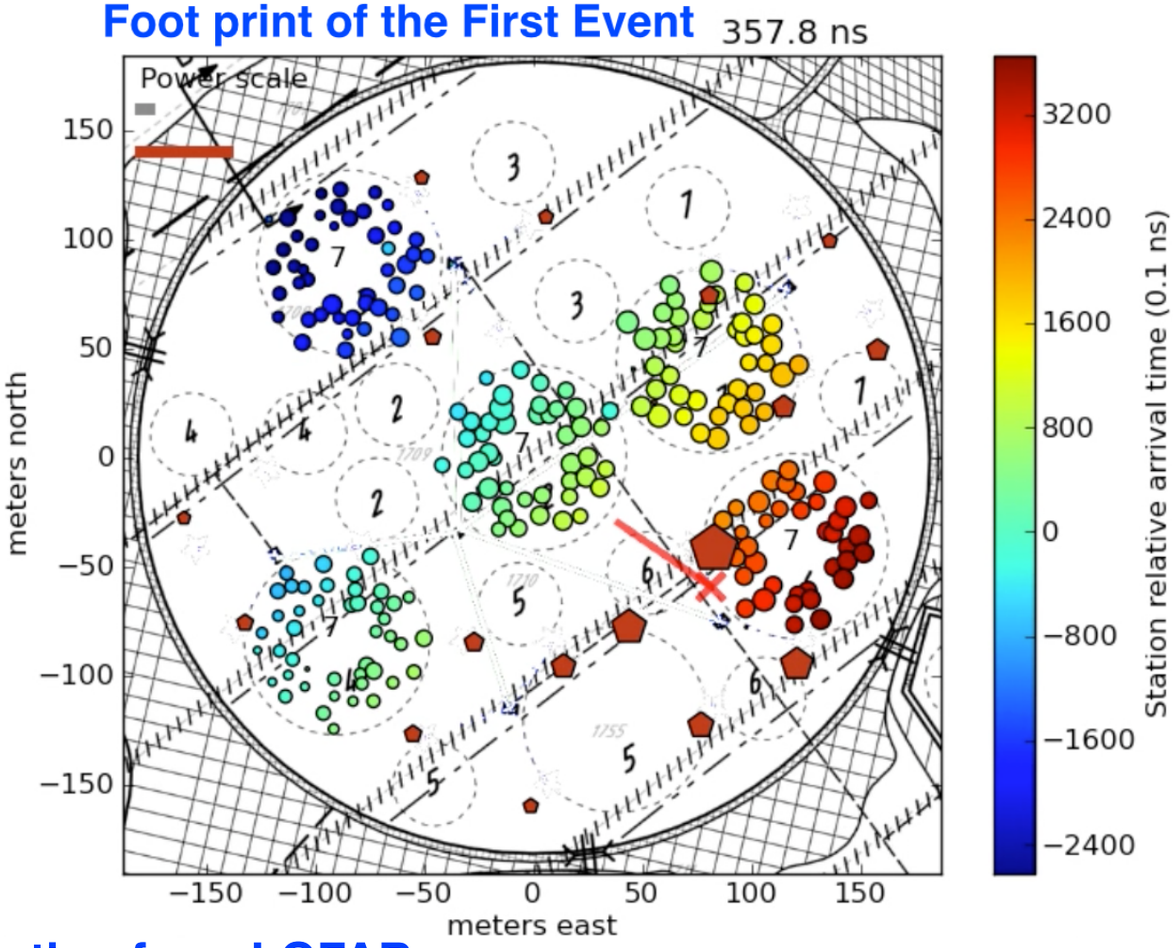
Schellart+ 2013, A&A, 560, A98

Credit: P. Schellart

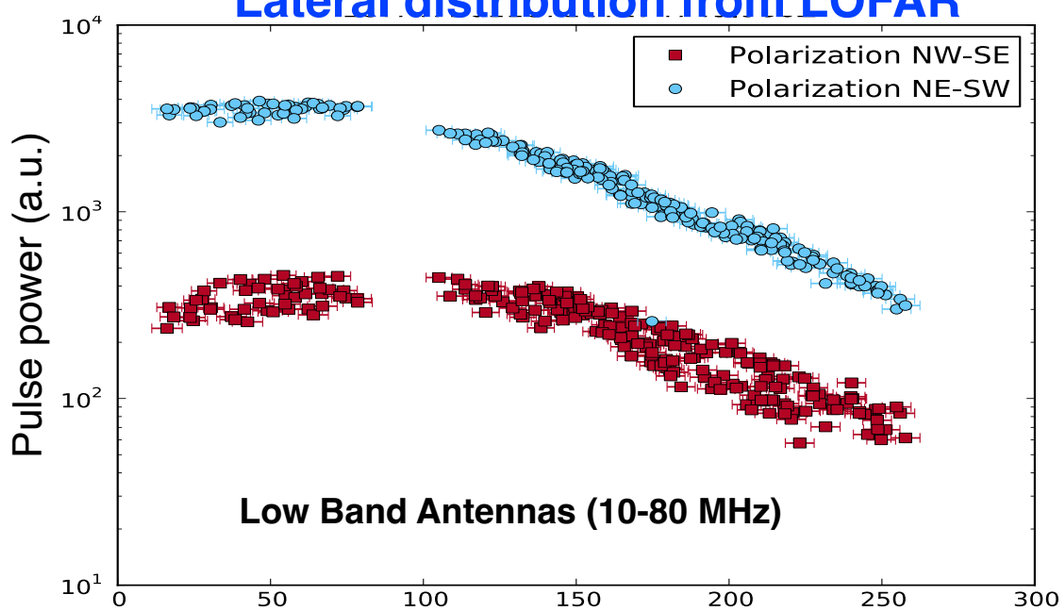
Sky map of the First Event



Foot print of the First Event



Lateral distribution from LOFAR



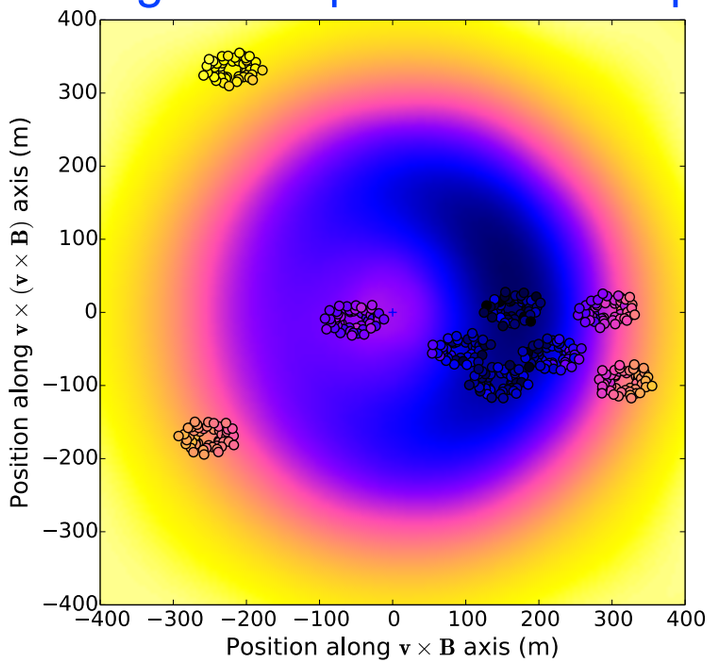
Some highlights from LOFAR cosmic-ray measurements:

Precision measurement of radio footprint of air showers

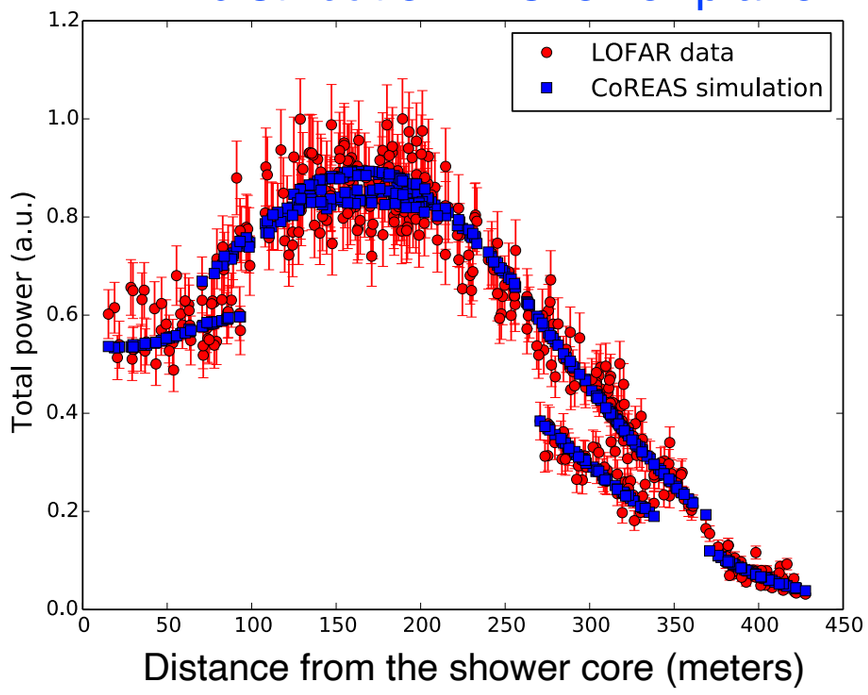
More events (comparison with simulation)

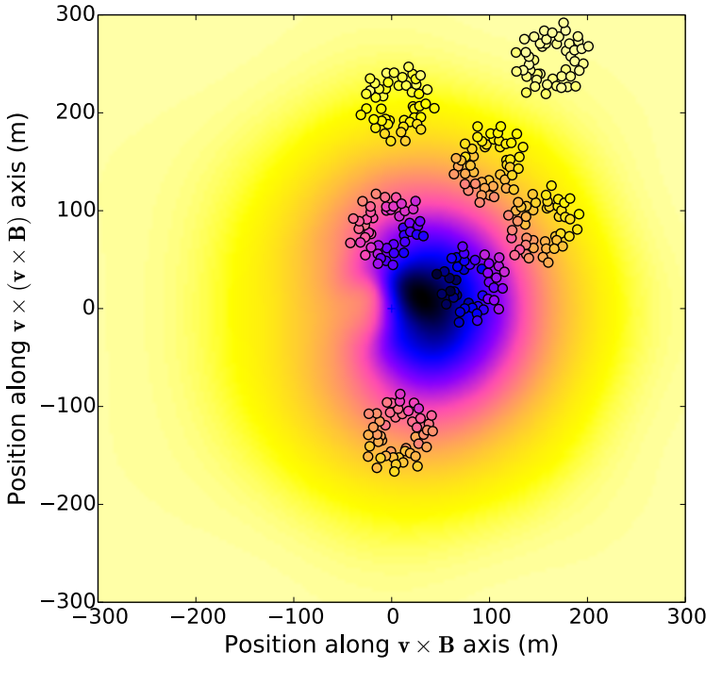
Buitink+ 2014, PRD, 90, 082003

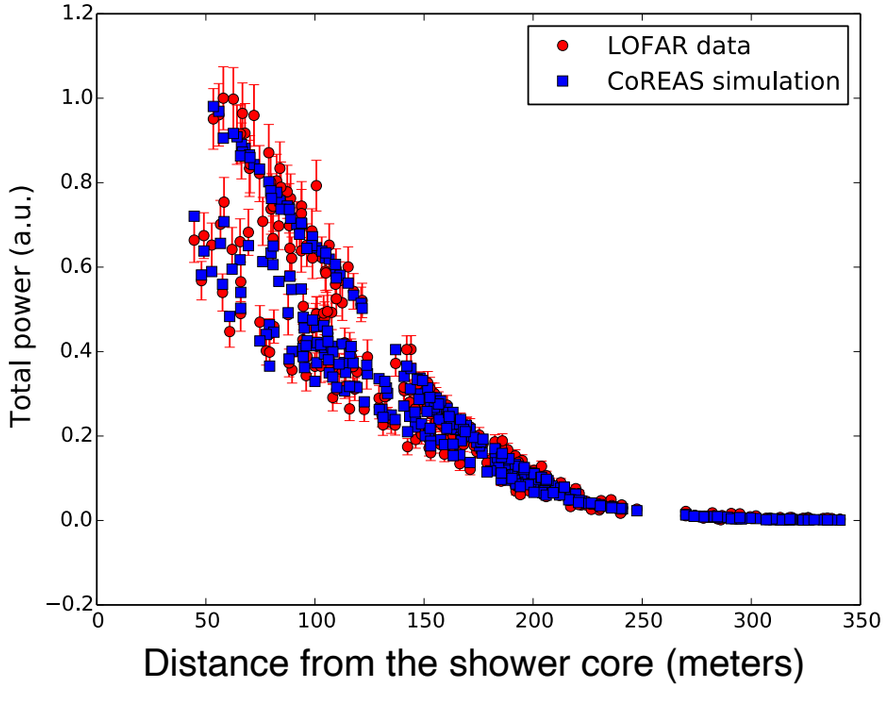
2D signal footprint in shower plane



1D distribution in shower plane

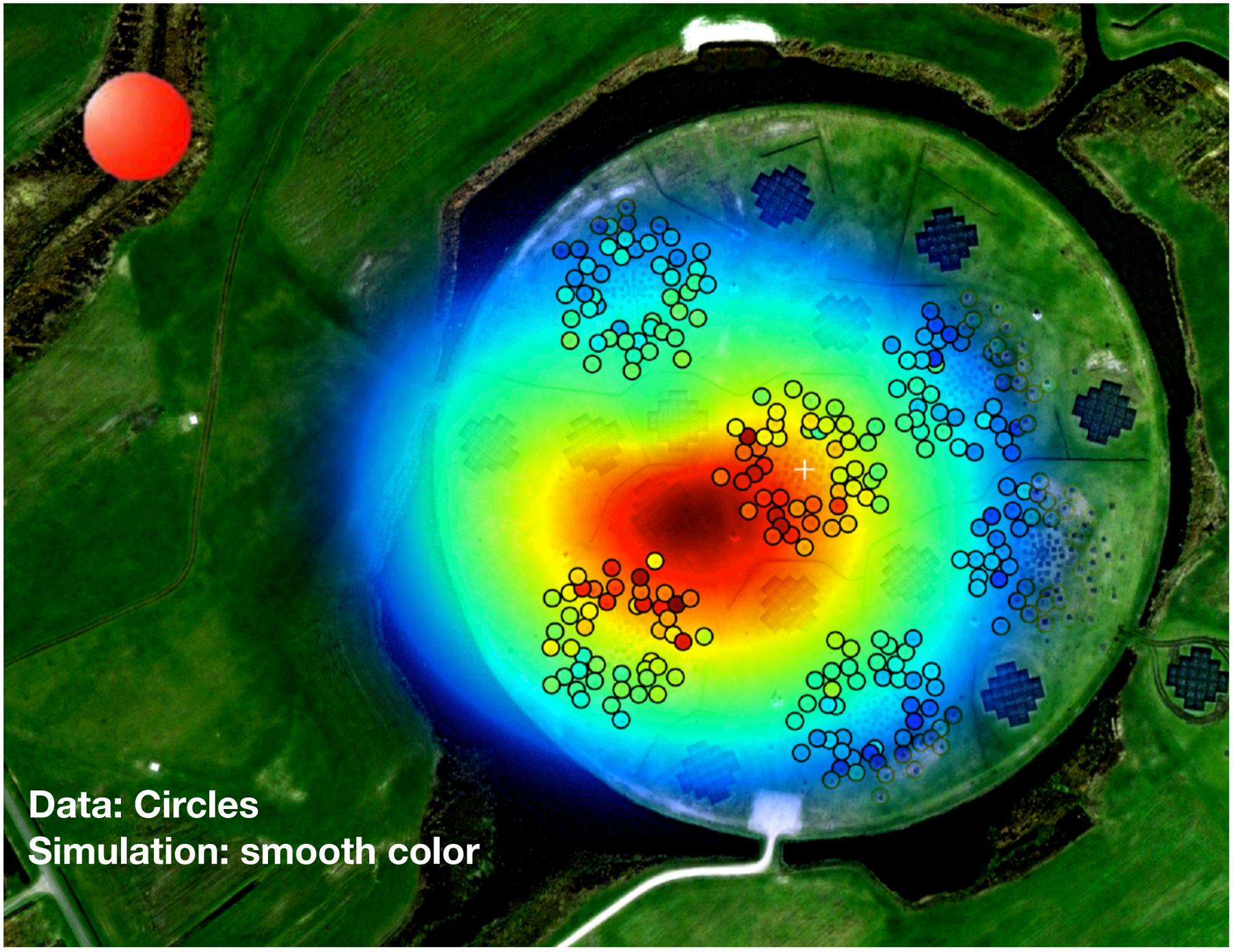


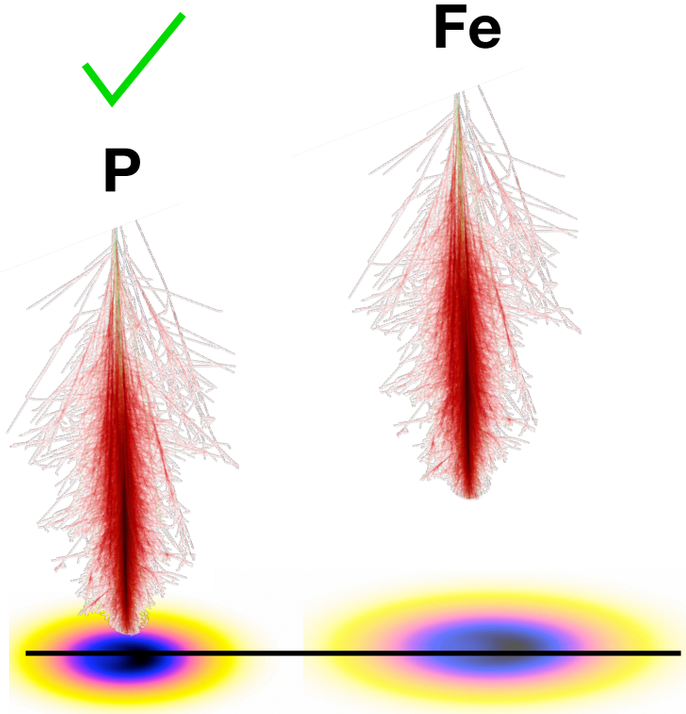




Some highlights from LOFAR cosmic-ray measurements:

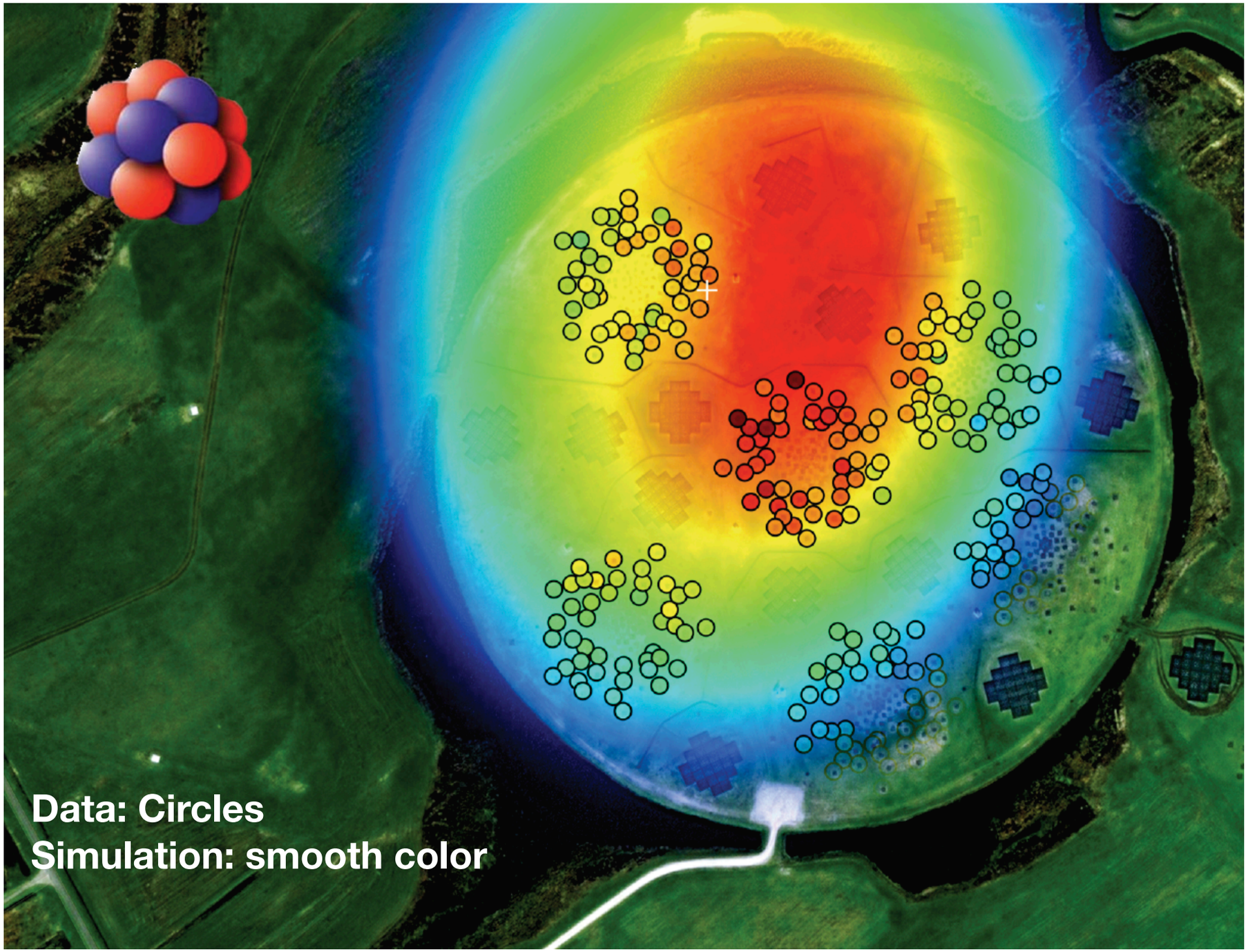
Radio footprint of a measured “Proton” event

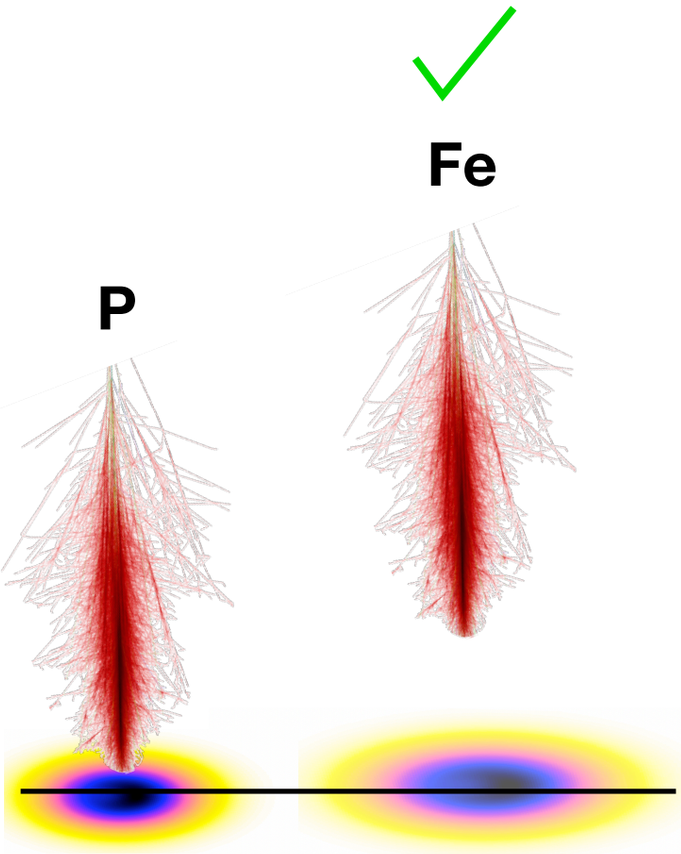




Some highlights from LOFAR cosmic-ray measurements:

Radio footprint of a measured “Iron” event



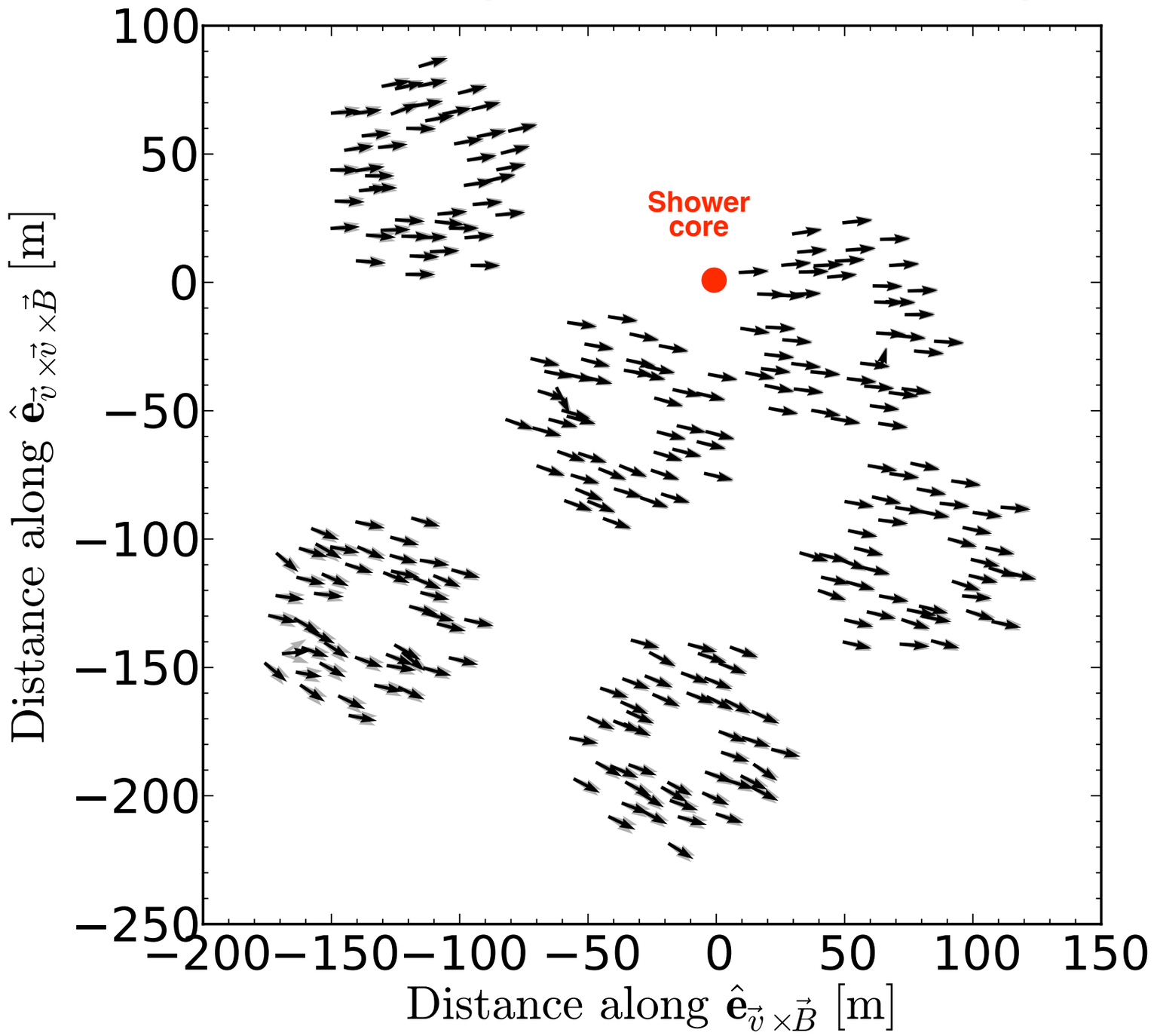


Some highlights from LOFAR cosmic-ray measurements:

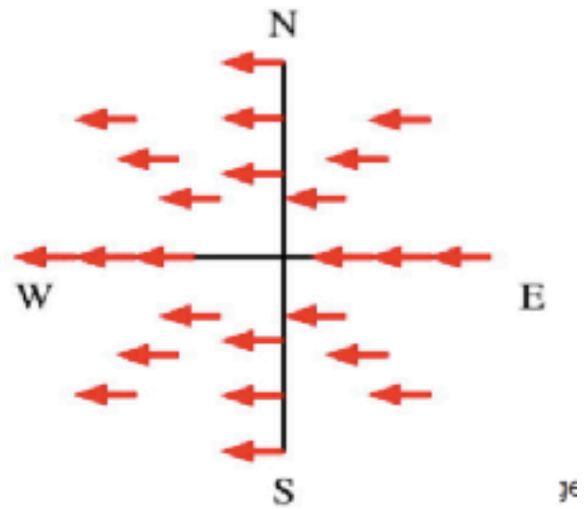
Polarization measurement of radio signals

Schellart+ 2014, JCAP, 10, 014

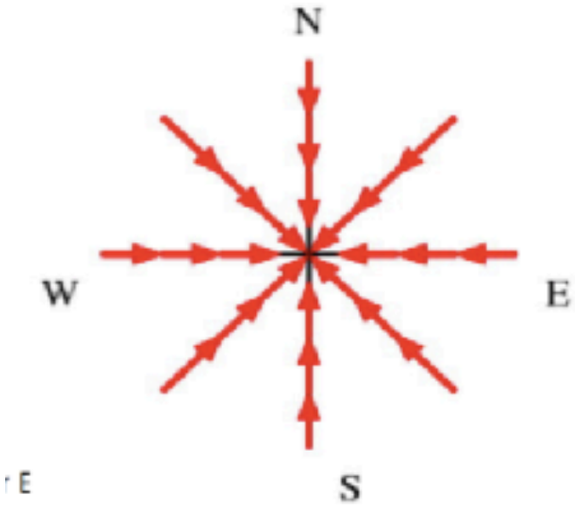
Polarisation footprint of CR shower in shower plane



Geo-synchrotron (~ 80%)



Charge-excess (~ 20%)



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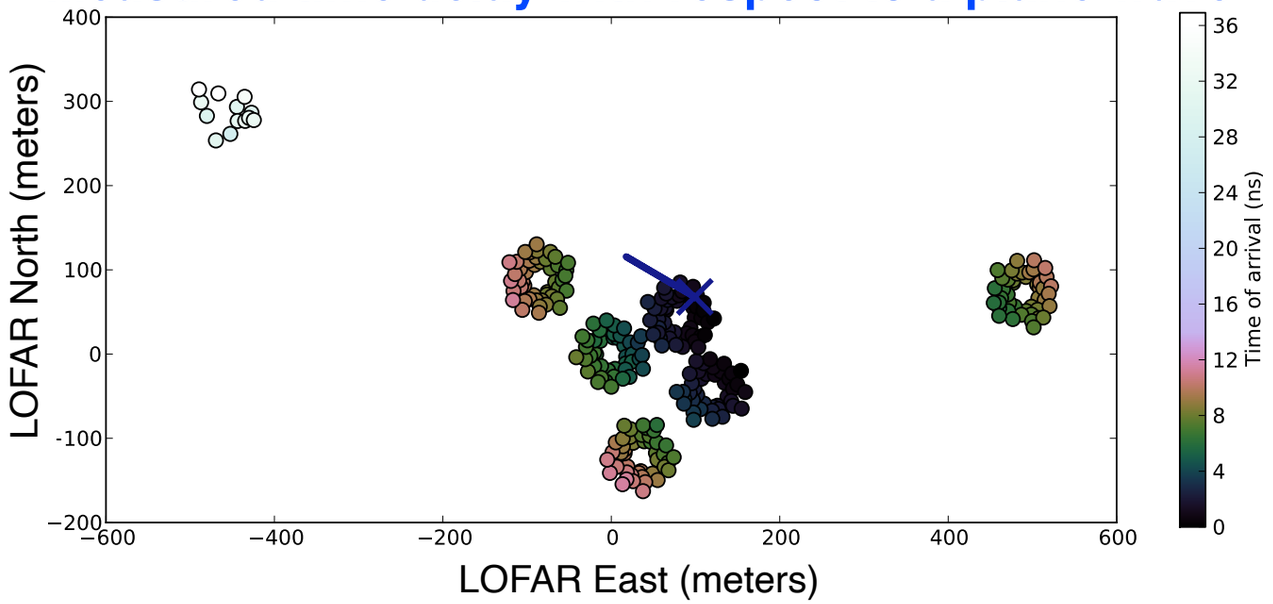
Some highlights from LOFAR cosmic-ray measurements:

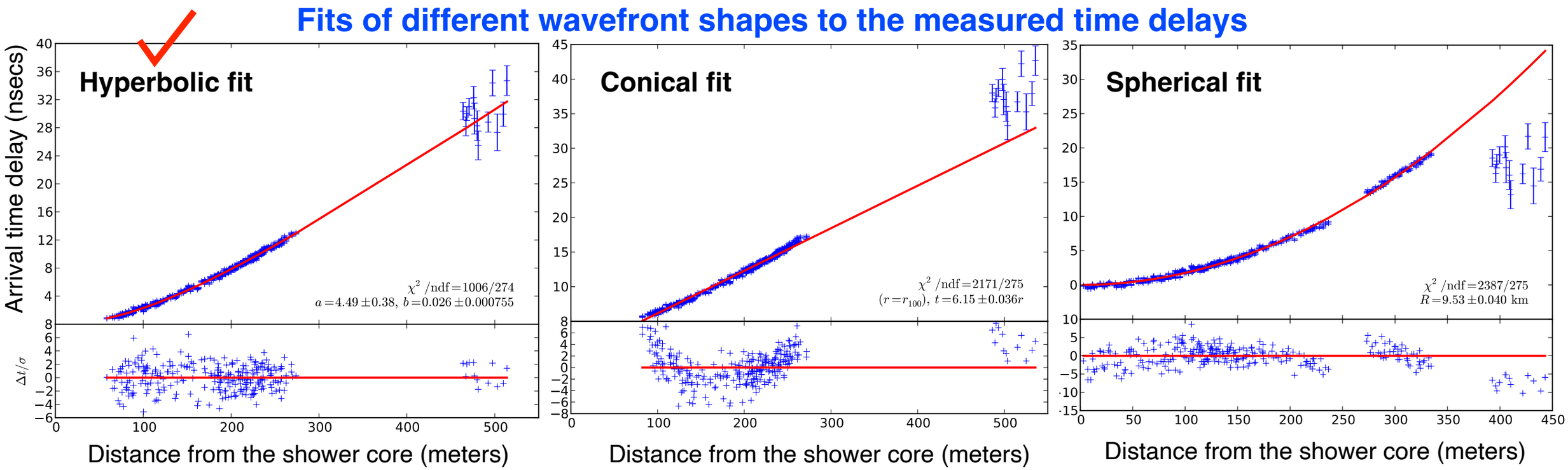
Measurement of radio wavefront of air showers

Corstanje+ 2015, APh, 61, 22

Measured time delay with respect to a plane wave



Fits of different wavefront shapes to the measured time delays



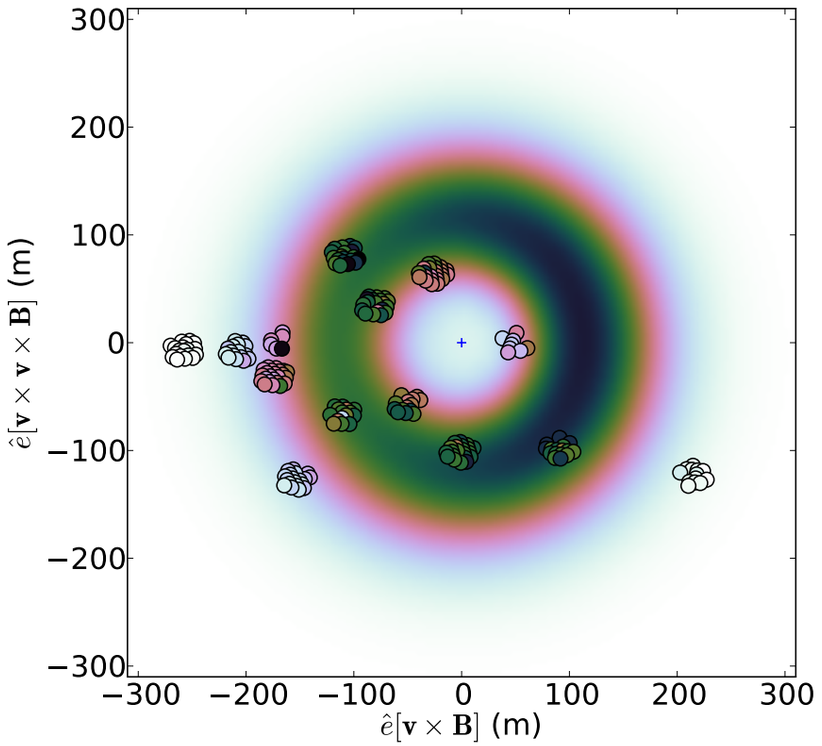
Some highlights from LOFAR cosmic-ray measurements:
Measurement of Cherenkov ring in radio emission above 100 MHz

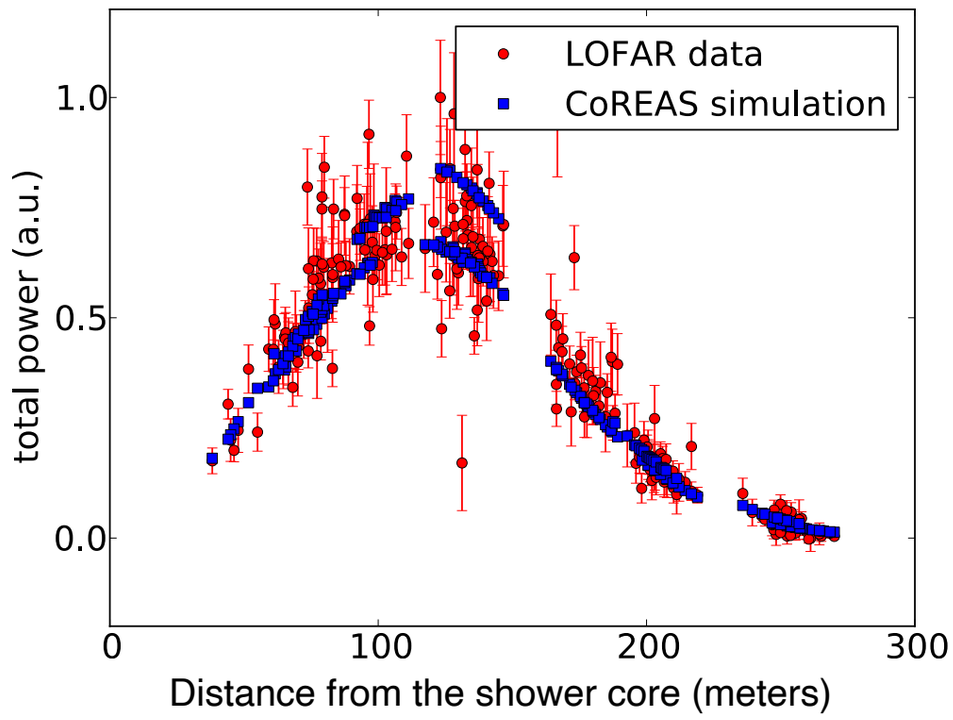
Nelles+ 2015, APh, 65, 11

2D signal footprint in shower plane

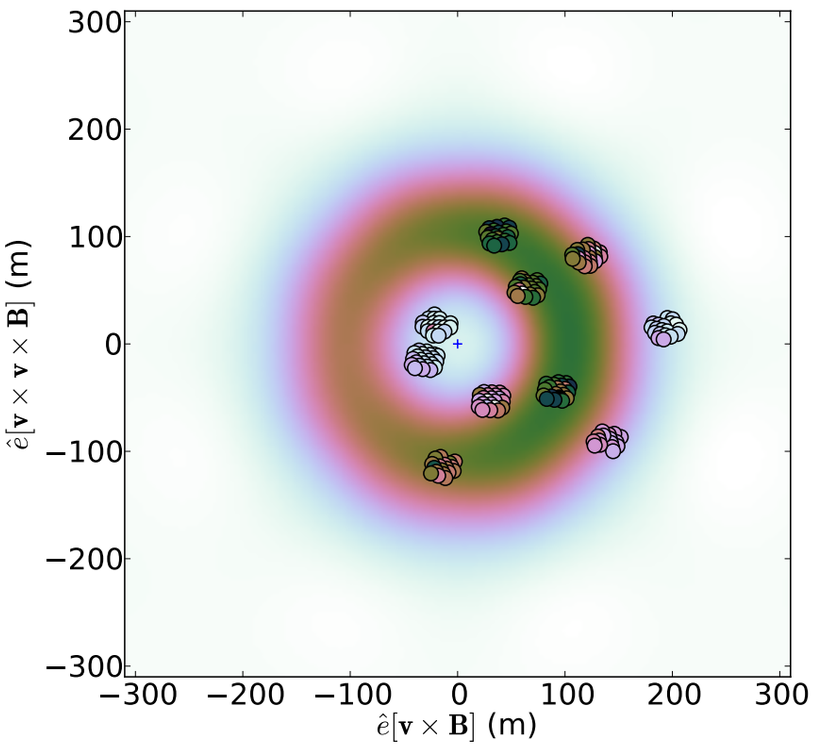
1D distribution in shower plane

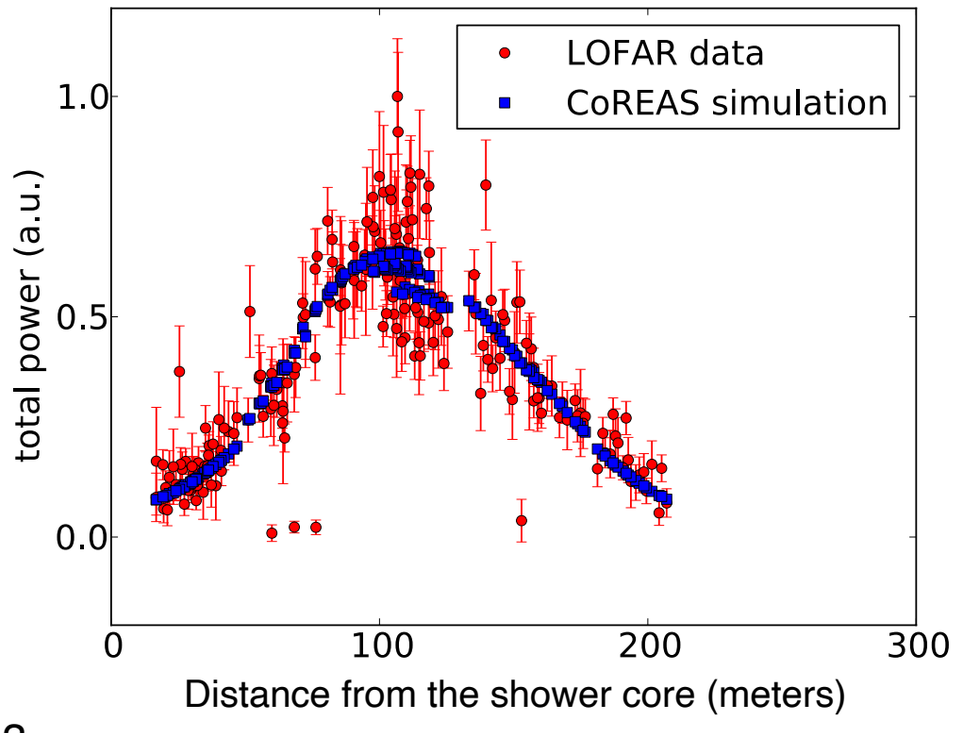
Zenith=43.4°
Radius=117.3 m





Zenith=40.5°
Radius=119.6 m

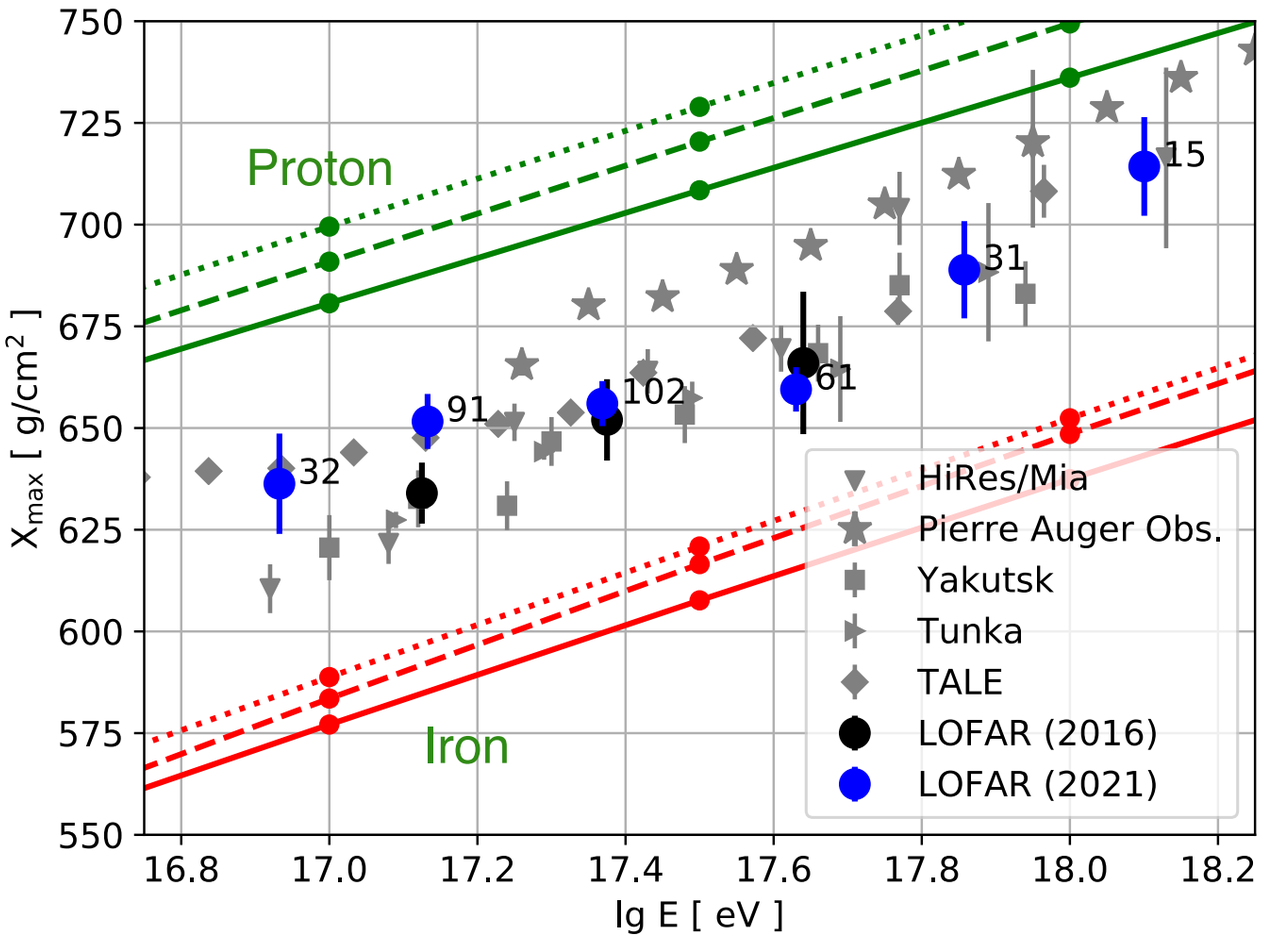




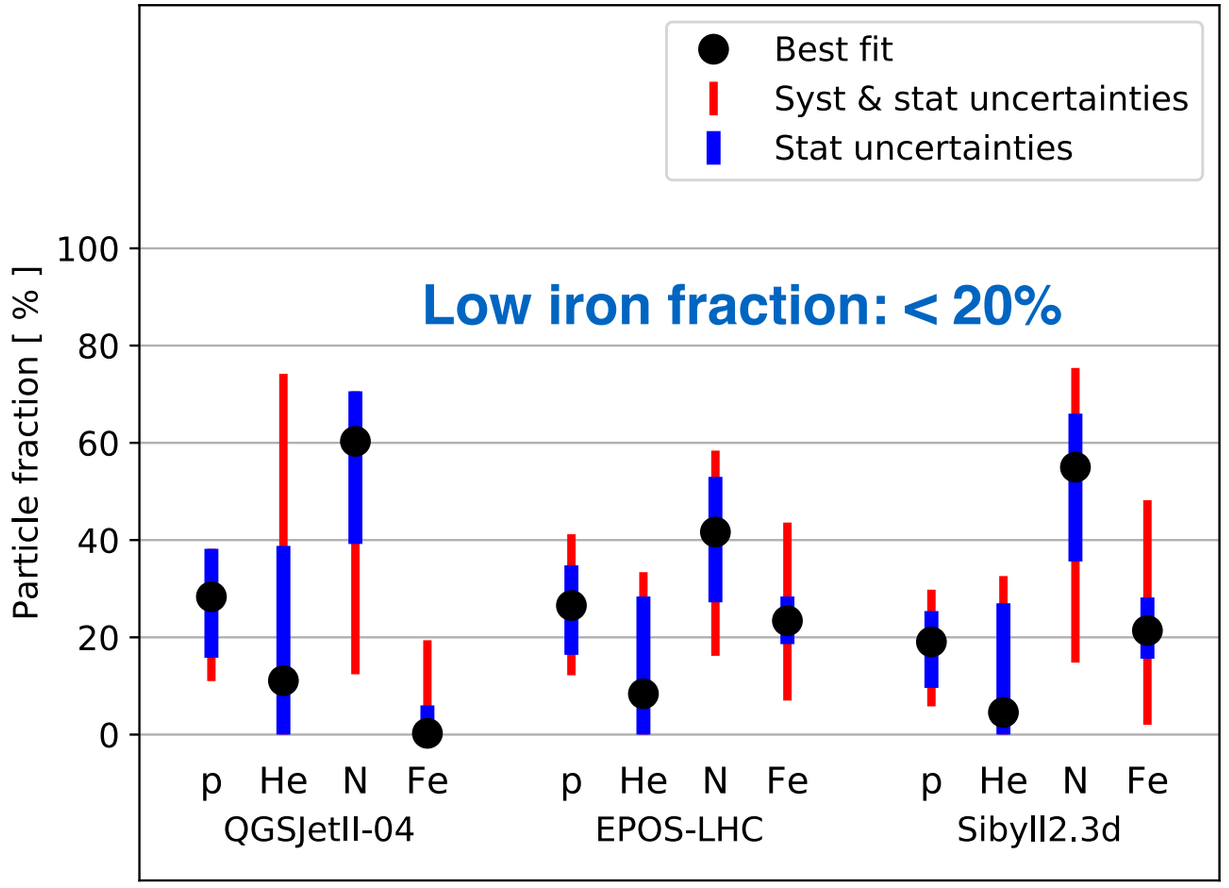
Some highlights from LOFAR cosmic-ray measurements:

Mass composition of cosmic rays at $10^{17\text{-}18}$ eV

Xmax measurement



Elemental fraction of cosmic rays



Xmax resolution ~ 20 g/cm²
Energy resolution ~ 9%
Core resolution ~ 5 m

Corstanje+ 2021, PRD , 103, 102006
Buitink+ 2016, Nature, 531, 70

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Some highlights from LOFAR cosmic-ray measurements:

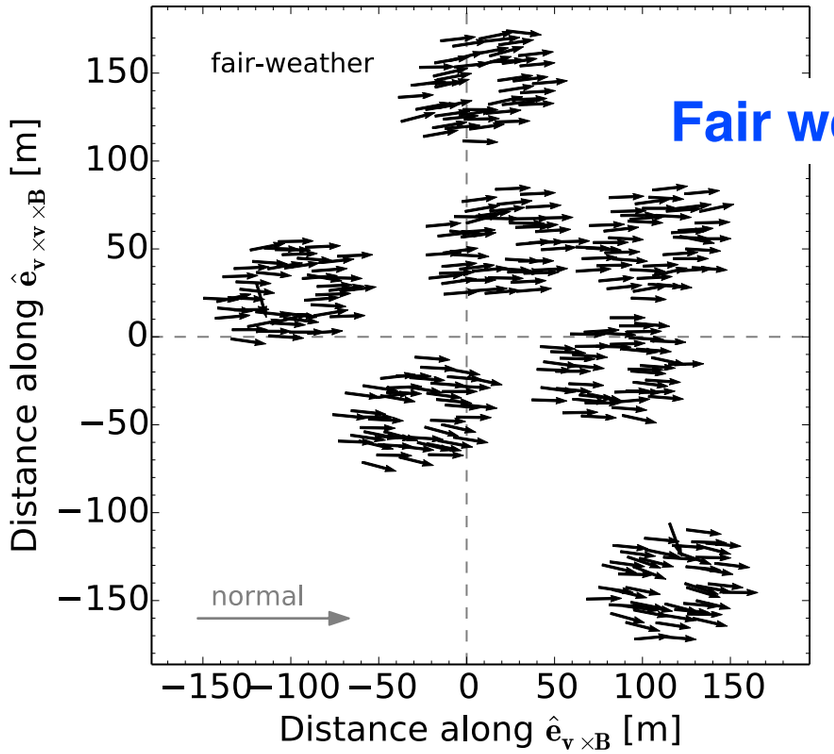
Lightning/Thunderstorm measurements

Atmospheric electric field measurements during thunderstorm

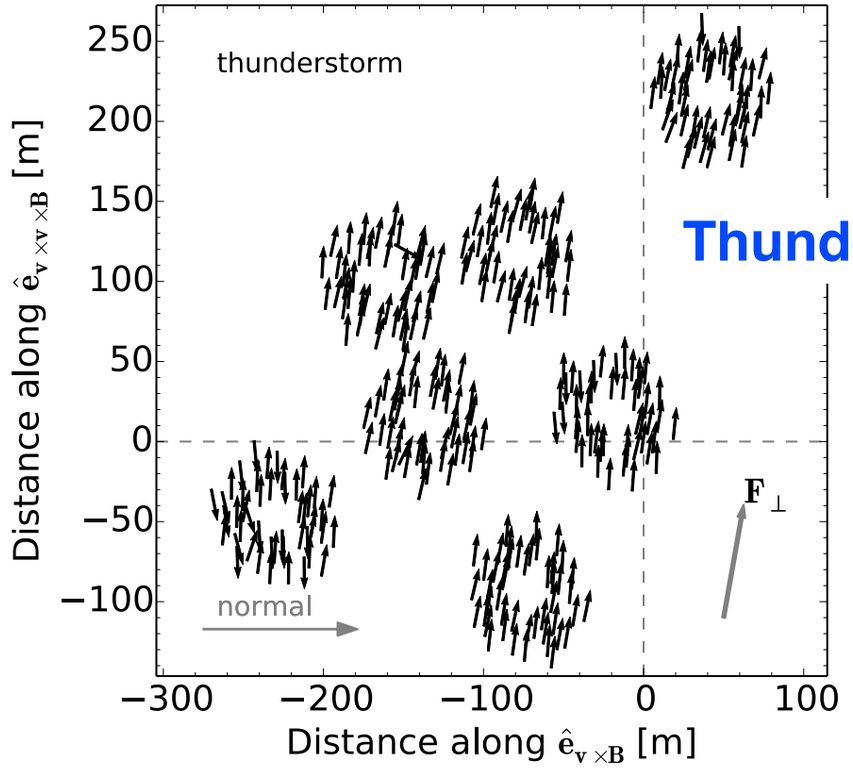
Schellart+ 2015, PRL, 114, 165001
Trinh+ 2017, PRD, 95, 083004
Trinh+ 2025, GRL, 52, e2025GL115586

Picture credit: Radboud University

Polarisation footprint of air showers during thunderstorm



Fair weather

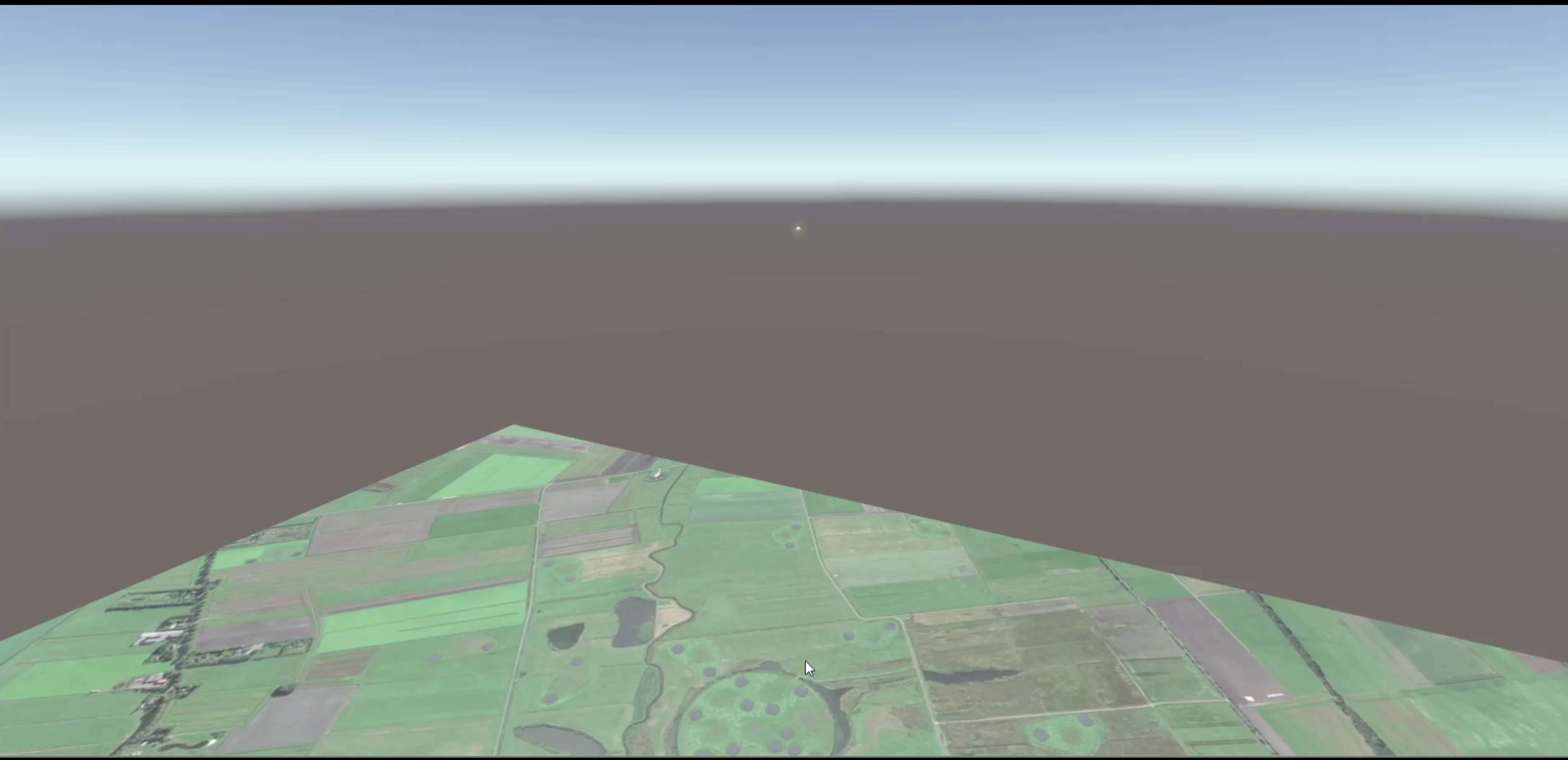


Thunderstorm

Some highlights from LOFAR cosmic-ray measurements:

Lightning/Thunderstorm measurements

LOFAR 3D lightning mapping



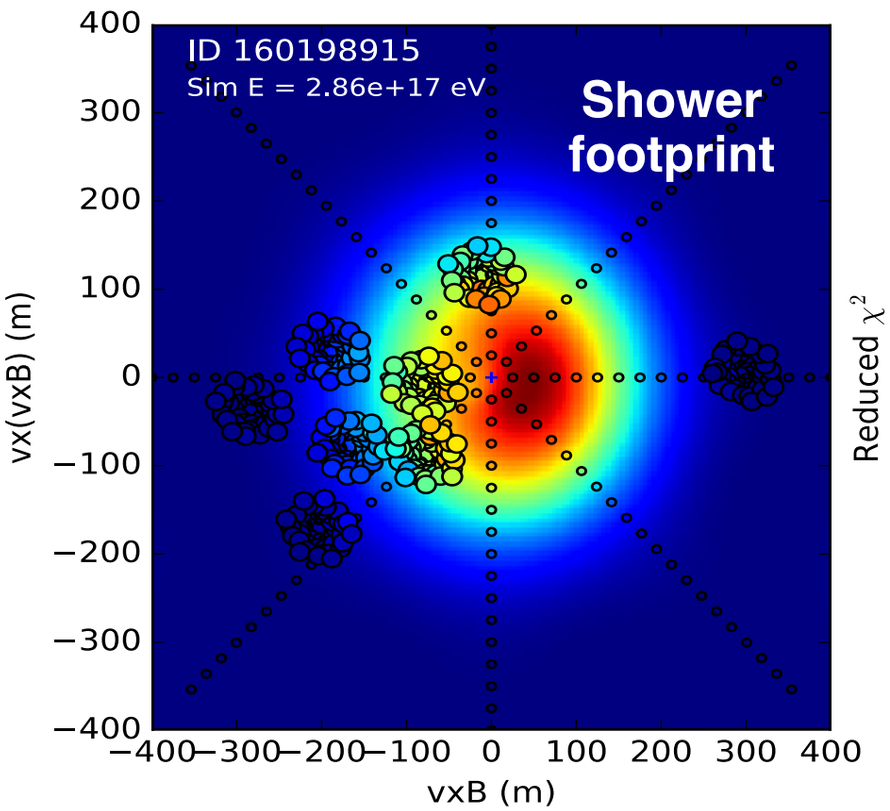
Hare+ 2018, JGRA, 123, 2876

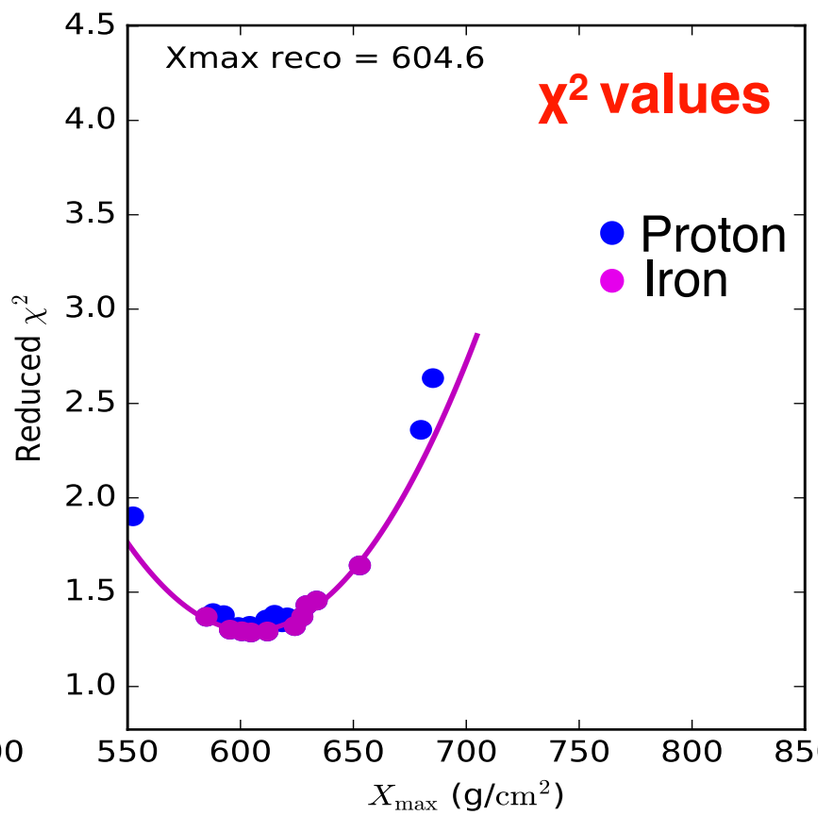
Credit: S. Buitink

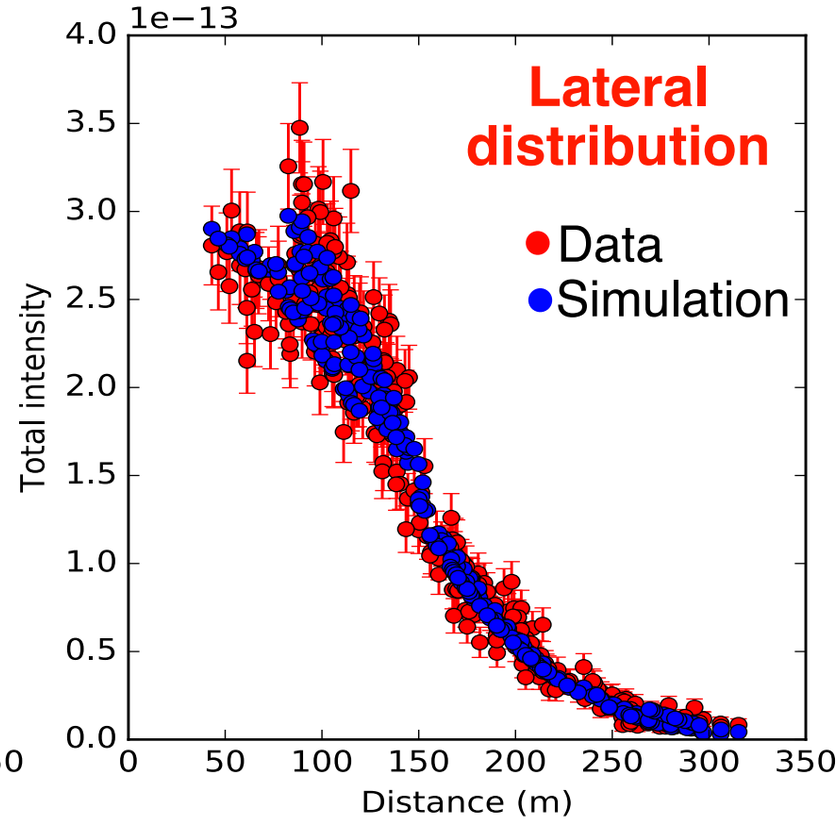
Standard LOFAR event reconstruction procedure

CoREAS simulation fit to a measured LOFAR CR event

$$\chi^2_{\text{radio}} = \sum_{\text{antennas}} \left(\frac{P_{\text{ant}} - f_r^2 \, P_{\text{sim}} \left(x_{\text{ant}} - x_0, \, y_{\text{ant}} - y_0 \right)}{\sigma_{\text{ant}}} \right)^2$$







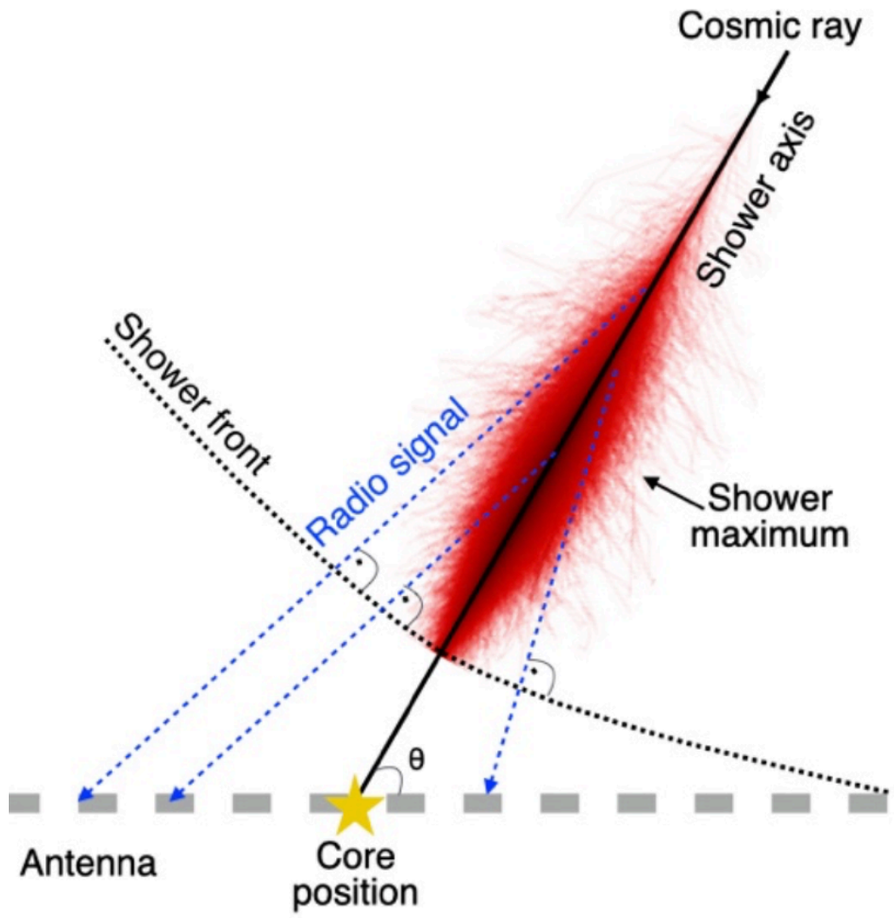
- A high-precision method, but computationally expensive.
- A single air shower simulation takes ~ 1-3 days of CPU time.
- Require ~ 30 simulations to reconstruct an event (~ 60 CPU days).
- **Need methods more efficient in time without losing precision.**
- **Several alternative methods are currently under investigation.**

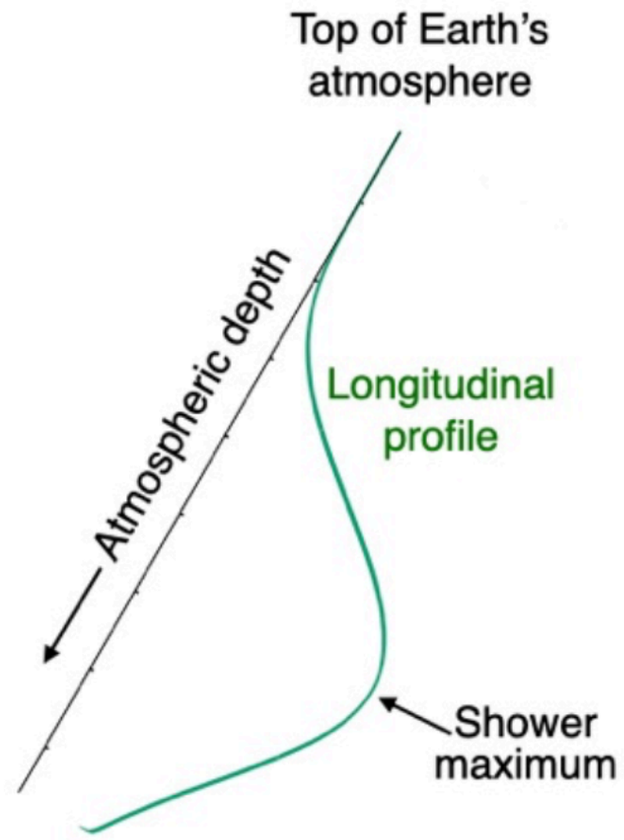
Corstanje+ 2021, PRD, 103, 102006

New shower reconstruction methods

Method based on Backtracking radio signals:

Reconstruction of longitudinal profile and shower Xmax





- Determine the shower core and the arrival direction (+ shower front).
- Backtrack signals to their emission points along the shower axis.
- Build longitudinal profile from the distribution of emission points.
- **Method independent of simulation inputs; Extremely fast: 30 CPU seconds!**

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Vuta, **Thoudam+** 2025, PRD, 111, 123051

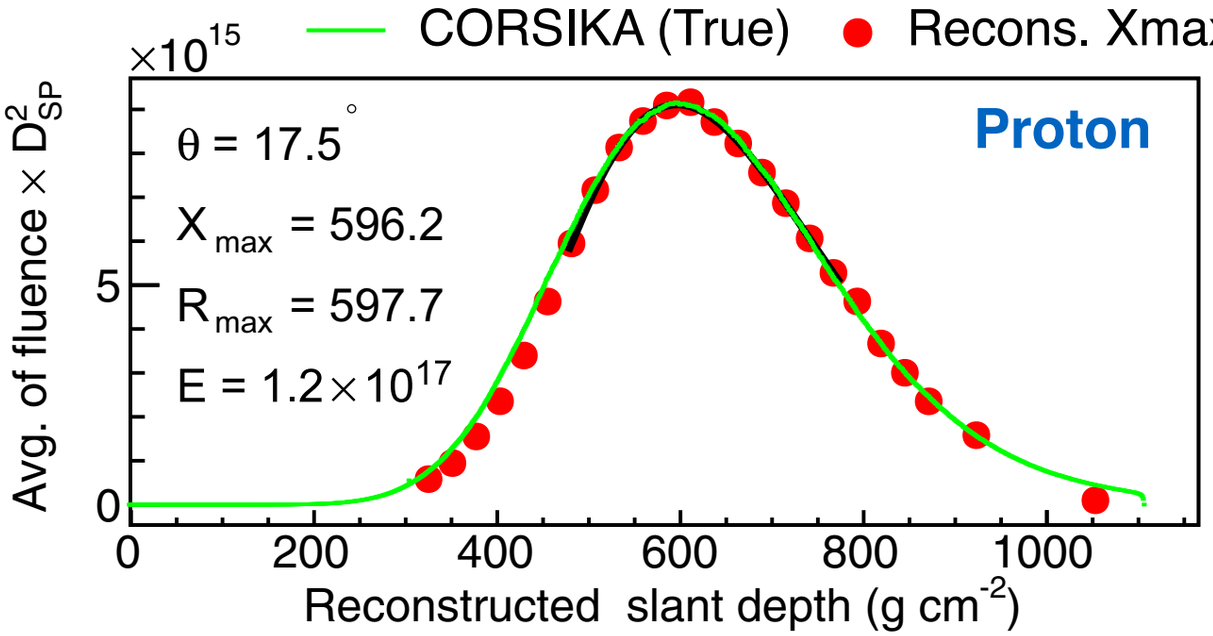
New shower reconstruction methods

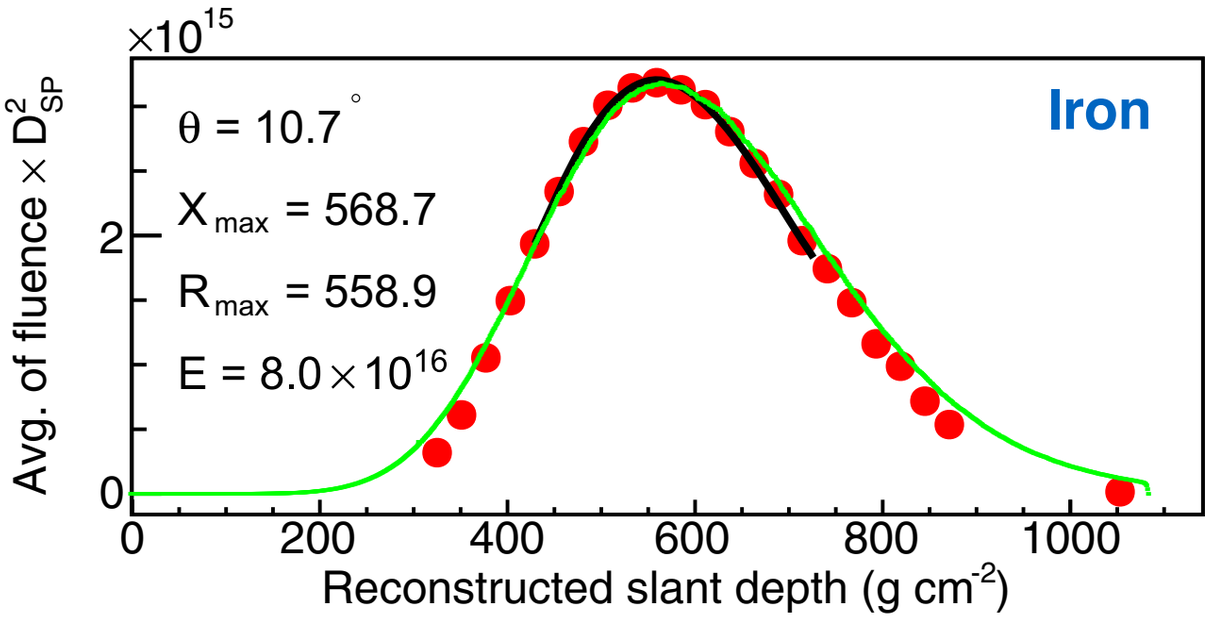
Method based on Backtracking radio signals:

Reconstruction of longitudinal profile and shower Xmax

Longitudinal profiles

Vuta, **Thoudam+** 2025, PRD, 111, 123051





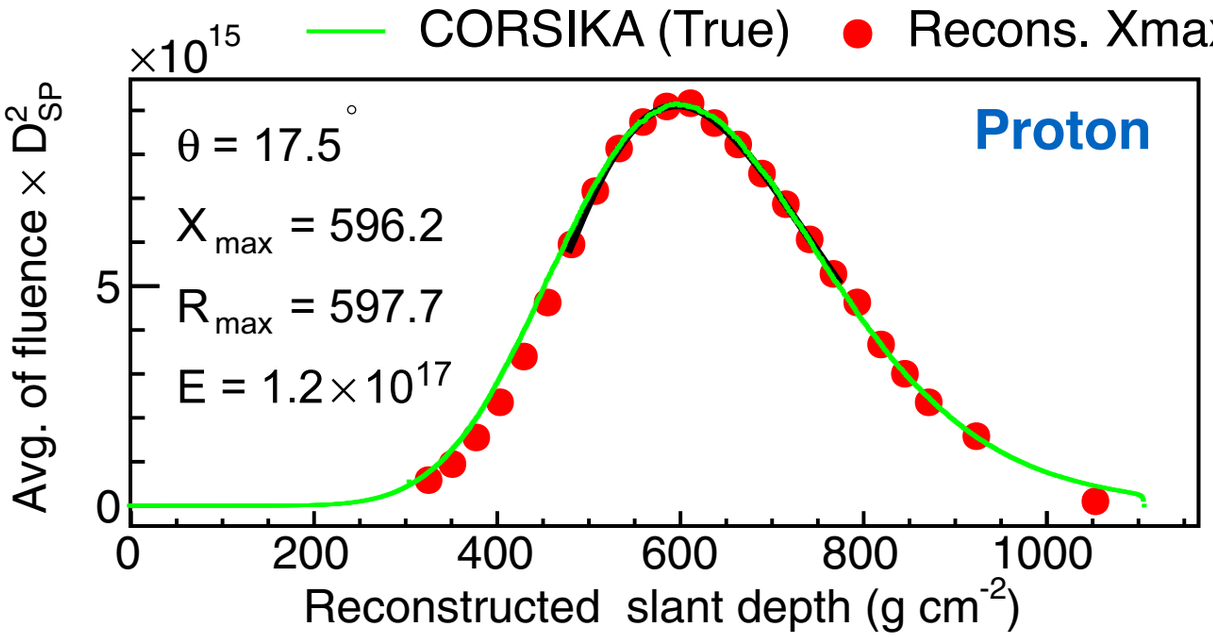
New shower reconstruction methods

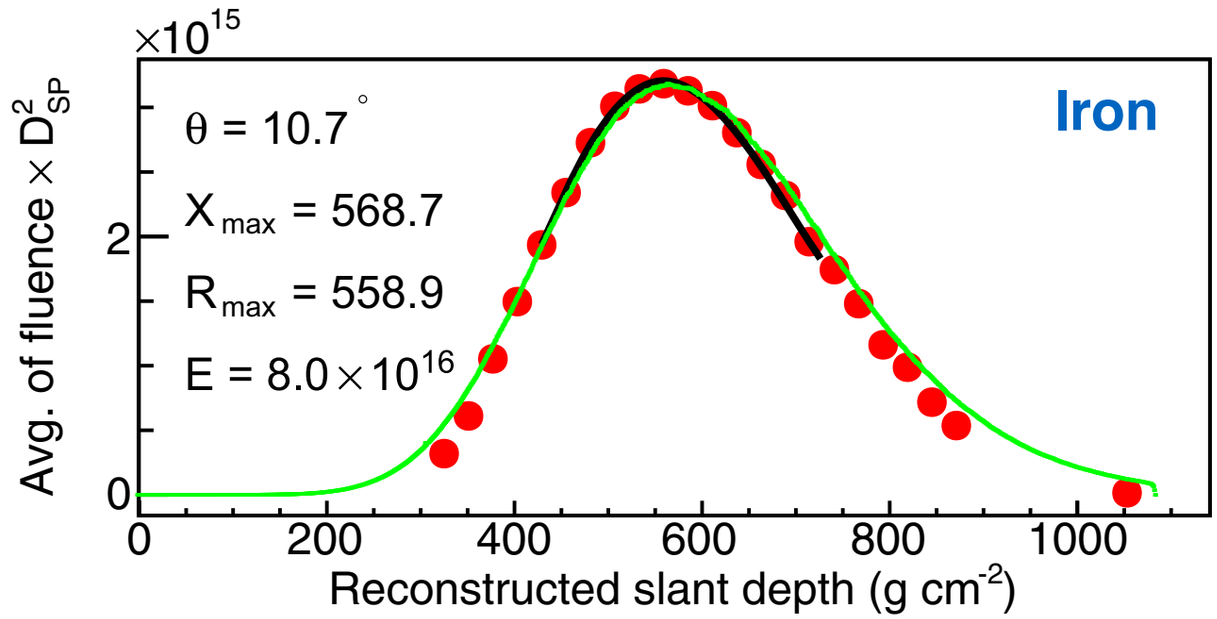
Method based on Backtracking radio signals:

Reconstruction of longitudinal profile and shower Xmax

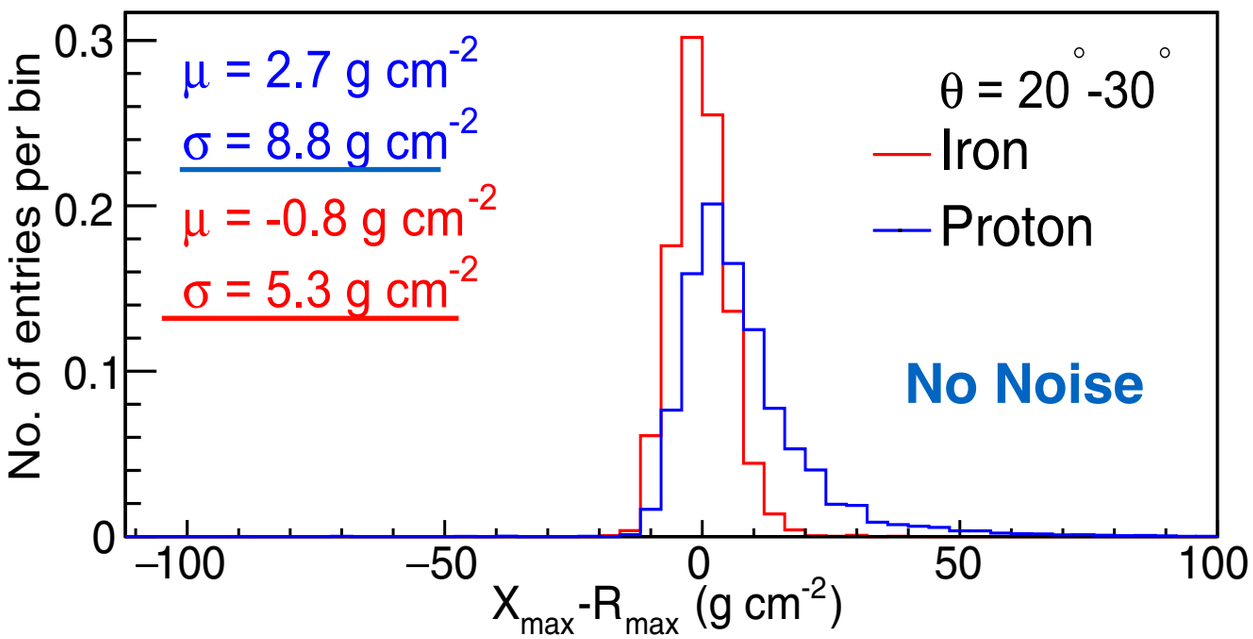
Longitudinal profiles

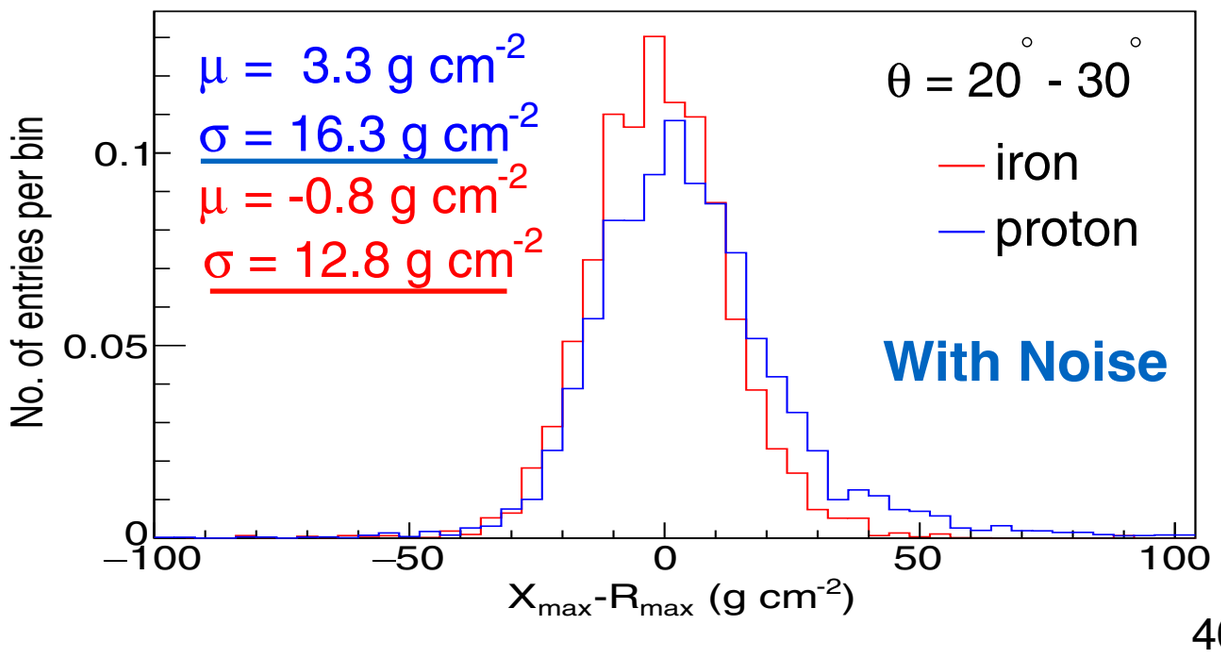
Vuta, **Thoudam+** 2025, PRD, 111, 123051





Xmax resolution

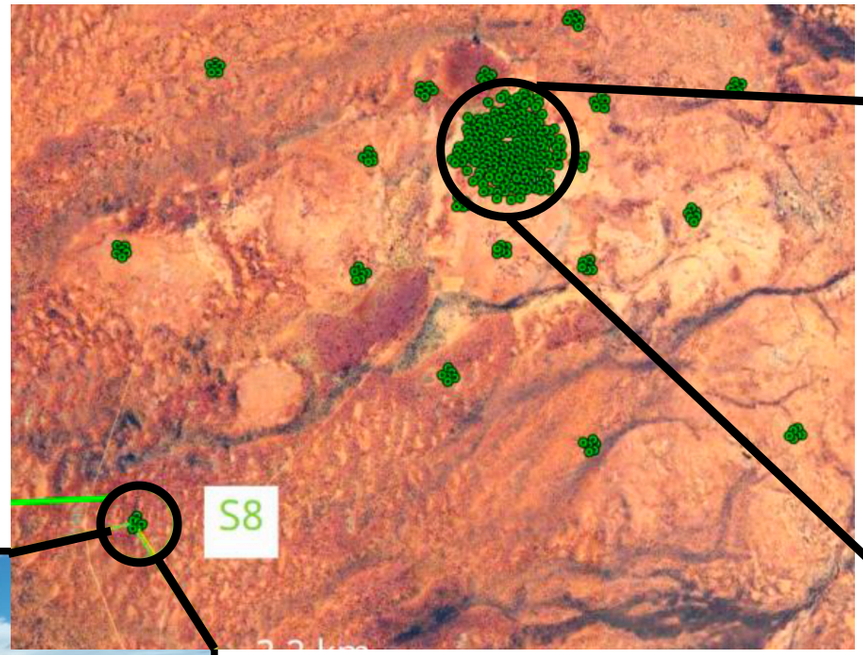


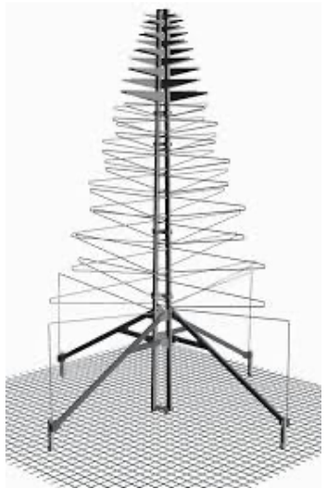


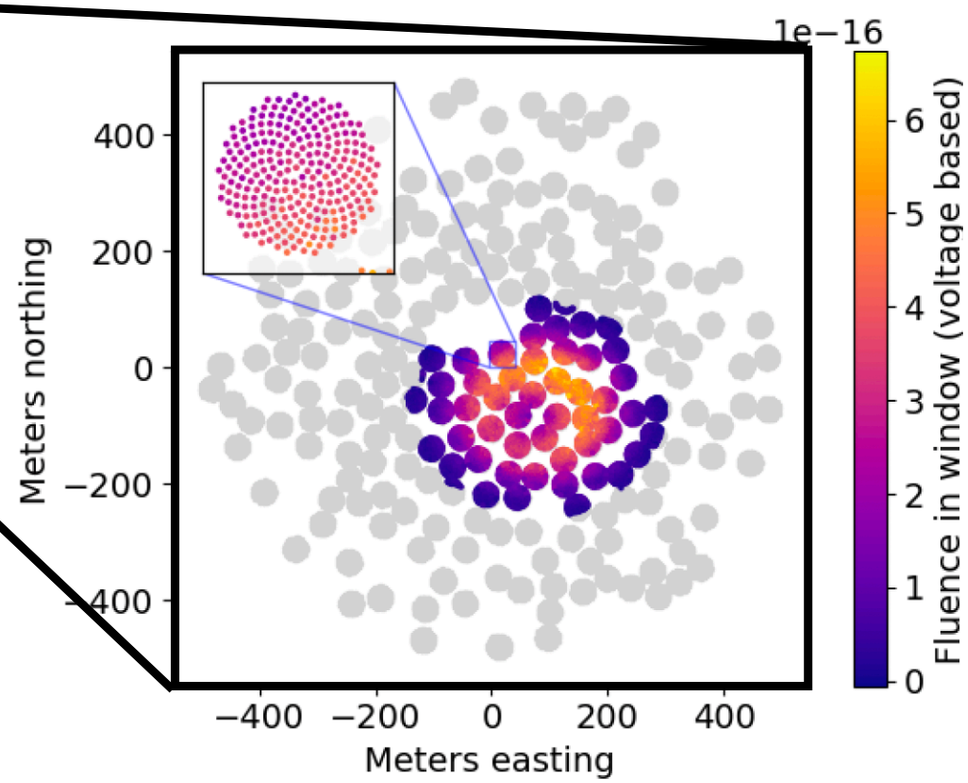
The future: LOFAR to SKA

Square Kilometer Array (SKA) telescope

Western Australia
700 km north of Perth







The densest antenna layout!

- Construction started; SKA-Low: 50-350 MHz.
- ~ 60,000 antenna @ 1 km² in cluster of 256 antennas.
- ~ 50 times higher antenna density than LOFAR.

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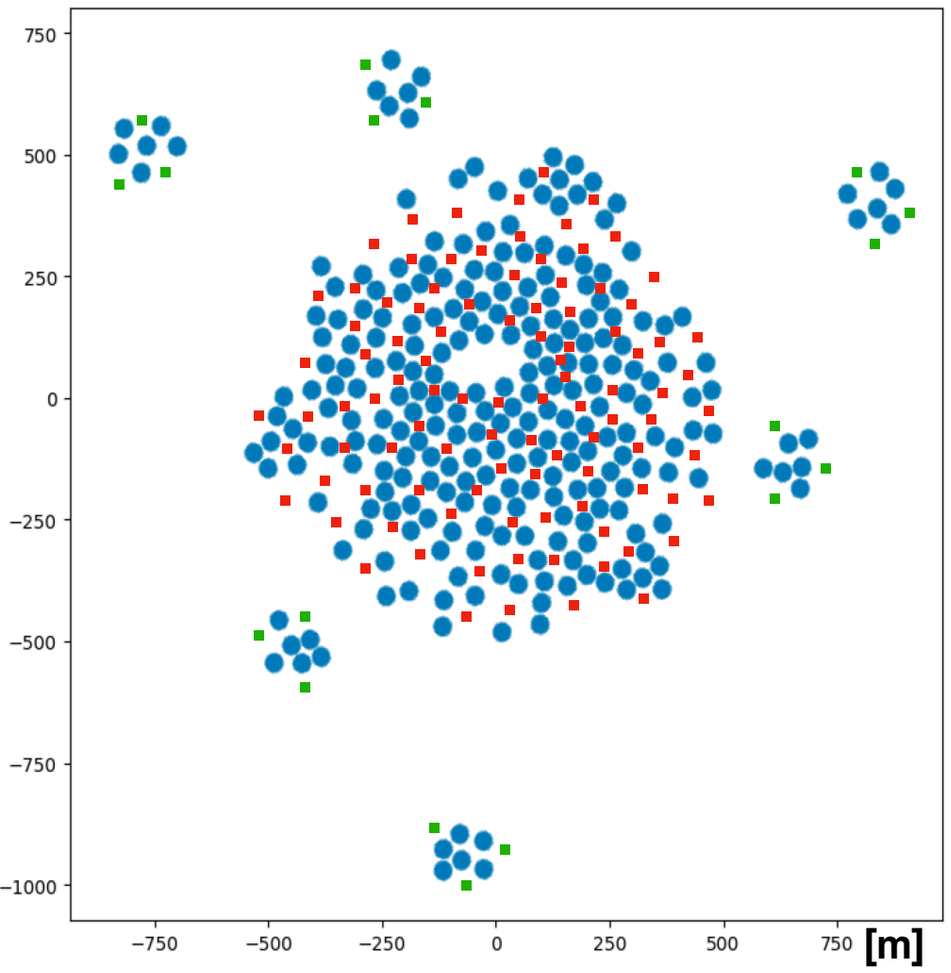
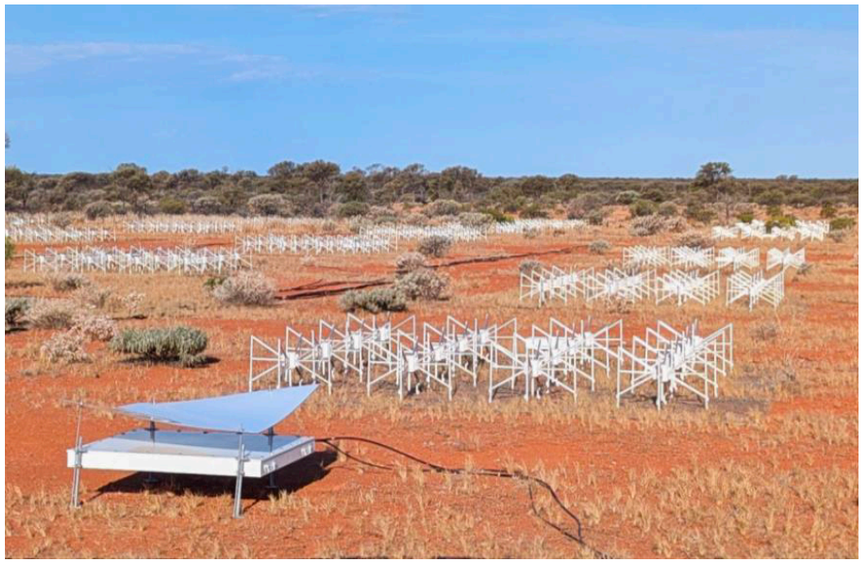
The future: LOFAR to SKA

SKA particle detector array



- **Antenna station**
- **Particle detectors dense array (~100 units)**
- **Particle detectors satellite (~18 units, optional)**

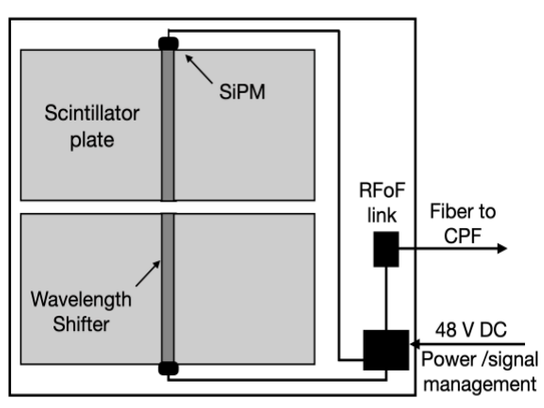


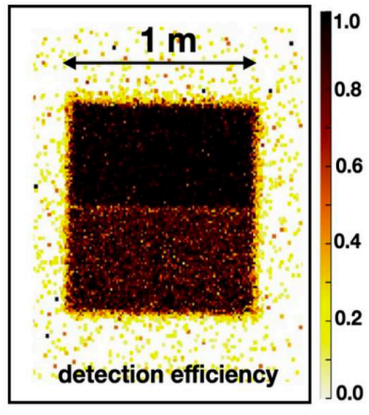
8-station prototype array installed at Murchison Widefield Array

Design: **Univ. of Manchester**
Deployment: **Curtin Univ.**
DAQ: **CSIRO**
J. Bray et al., NIMPA 973, id. 164168 (2020)

Construction of scintillator array now fully funded:

FWO Belgium, **NOVA** Netherlands + in-kind contributions **KIT** (Karlsruhe, Germany)
3rd generation design developed at **KIT** and **Radboud Nijmegen**





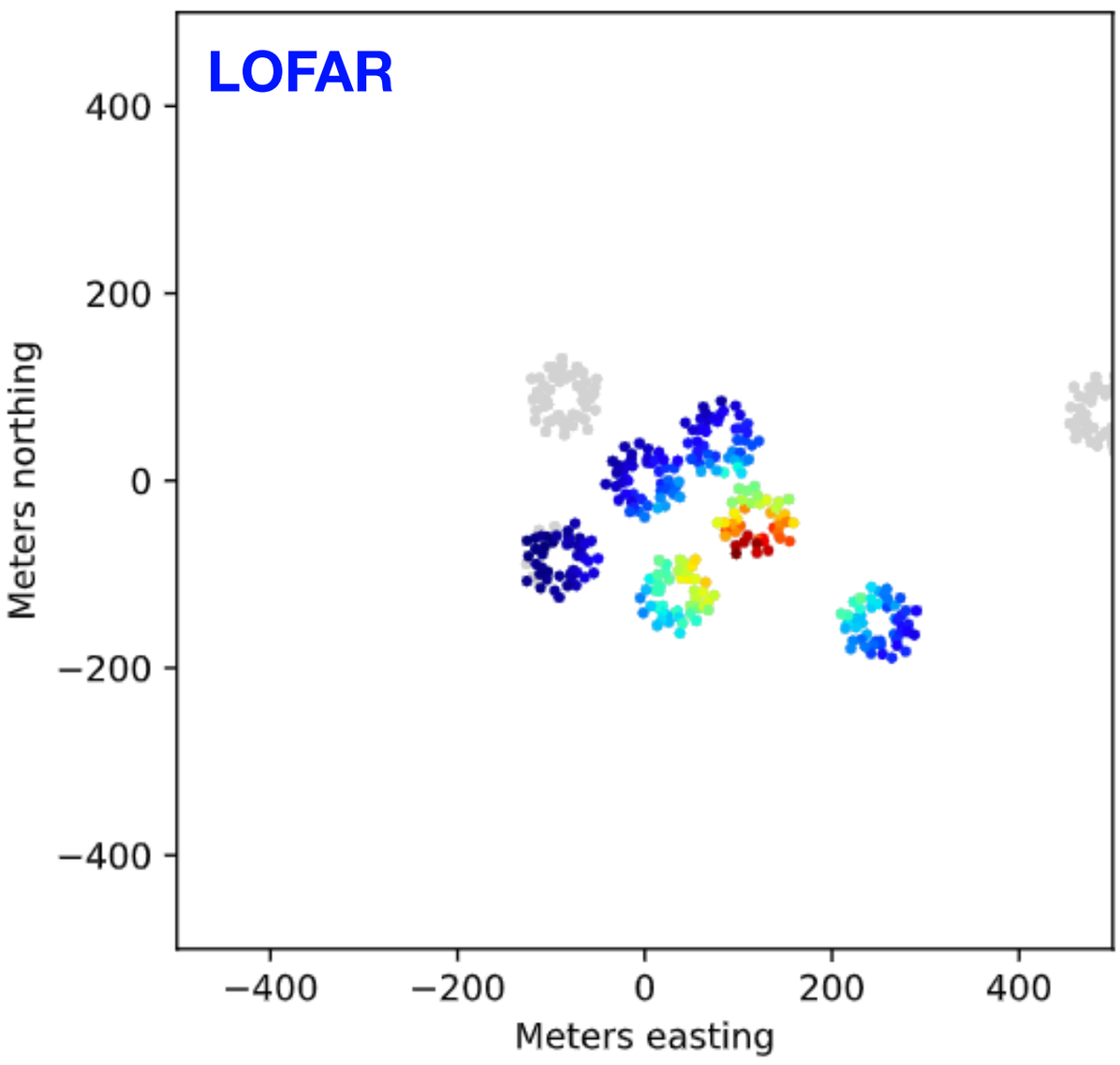
first light at KIT muon tower

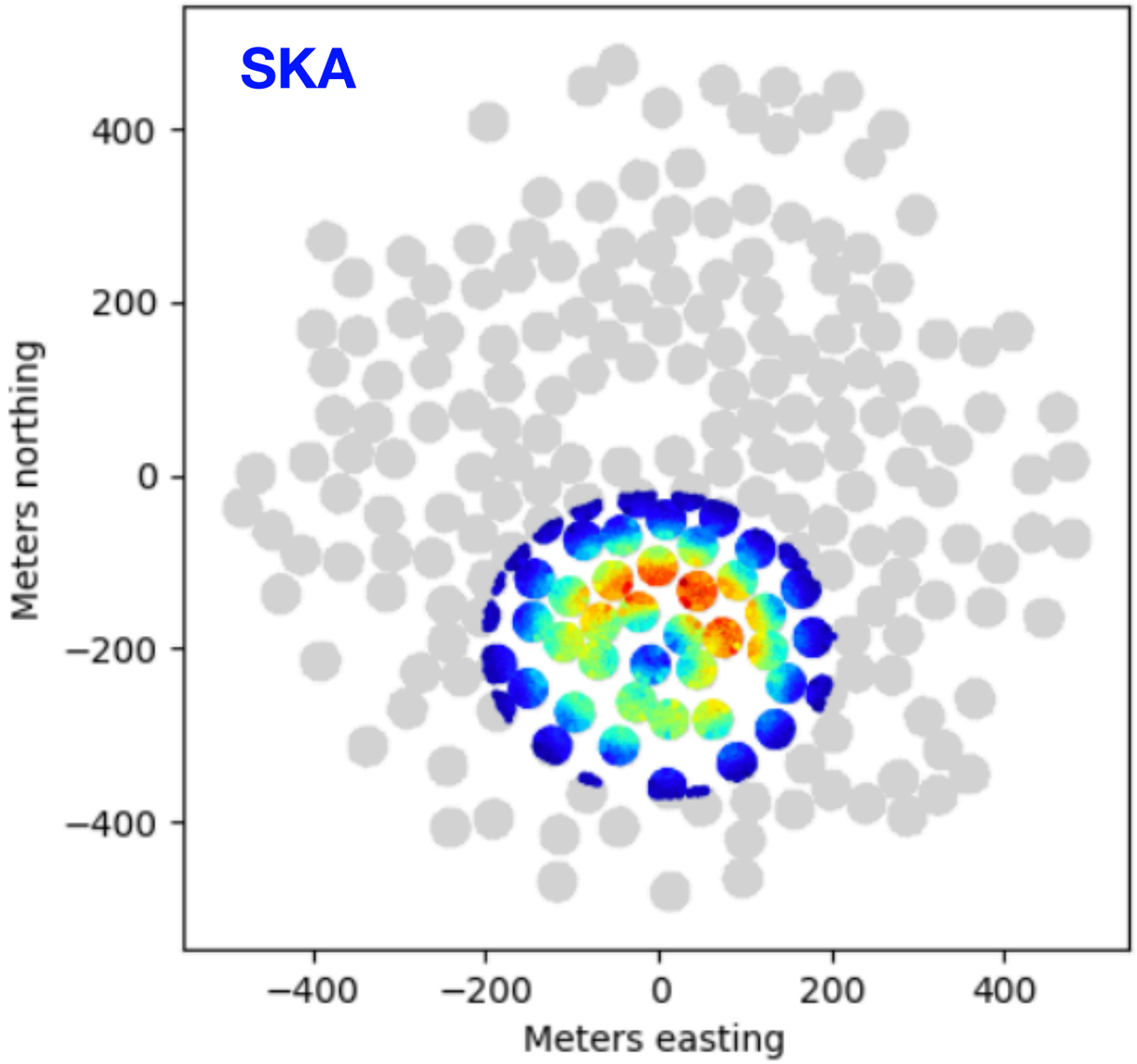
S. Buitink
ICRC 2025, Geneva

- Detector and electronics testings underway.
- Preparation to integrate particle detector set-up into SKA framework.
- Ongoing efforts to build faster data analysis methods.

Improvement from LOFAR to SKA: Shower footprint

Simulated air shower footprints

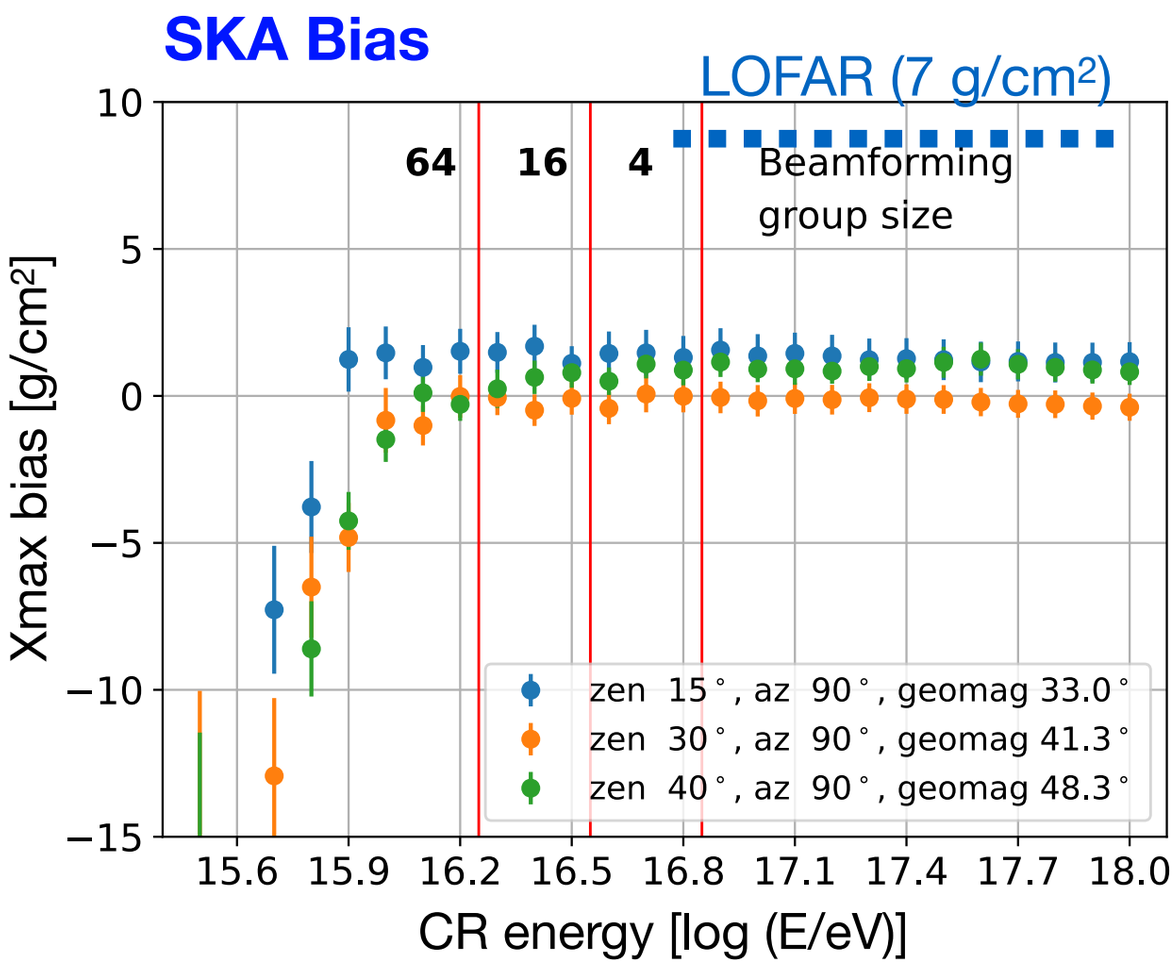


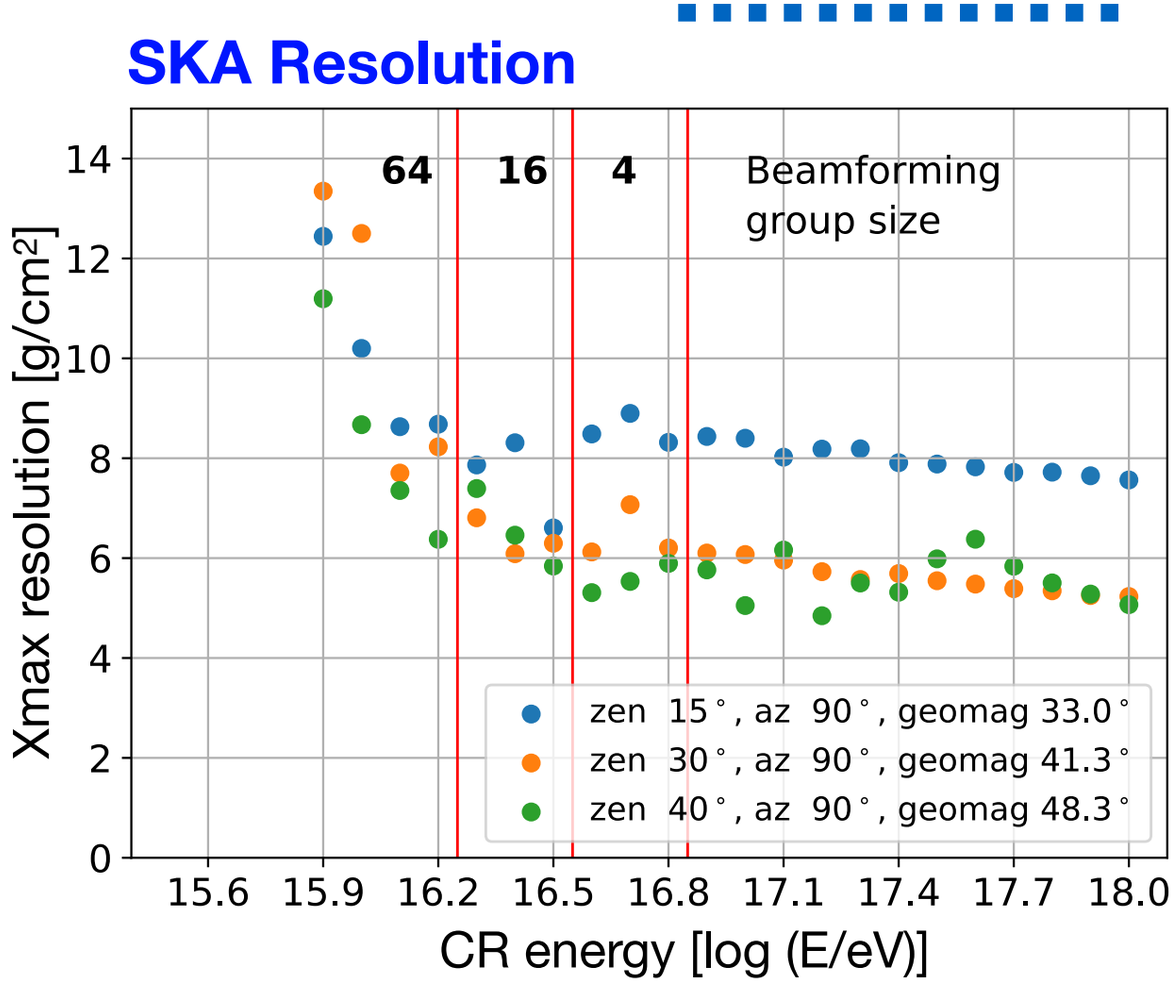


- SKA will sample air shower footprint at unprecedented details.
- 600 antennas at LOFAR compared to 60,000 at SKA.

Improvement from LOFAR to SKA: Shower Xmax

SKA performance for Xmax reconstruction





- **Lower energy threshold: Below 10^{16} eV**
- **Higher Xmax precision: 2-3 times better.**

	SKA (simulated)	**LOFAR**
X_{max} resolution	: **6 - 8 g/cm²**	20 g/cm²
Energy resolution	: **3 %**	9 %
Core resolution	: **50 cm**	~ 5 m

Corstanje+ 2025, PRD, 112, 023017
Corstanje+, PoS(ARENA2022)024

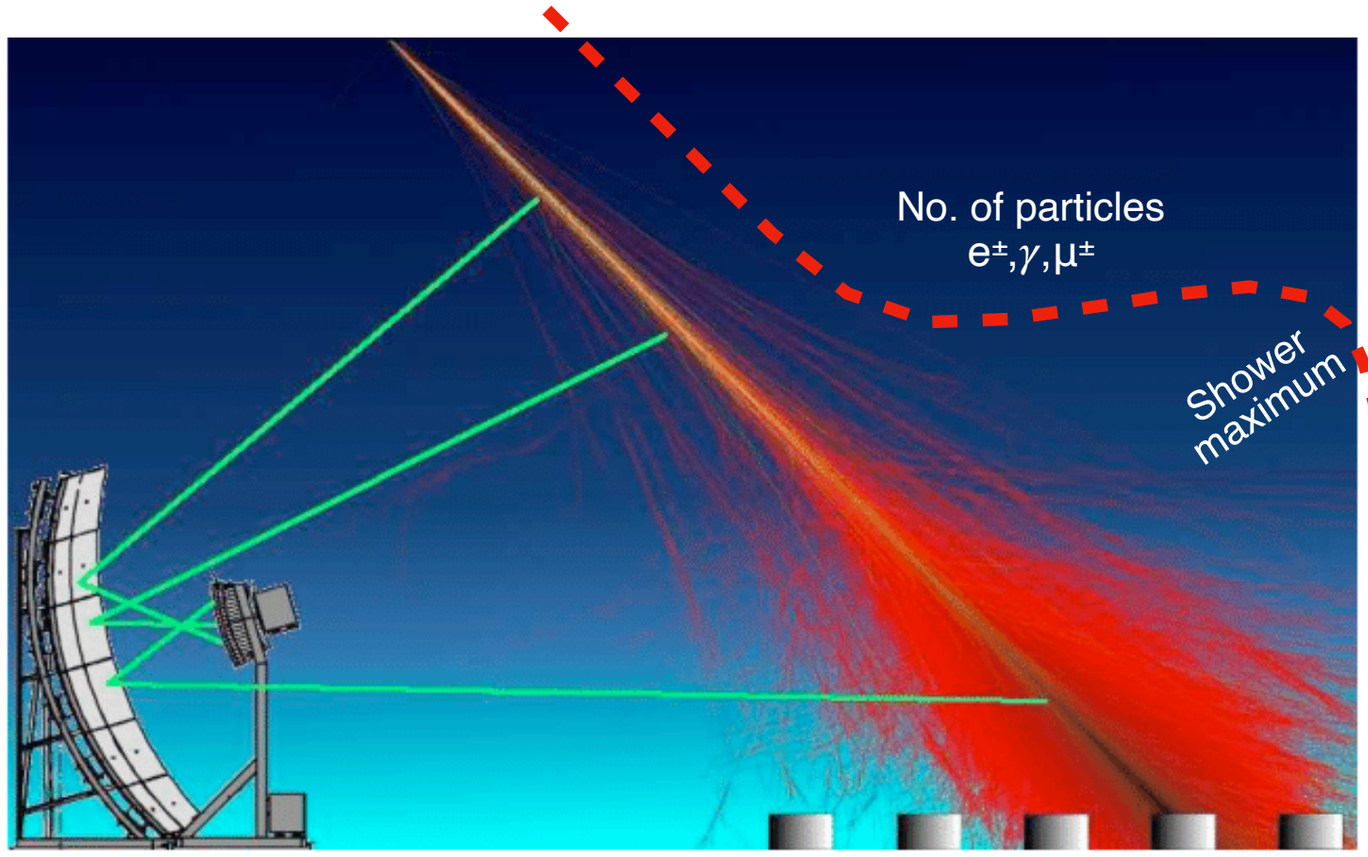
Cosmic-ray experiments: (3) Fluorescence telescope

(3) Fluorescence telescope

The charged particles in an air shower also interact with atmospheric nitrogen, causing it to emit ultraviolet light via a process called fluorescence, which is invisible to the human eye. Fluorescence detectors can be used to observe these trail of nitrogen fluorescence and track the development of air showers by measuring the brightness of the emitted light. This technique is currently the best method to measure the depth of the shower maximum (X_{max}) which is a measure of the type of the CR particle.

Using a grid of focusing mirrors to collect the light, fluorescence cameras can view the air shower up to 15 kilometers away. The Auger Observatory's fluorescence detectors are much more sensitive than the human eye and can "see" distant air showers develop.

Occasionally, a shower cascade can occur in a place where two fluorescence detectors can record it, which allows for very precise measurements of the direction the cosmic-ray particle.



Letessier-Selvon & Stanev, 2011, Reviews of Modern Physics, 83

https://www.auger.org/observatory/auger-hybrid-detector

Cosmic-ray experiments: (3) Fluorescence telescope

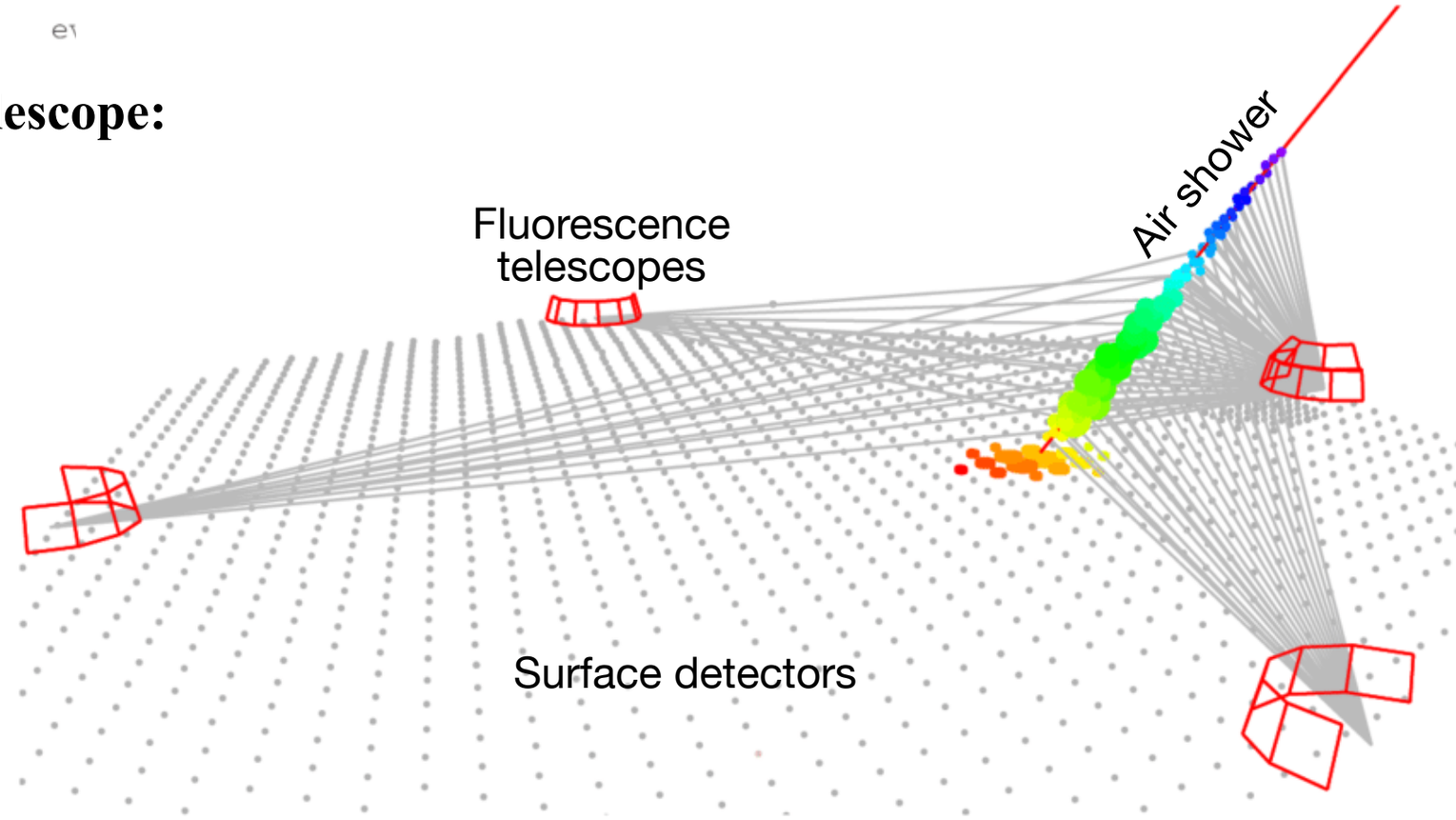
(3) Fluorescence telescope

- AUGER observatory uses 27 fluorescence telescopes in 4 different places.

- Specifications of a single fluorescence telescope:

- Mirror surface: 3.6 m x 3.6 m (spherical)
- Opening: 30° x 30°
- Camera: 80 cm x 80 cm

- 440 Photomultipliers
- Detection of fluorescence light
- Range: > 30 km (for 10^{20} eV air shower)



Example of an ultra-high energy event measured by the Pierre Auger Observatory with the surface detector and the four fluorescence stations. Colors denote the times at the shower and ground respectively. The size of the markers correspond to the magnitude of measured signals at ground and reconstructed energy deposits at the shower. Red squares illustrate the field of view of the 27 fluorescence telescopes, the viewing directions of triggered pixels are shown as gray lines and the surface detector stations are displayed as gray dots. The averaged energies and shower maxima from the four fluorescence sites yield $E = (4.7 \pm 0.1) \times 10^{19}$ eV and $X_{max} = (730 \pm 3)$ g/cm 2 respectively.

Unger, M.. (2013), EPJ Web of Conferences. 53. 04009

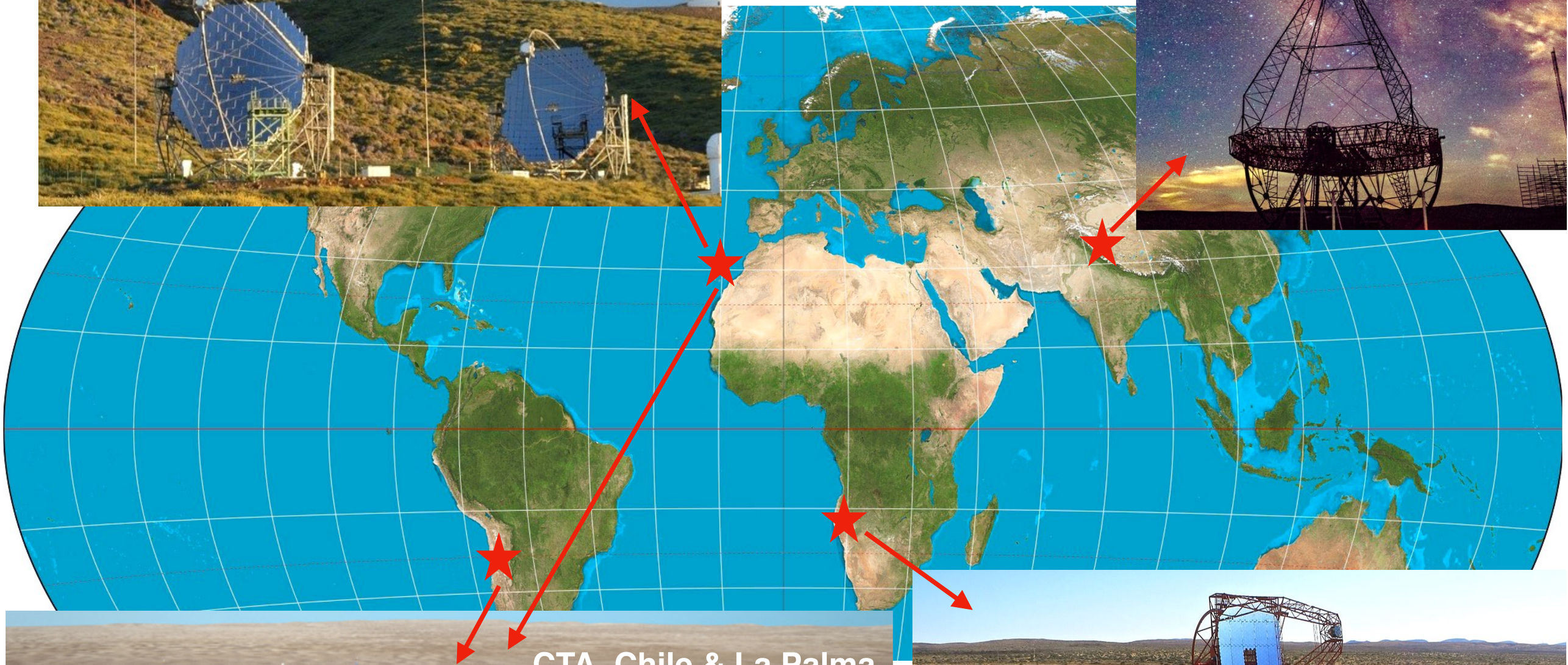
Gamma-ray experiments

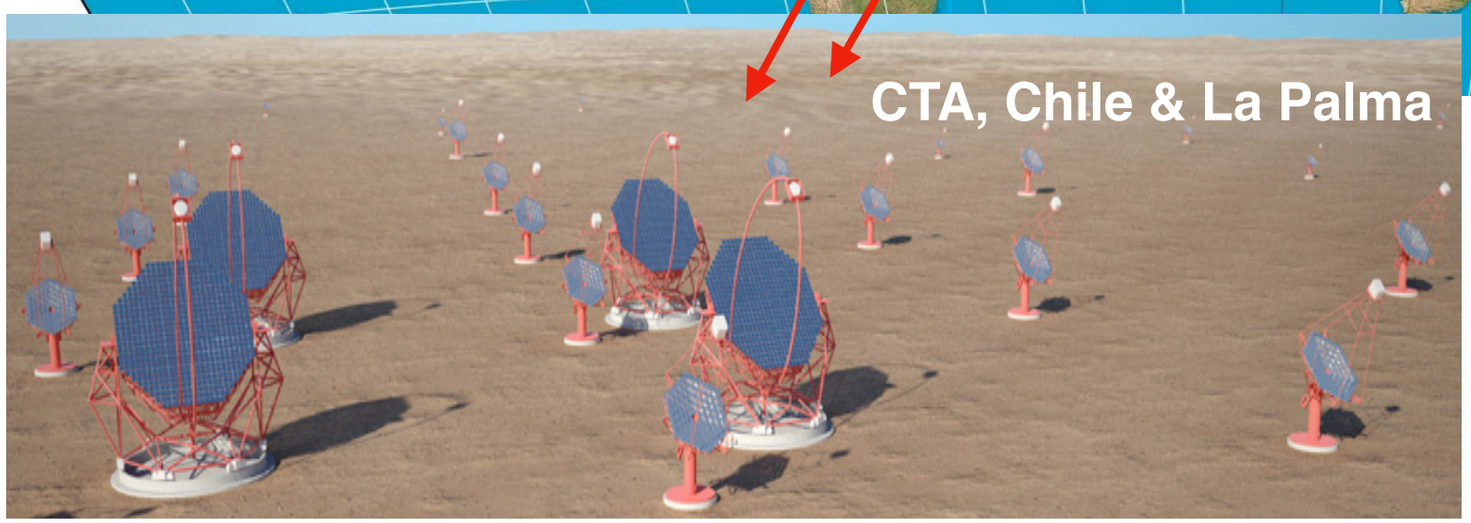
(1) Cherenkov telescopes

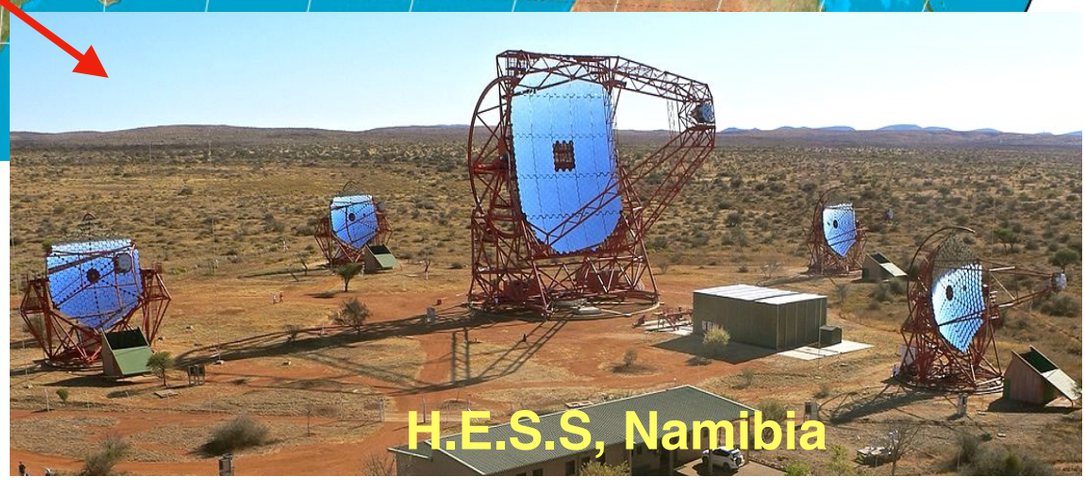
(2) Particle detector Array (Water Cherenkov detectors)

Gamma-ray experiments: (1) Cherenkov Telescopes

Imaging Atmospheric Cherenkov Telescopes (IACTs)

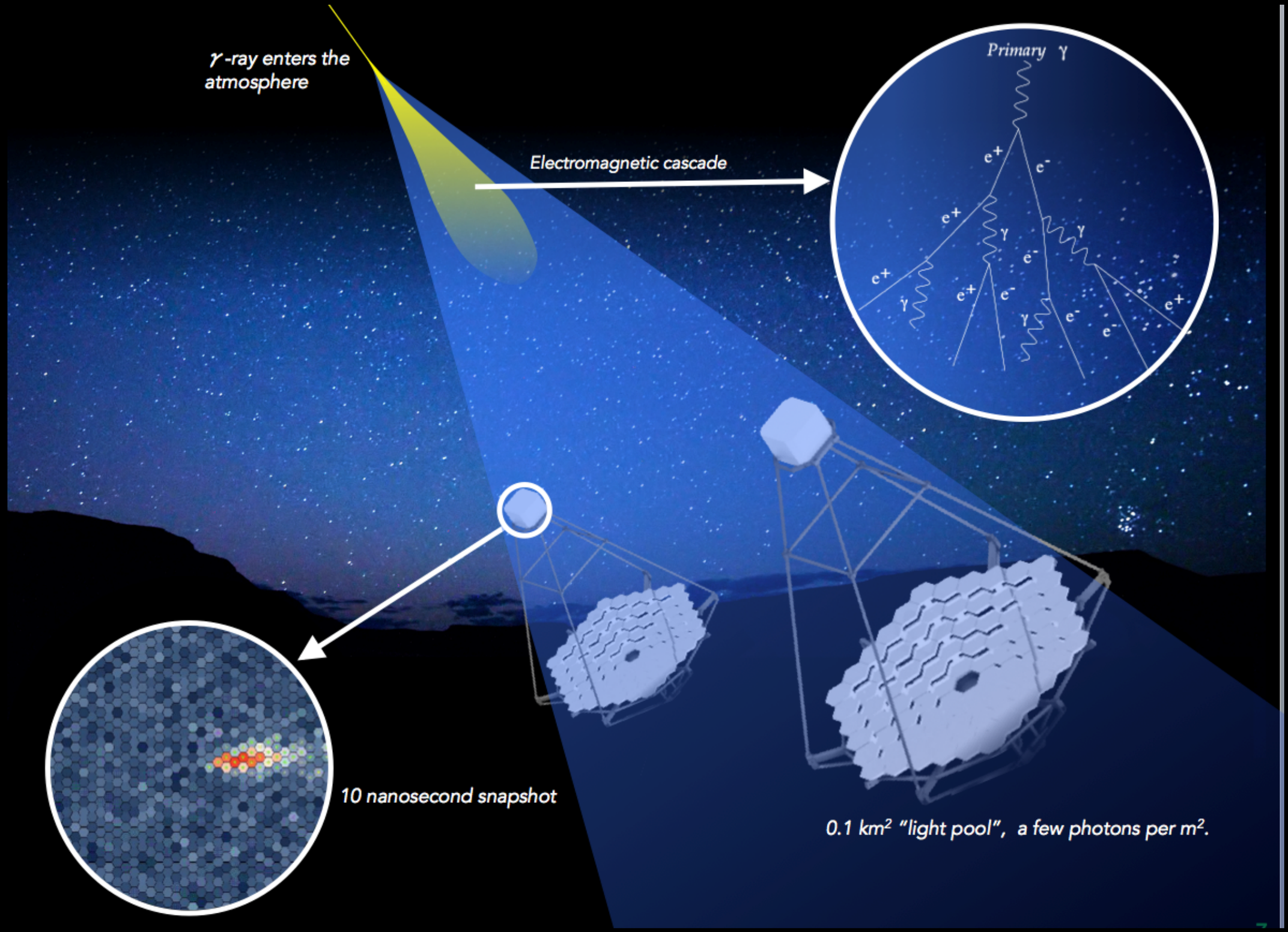






Gamma-ray experiments: (1) Cherenkov Telescopes

How Imaging Atmospheric Cherenkov Telescopes (IACTs) work?

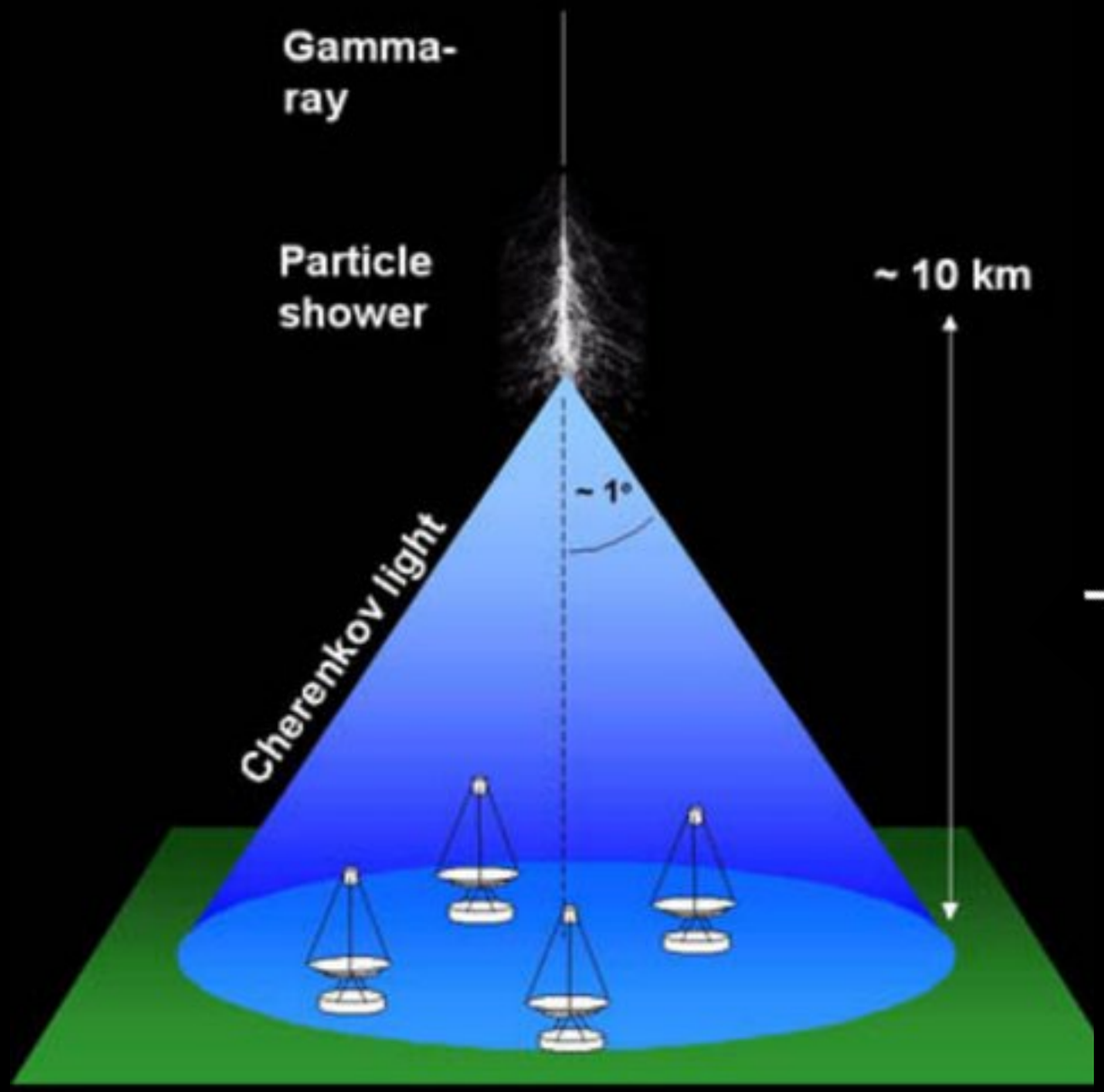


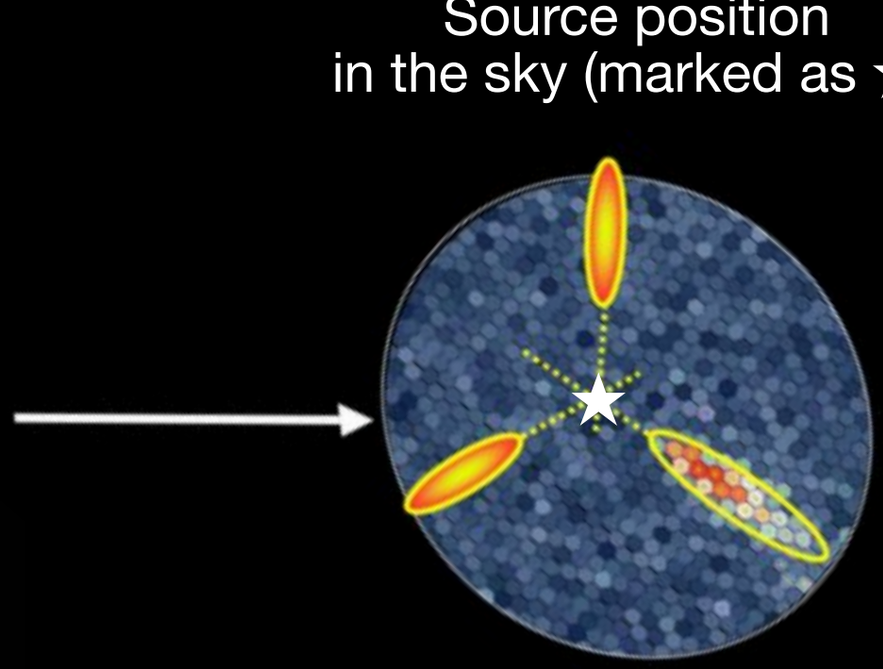
https://www.cta-observatory.org/about/how-cta-works/cta47/

Gamma-ray experiments: (1) Cherenkov Telescopes

How Imaging Atmospheric Cherenkov Telescopes (IACTs) work?

If multiple telescopes are used (the so-called stereoscopic system), the position of the source in the sky can be reconstructed, which is given by the intersection point of the image axes in the figure on the right.

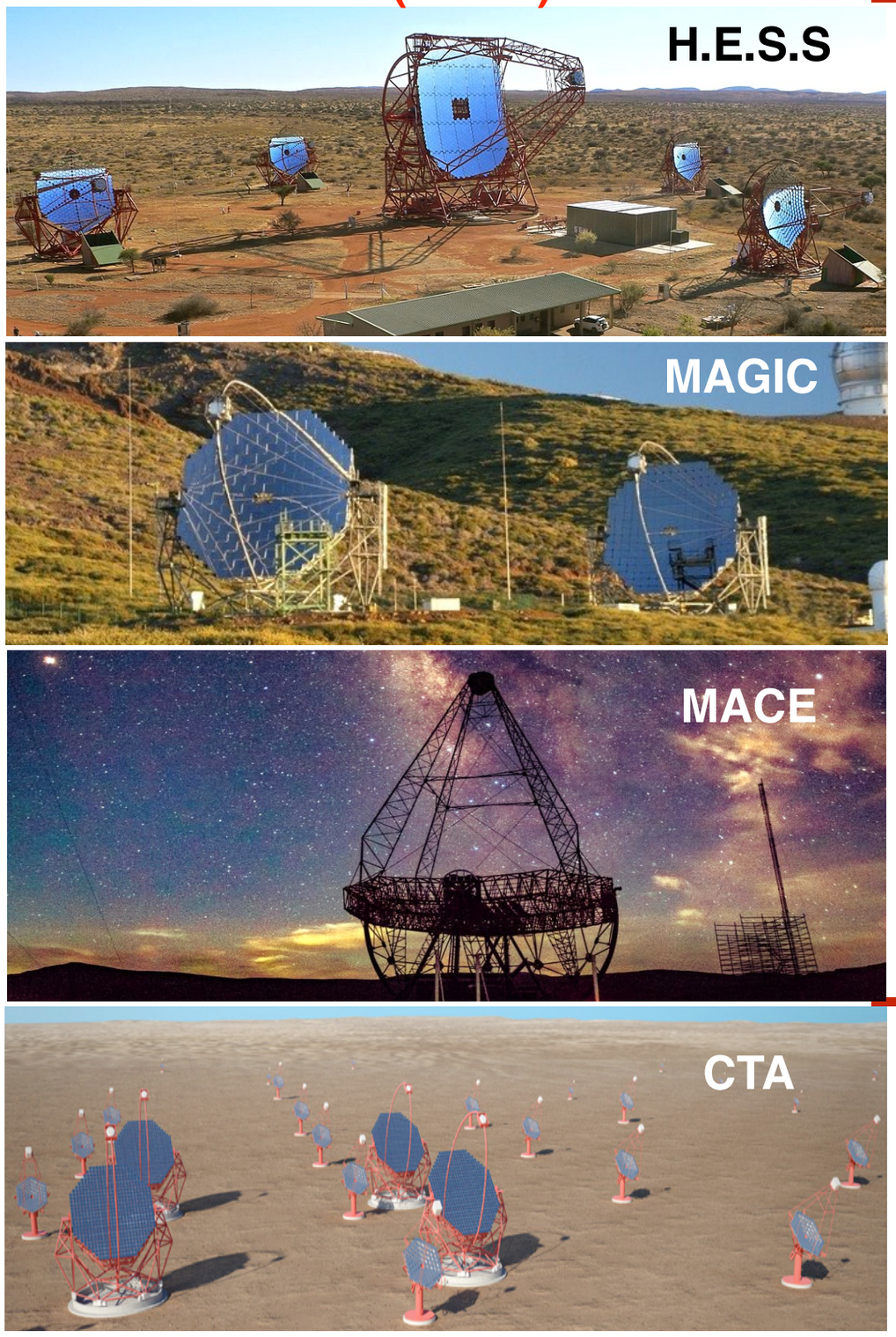




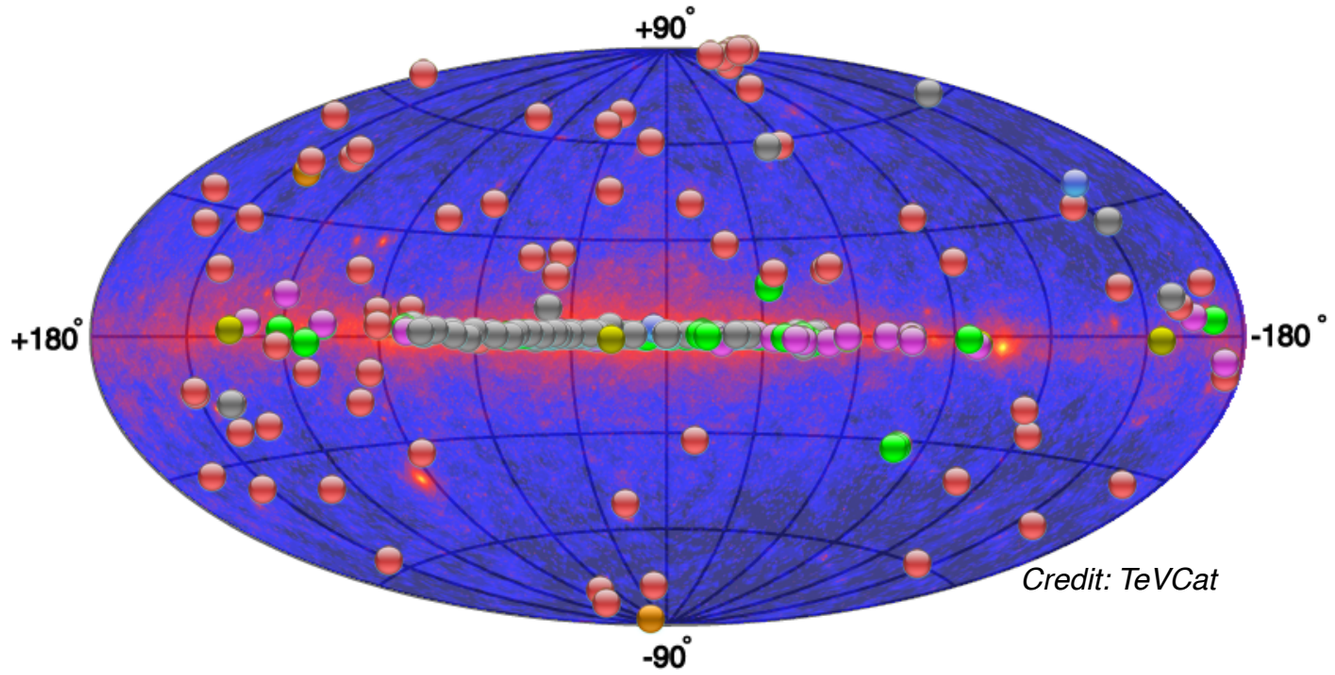
https://www.cta-observatory.org/about/how-cta-works/cta47/

Gamma-ray experiments: (1) Cherenkov Telescopes

Imaging Atmospheric Cherenkov Telescopes (IACTs)



TeV gamma-ray sources (overlaid over the *Fermi* GeV sky)



Main limitations of IACTs

- **Limited Field-of-View** ~ 4°
- **Limited duty cycle**: Dark moonless night

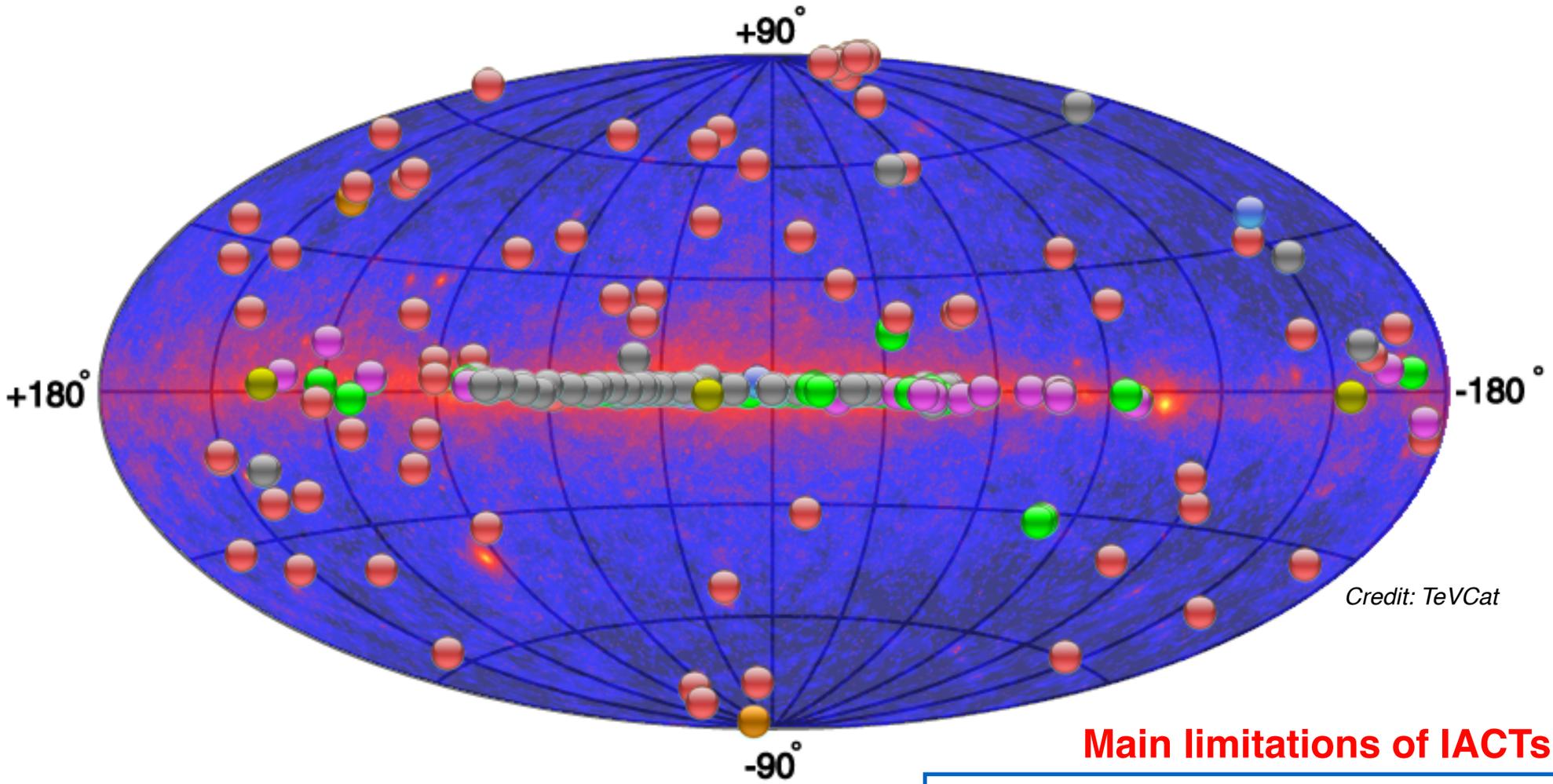
- LST: 20-200 GeV; FoV ~ 4.5°
- MST: 100 GeV - 10 TeV; FoV ~ 7°-8°
- SST: Few to 300 TeV; FoV ~ 9°

Water Cherenkov Detector arrays

FoV ~ 60°, ~100% duty cycle

Gamma-ray experiments: (1) Cherenkov Telescopes

TeV gamma-ray sources (overlaid over the *Fermi* GeV sky)



Credit: TeVCat

- *Fermi* has detected over 6000 sources above 50 MeV up to above few hundreds GeV.
- At TeV energies, only a little over 100 sources detected so far!
- This tells us how much we still need to explore the sky at TeV energies.

Main limitations of IACTs

- **Limited Field-of-View** ~ 4° for existing telescopes such as H.E.S.S, MAGIC…
- **Limited duty cycle**: Dark moonless night

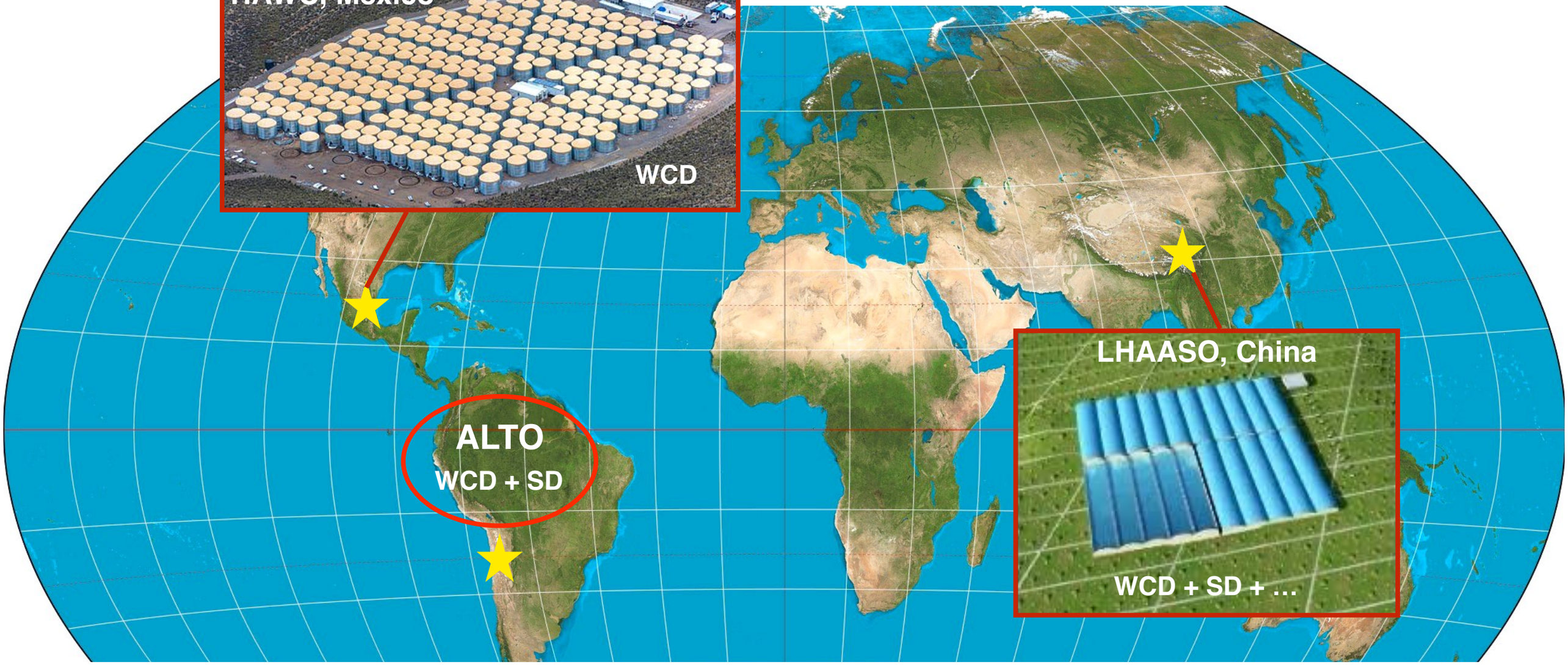
Even for the upcoming CTA

- LST: 20-200 GeV; FoV ~ 4.5°
- MST: 100 GeV - 10 TeV; FoV ~ 7°-8°
- SST: Few to 300 TeV; FoV ~ 9°

Gamma-ray experiments: (2) Particle detector array

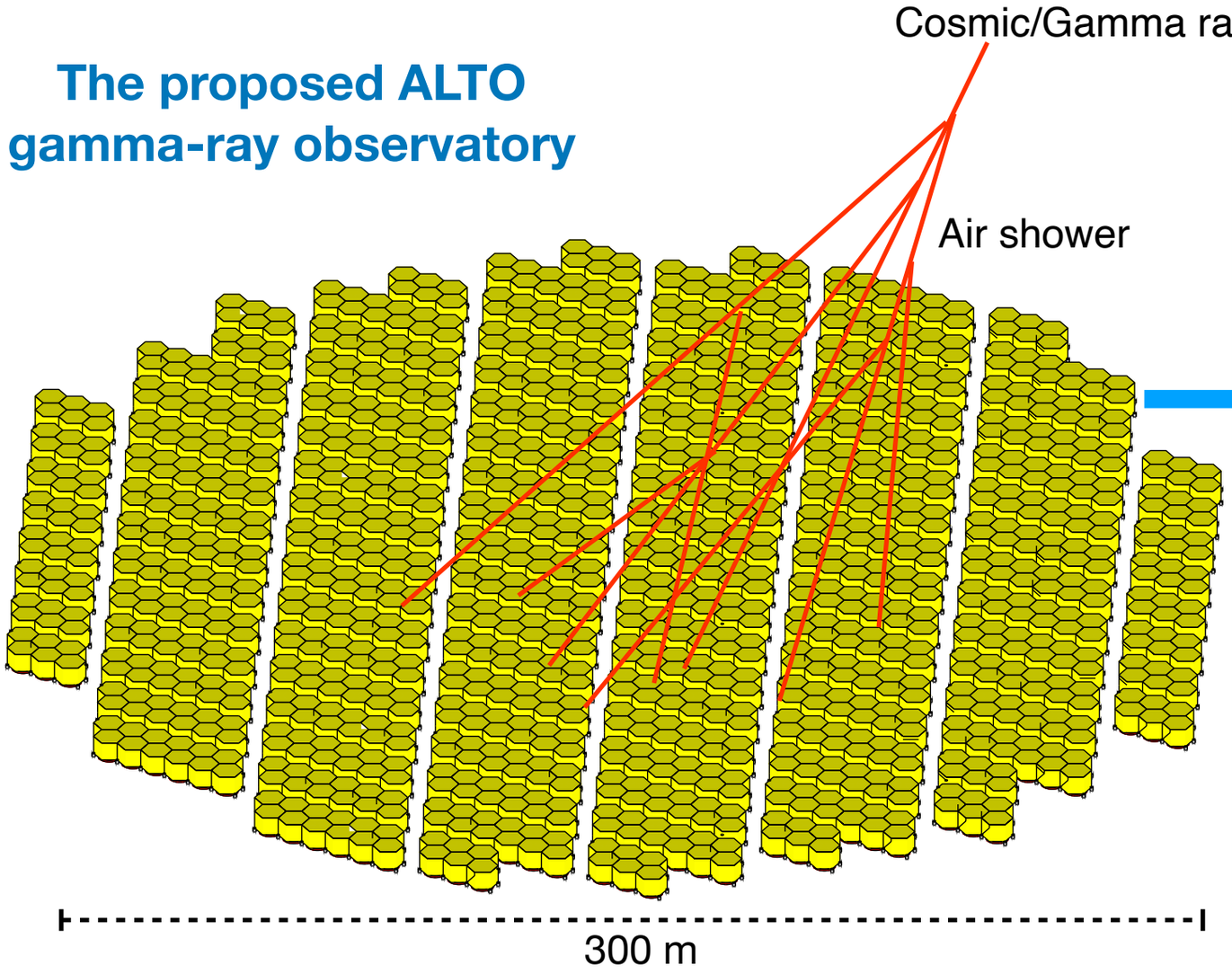
Detector type:
- Water Cherenkov Detector (WCD)
- Plastic or liquid Scintillation Detector (SD)

Particle Detector arrays
FoV ~ 60º, ~100% duty cycle



Gamma-ray experiments: (2) Particle detector array

The proposed ALTO gamma-ray observatory



- Altitude ~ 5.1 km above sea level
- Energy range ≳ 200 GeV

—> **Lower energy threshold**

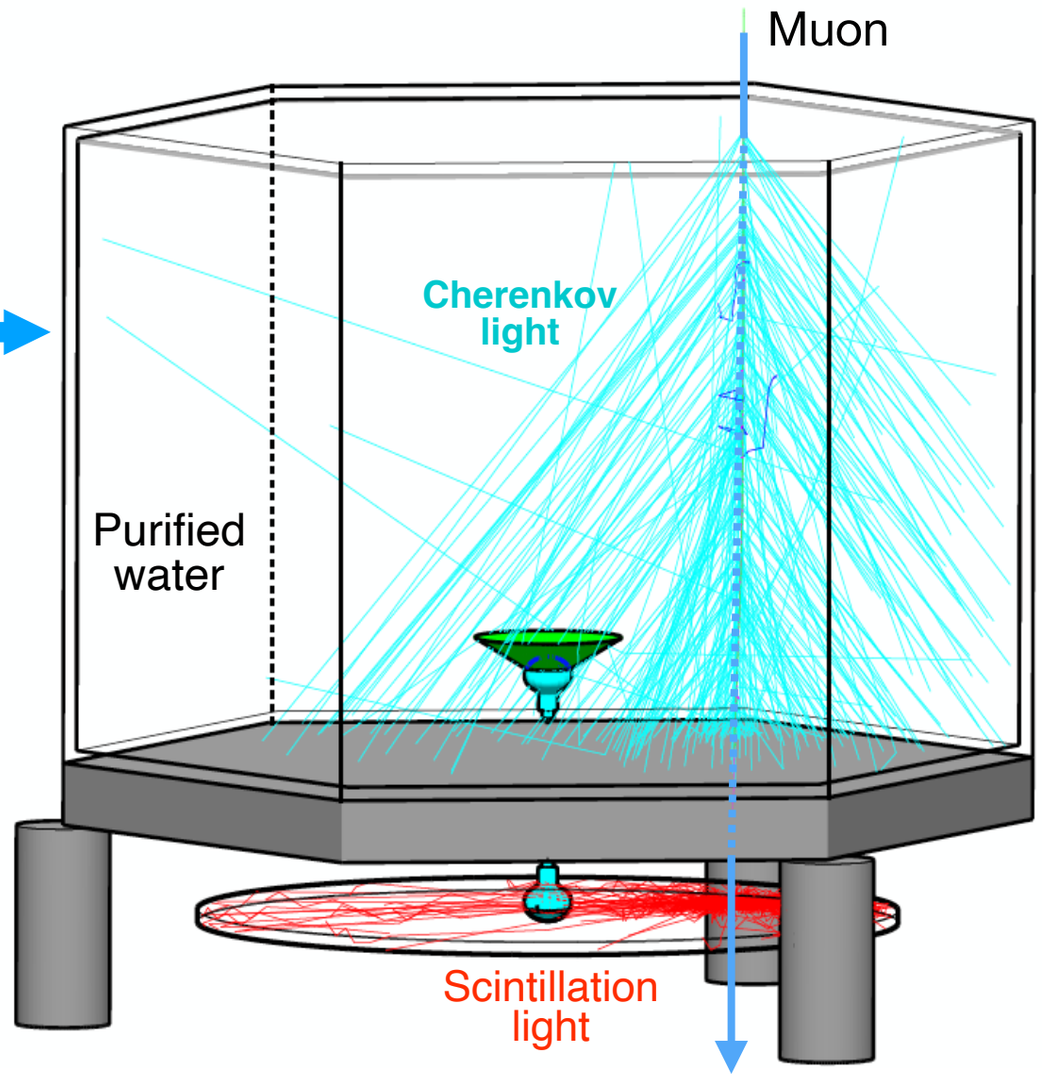
- ~ 1200 small-size detector units
- Water Cherenkov Detector + Scintillator

—> **Better sensitivity**

- Advanced electronics:
  - Sub-ns timing precision

—> **Better angular resolution**

An ALTO detector unit

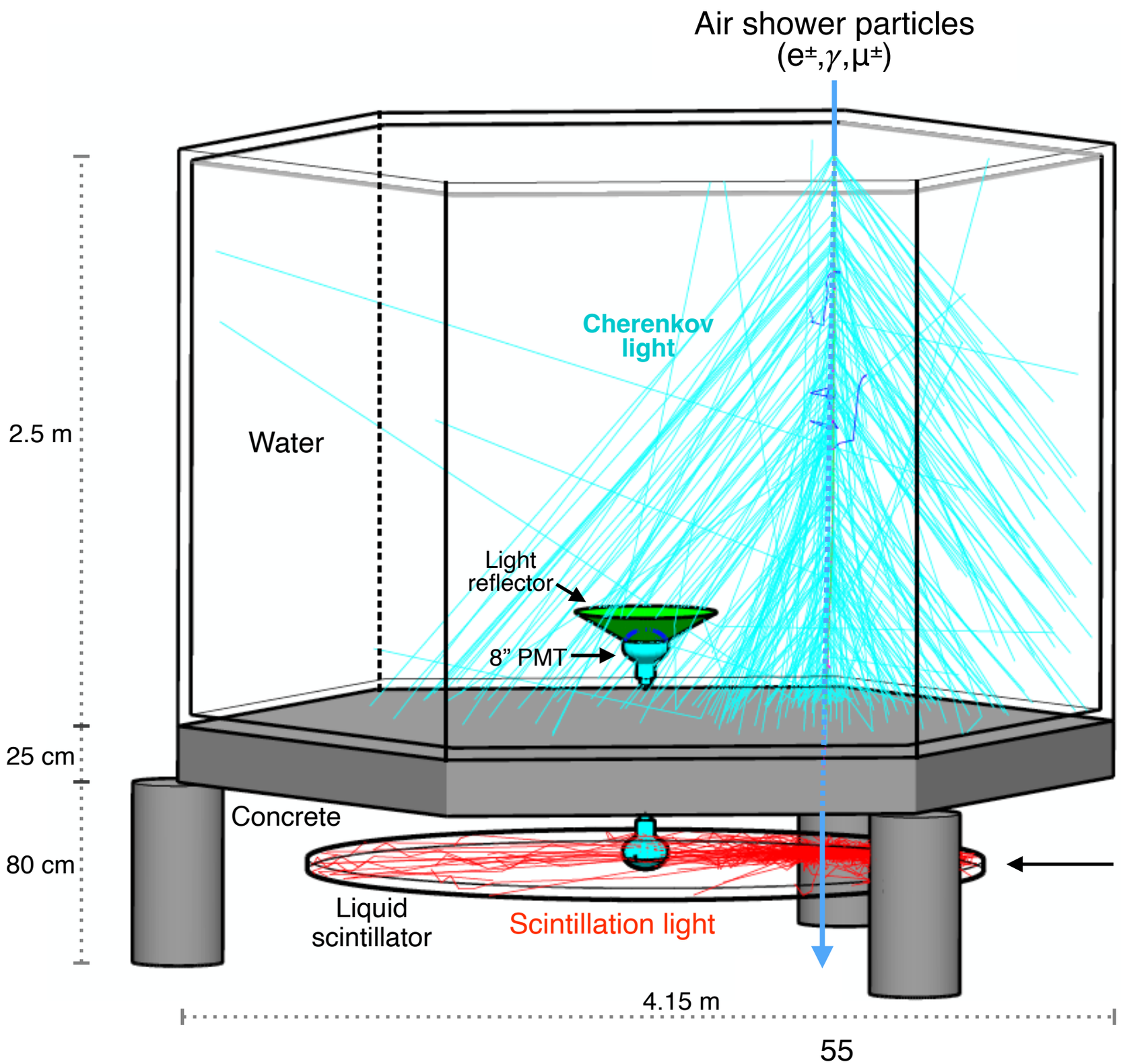


Thoudam+, 2017, ICRC, Busan

Credit: https://alto-gamma-ray-observatory.org

Gamma-ray experiments: (2) Particle detector array

Working principle of an ALTO detector



Water Cherenkov Detector

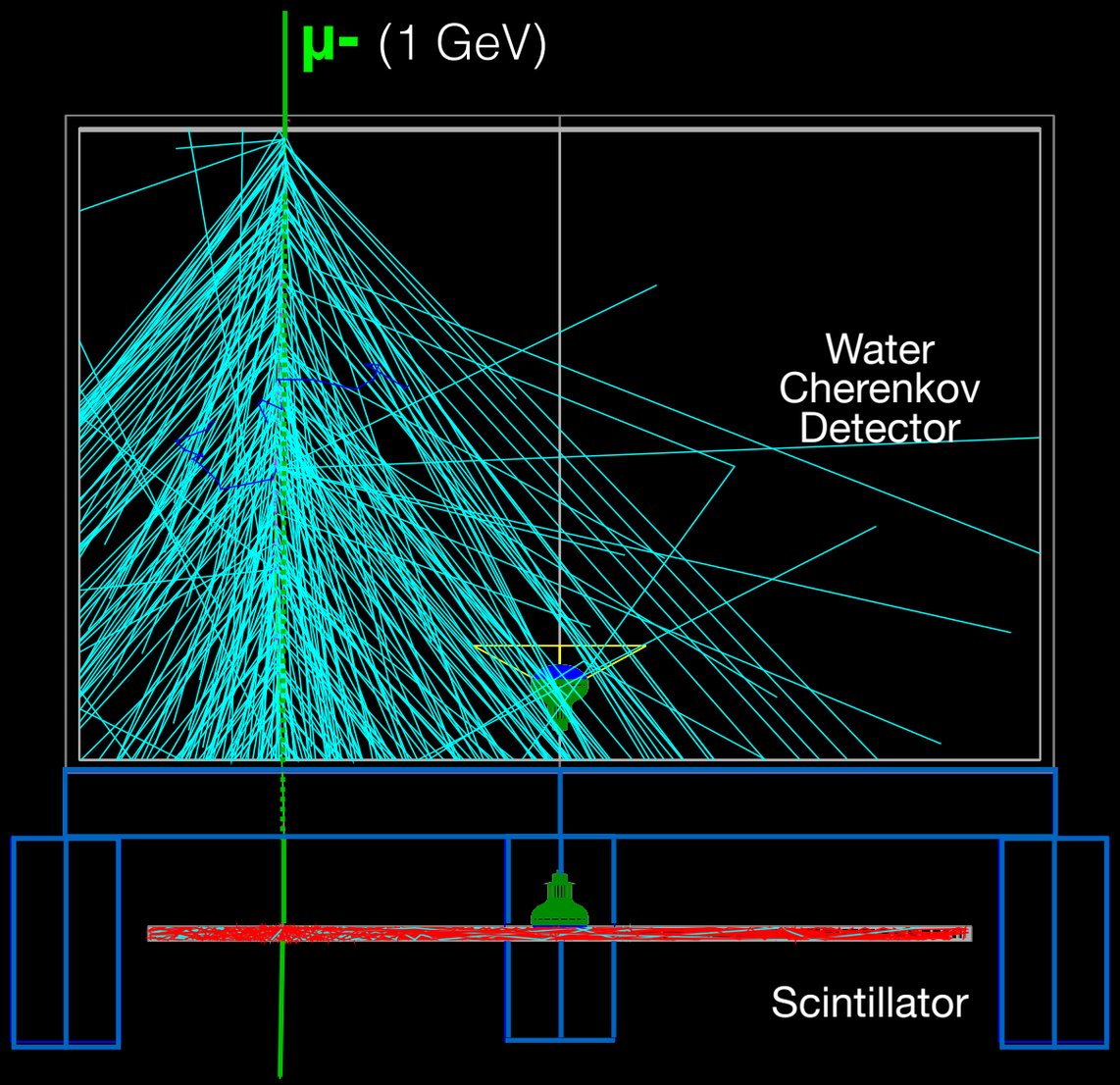
- Carbon fibre + Poly Vinyl Chloride
- Non-reflective walls
- Highly purified water
- Sensitive to e±,γ,μ±

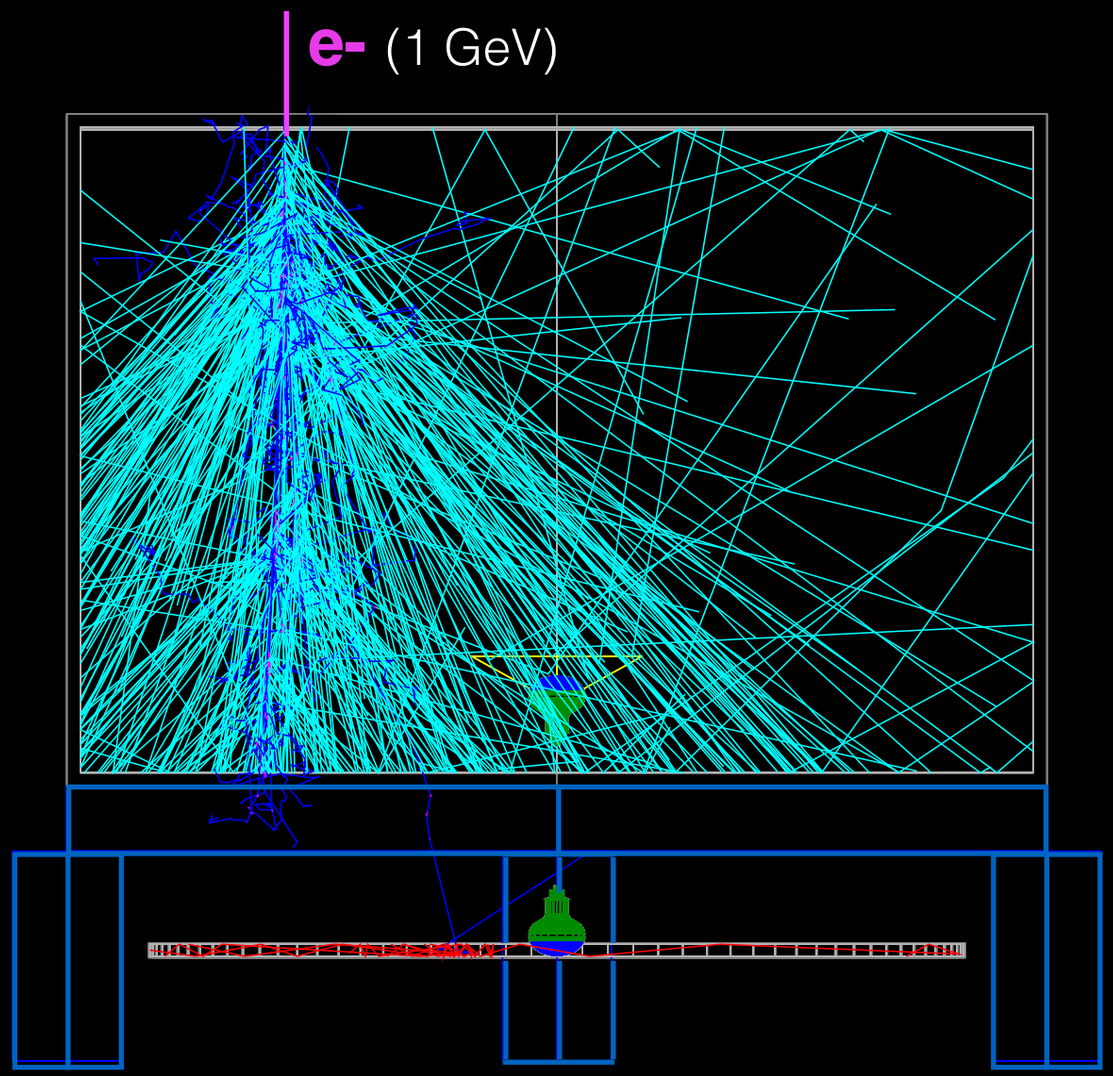
Scintillation Detector

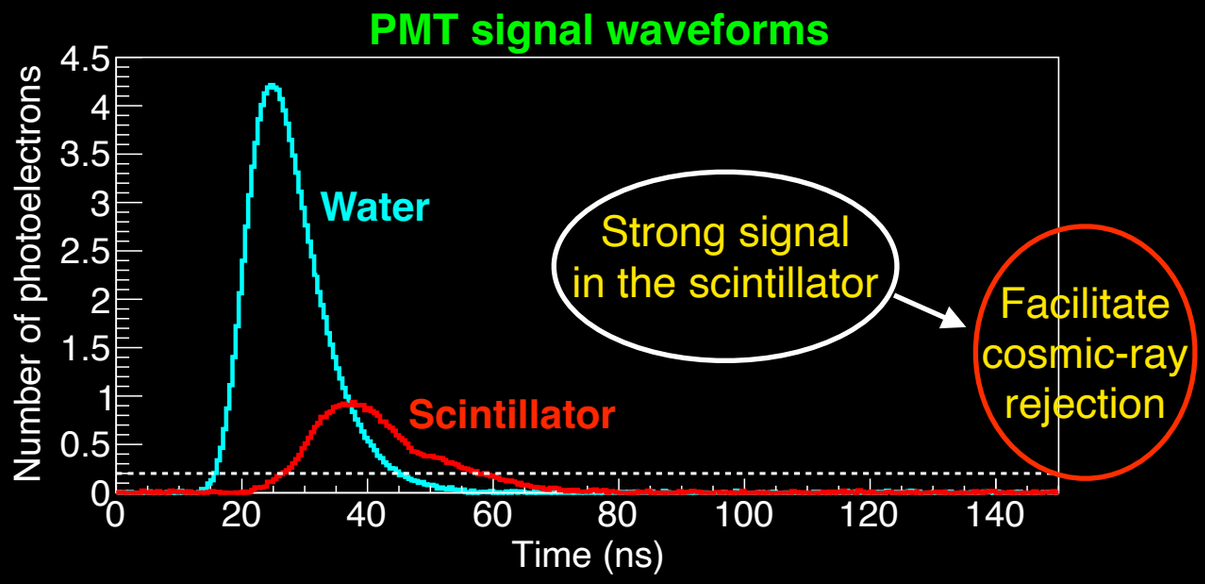
- Aluminium
- Highly-reflective walls
- Liquid scintillator: LAB+PPO+POPOP
- μ± detector

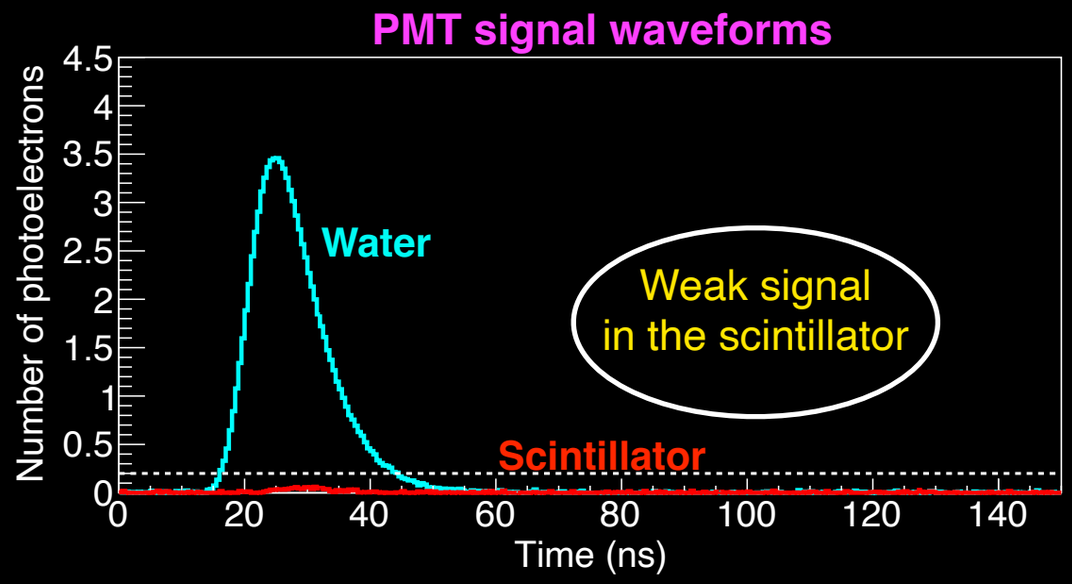
Thoudam+, 2017, ICRC, Busan

ALTO detector response to different types of particles







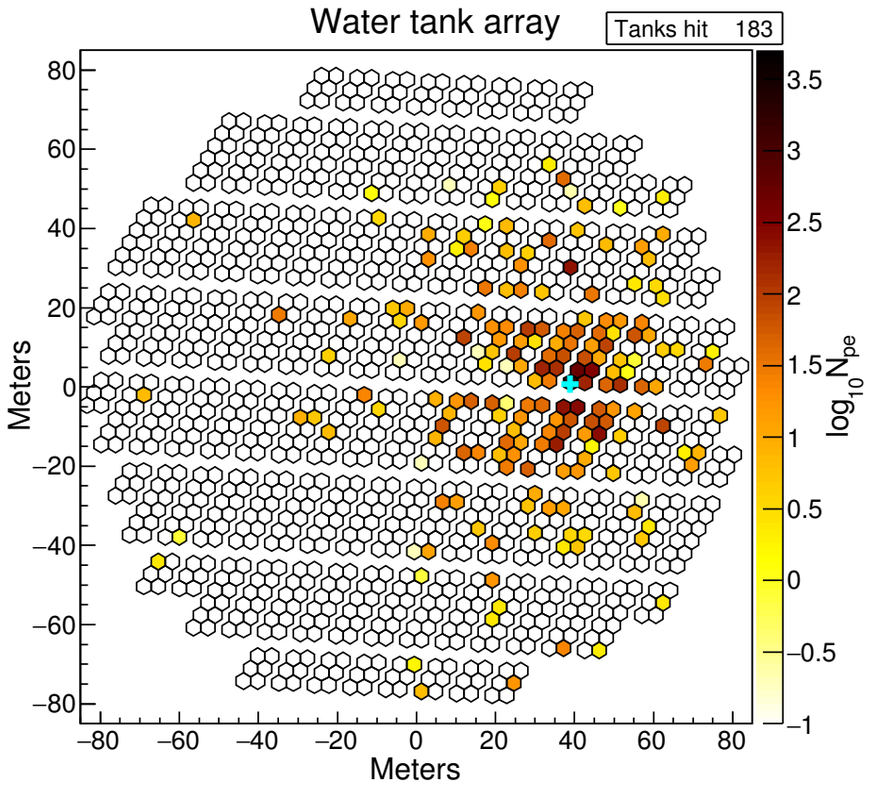


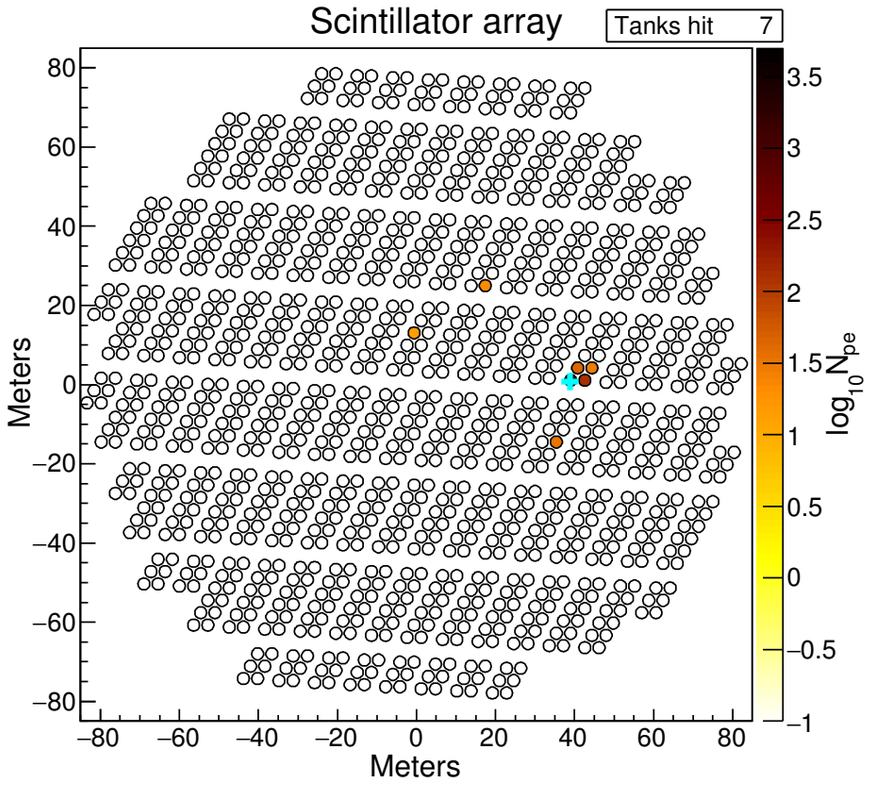
Thoudam+, 2017, ICRC, Busan

Reconstruction of the primary particle properties for particle detectors

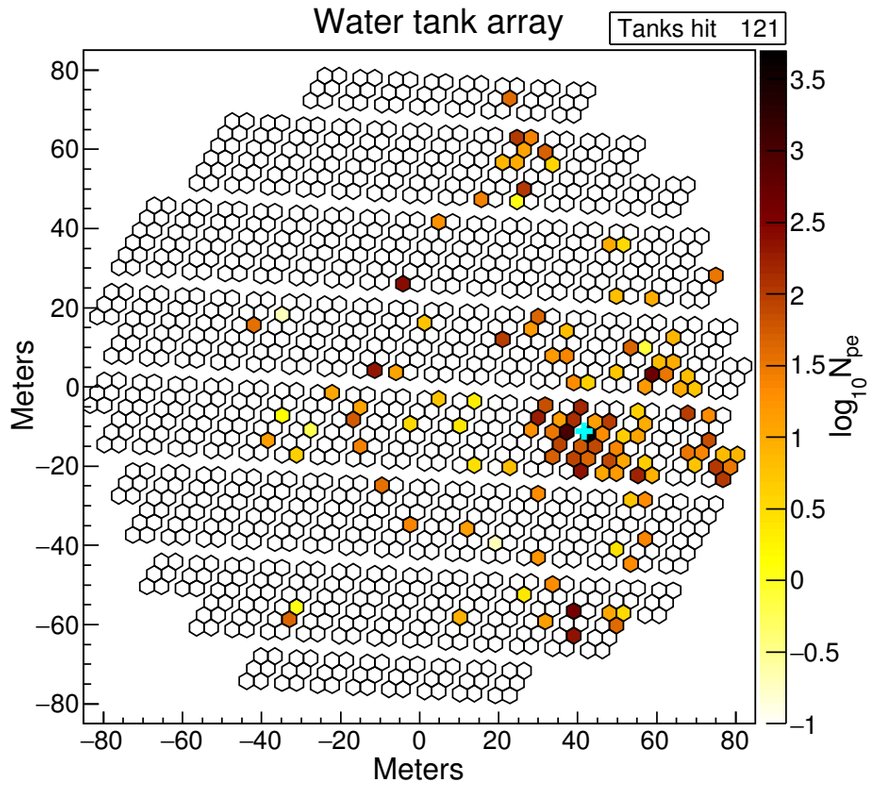
- Primary type (γ/CR) can be determined from the shower footprint.

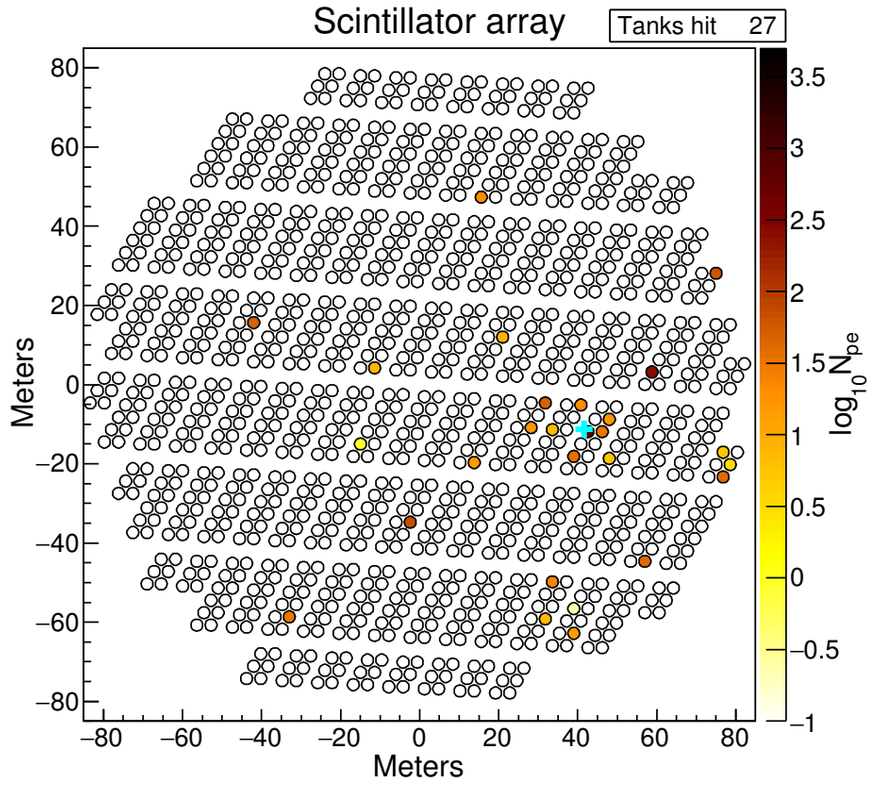
Gamma ray (1 TeV, 18°)





Proton (1 TeV, 18°)





Gamma ray

- More compact.
- Regular pattern.

Cosmic ray

- Clumpy.
- Irregular, hot spots far away from the core.

Gamma ray

- Less scintillator trigger.

Cosmic ray

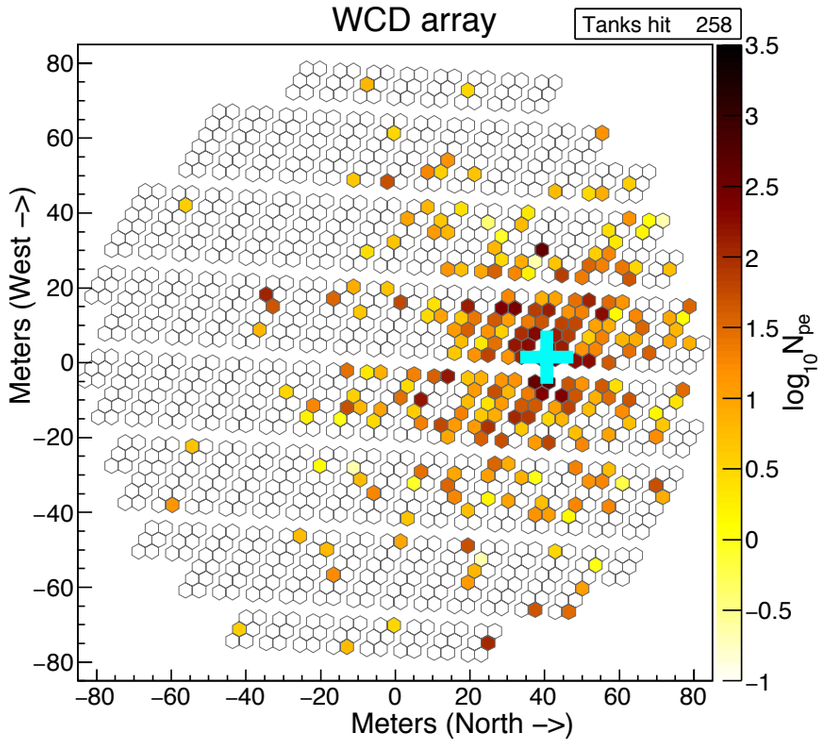
- More scintillator trigger due to muons.

Thoudam+, 2017, ICRC, Busan

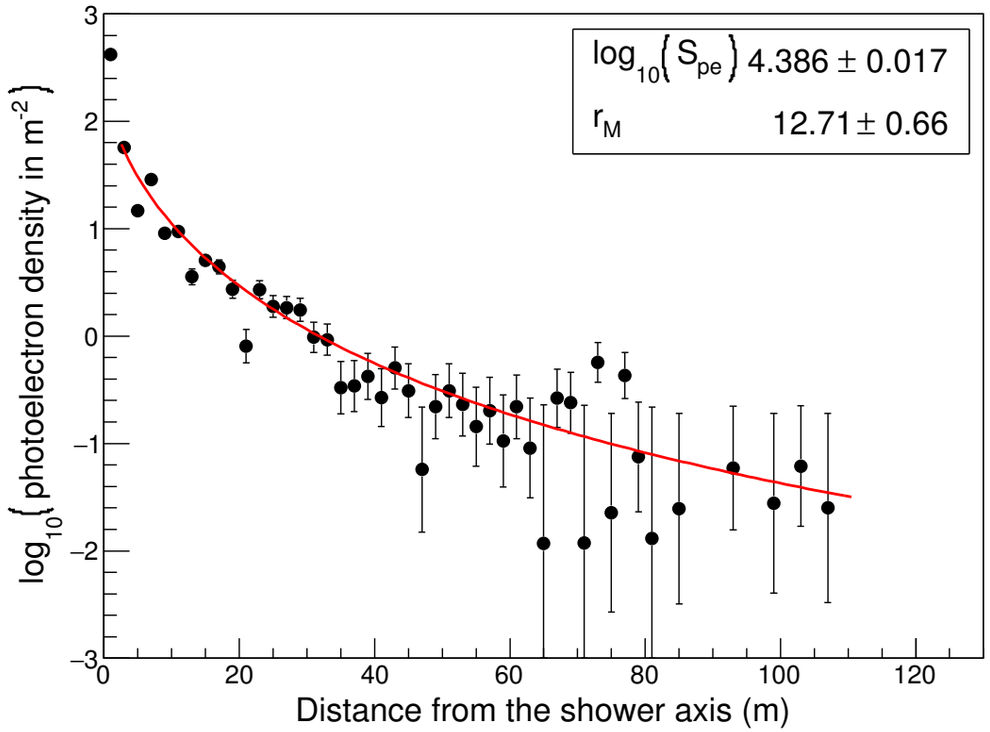
Reconstruction of the primary particle properties for particle detectors

Energy is determined from the shower size (the no. of particle reaching the ground)

Gamma ray (1 TeV, 18°)



Lateral distribution of charge deposits



True core = (40, 0) m

Reconstructed core= (39.3±0.3, 1.5±0.9)m

Accuracy = 2.3 m

Fit the lateral distribution with NKG function:

$$\rho(r') = S_{\mathrm{pe}} C(s) \left(\frac{r'}{r_{\mathrm{M}}}\right)^{s-2} \left(1+\frac{r'}{r_{\mathrm{M}}}\right)^{s-4.5}$$

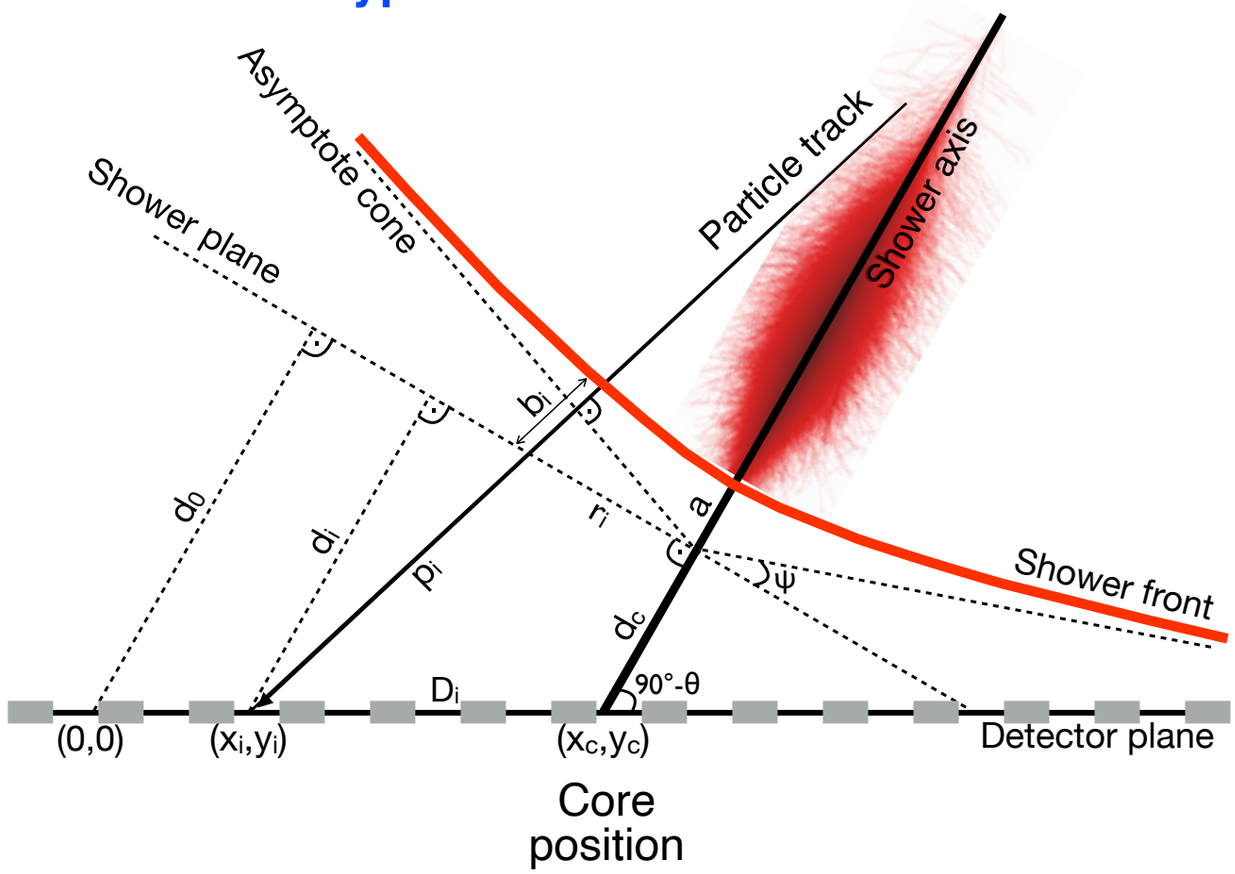
$$C(s) = \frac{\Gamma(4.5-s)}{2\pi r_{\mathrm{M}}^2 \Gamma(s)\Gamma(4.5-2s)}.$$

- S_{pe}: Shower Size—> **Energy**
- r': Shower core position
- r_{M}: Moliere radius
- s: Shower age

Reconstruction of the primary particle properties for particle detectors

Arrival direction is determined from the relative signal arrival time at the different detectors

Hyperbolic fit to the wavefront



Minimize the signal delay times with:

$$ct_\mathrm{i} = \frac{d_\mathrm{i}}{\cos\psi} + \frac{r_\mathrm{i}}{\sin\psi} - \frac{\cos\psi}{\left(\cot^2\psi - 1\right)} \times \left\{ r_\mathrm{i}\cot^3\psi - \sqrt{a^2\cot^4\psi + r_\mathrm{i}^2\cot^2\psi - a^2} \right\},$$

$$d_\mathrm{i} = d_0 - lx_\mathrm{i} - my_\mathrm{i},$$

$$r_\mathrm{i} = \sqrt{D_\mathrm{i}^2 - \left(d_\mathrm{i} - d_\mathrm{c}\right)^2} - \frac{d_\mathrm{i}}{\cot\psi},$$

$$D_\mathrm{i} = \sqrt{(x_\mathrm{c} - x_\mathrm{i})^2 + (y_\mathrm{c} - y_\mathrm{i})^2},$$

$$d_\mathrm{c} = d_0 - lx_\mathrm{c} - my_\mathrm{c}.$$

Gamma ray (1 TeV, 18°)

True direction (θ, ϕ)= (18°, 0°)
Reconstructed: (θ, ϕ)= (17.91°±0.06°, 0.48°±0.15°)

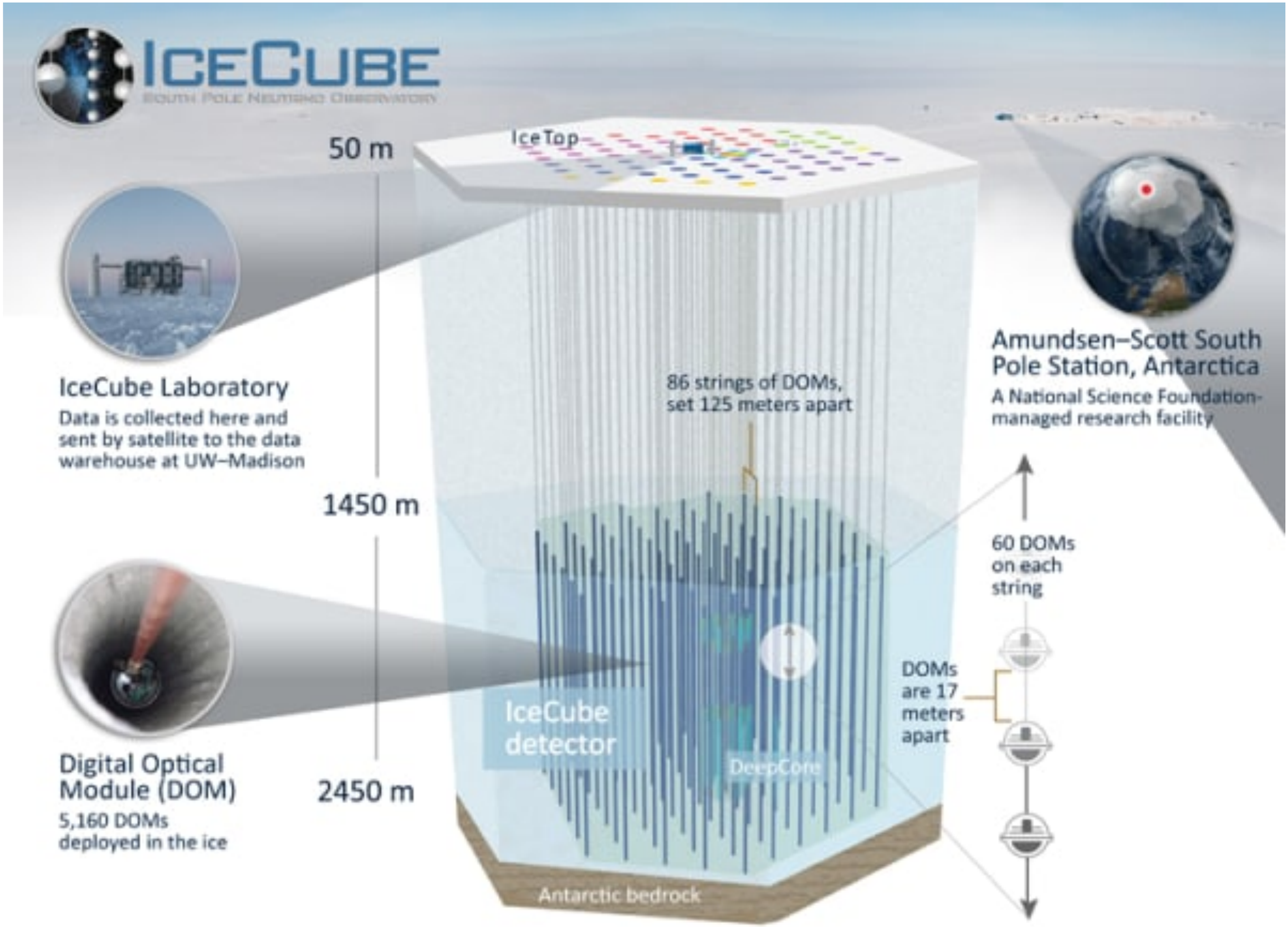
Accuracy = 0.17°

Neutrino experiments

(1) IceCube neutrino telescope in Antarctica.

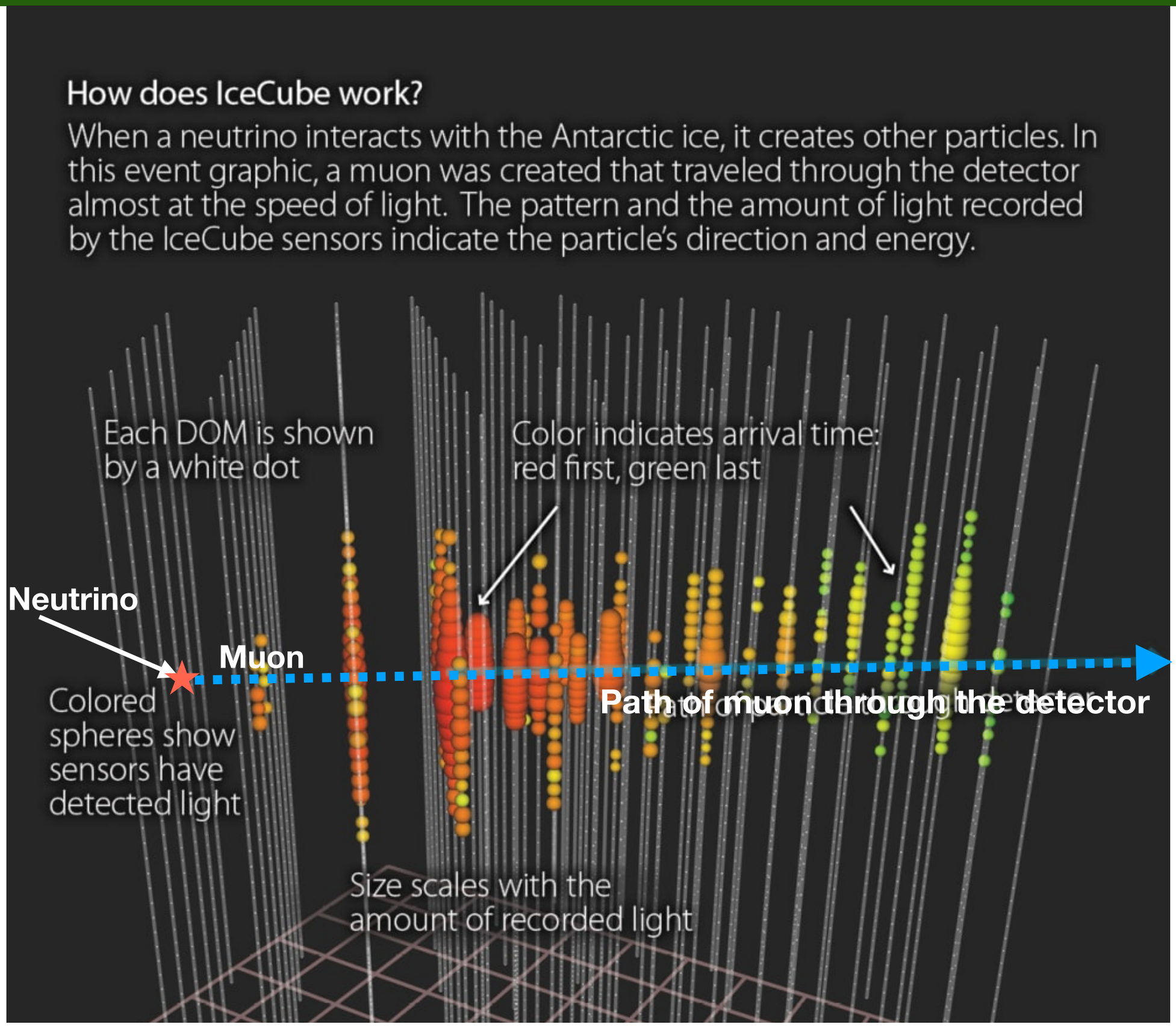
(2) Km3NeT in the Mediterranean Sea.

Neutrino experiments: IceCube experiment in Antarctica



Source: IceCube Neutrino Observatory

Neutrino experiments: IceCube experiment in Antarctica

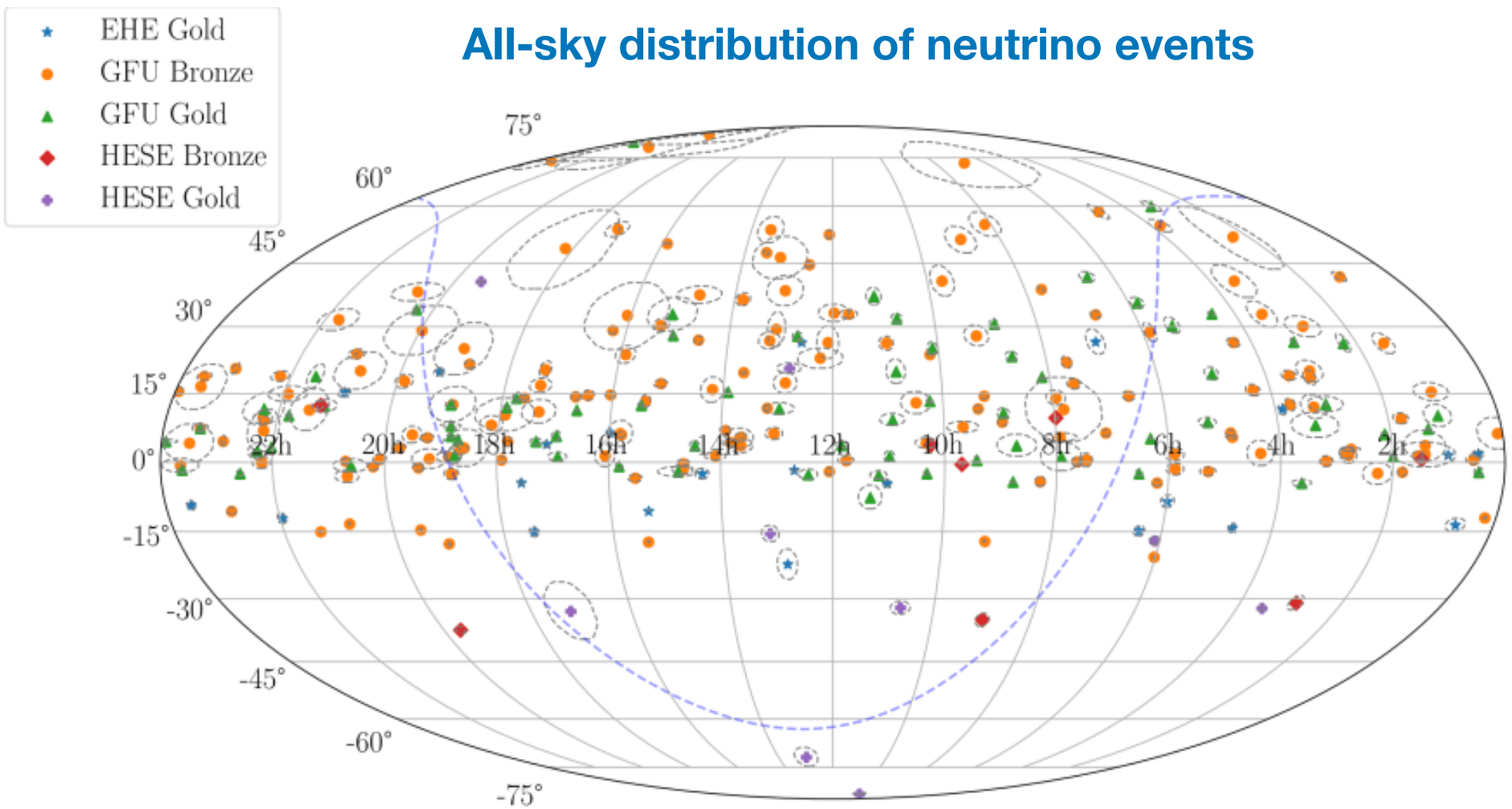


Source: IceCube Neutrino Observatory

Neutrino experiments: IceCube experiment in Antarctica

Some highlights from IceCube experiment

All-sky distribution of neutrino events



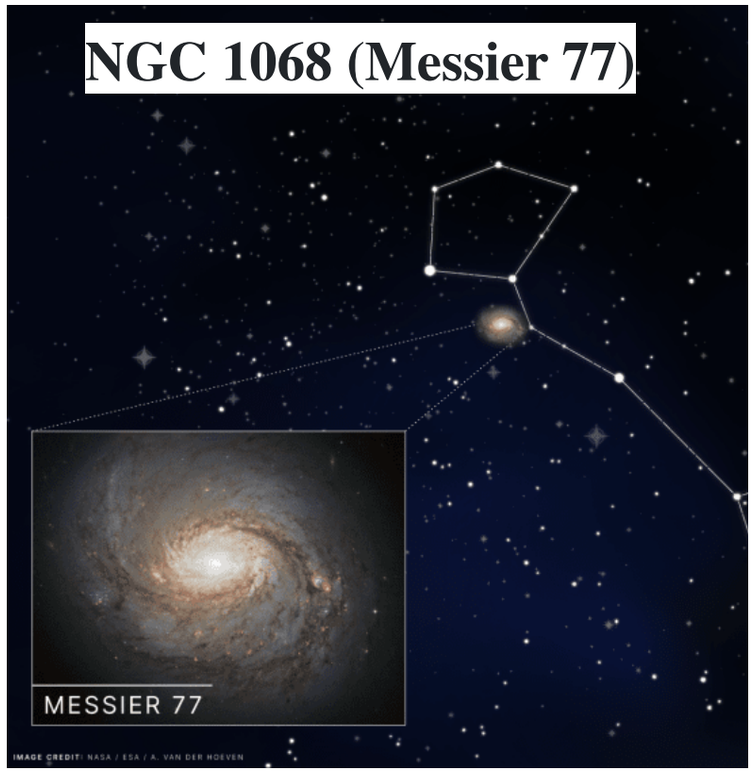
The all-sky distribution of the alerts in the catalog in equatorial coordinates. The blue stars denote extremely high energy gold alerts, the orange circles show gamma-ray follow-up (GFU) bronze, the green triangles show GFU gold, the red diamonds show high-energy starting event (HESE) bronze, and the purple plus signs show HESE gold alerts.The 90% uncertainty contours at the location of each alert are shown by the dashed ellipses.

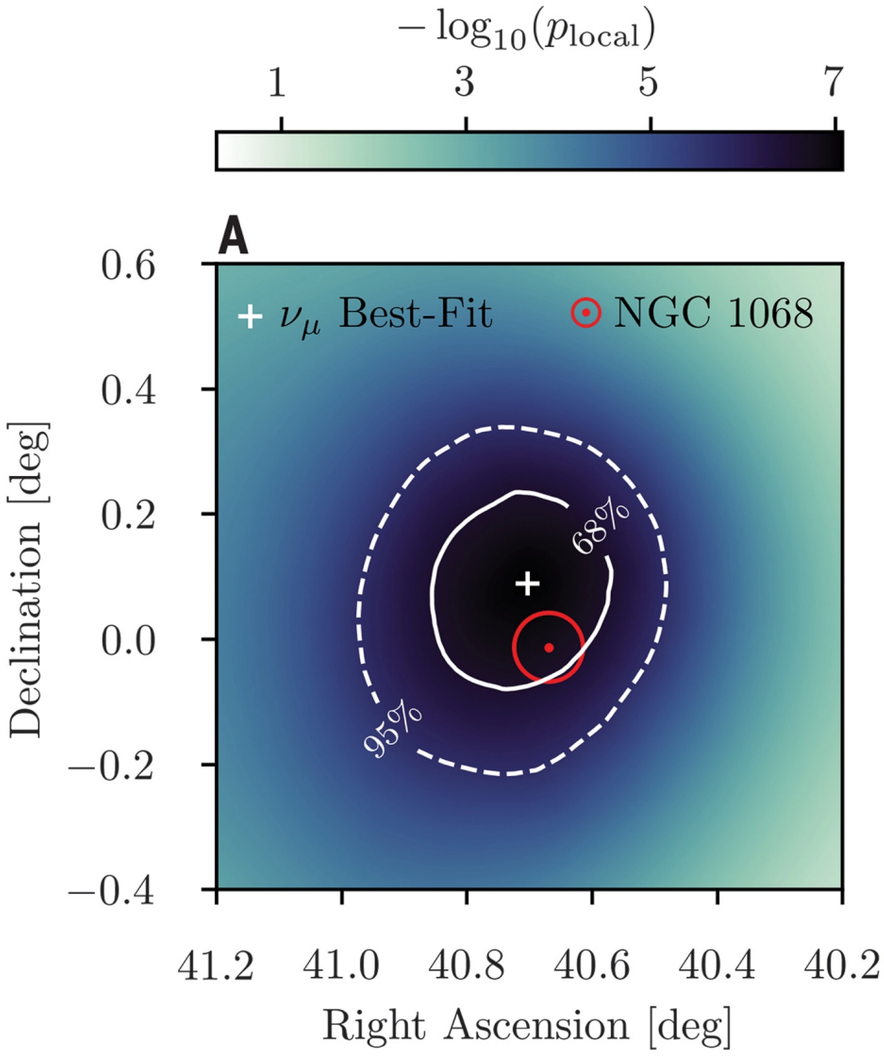
Source: IceCube Neutrino Observatory

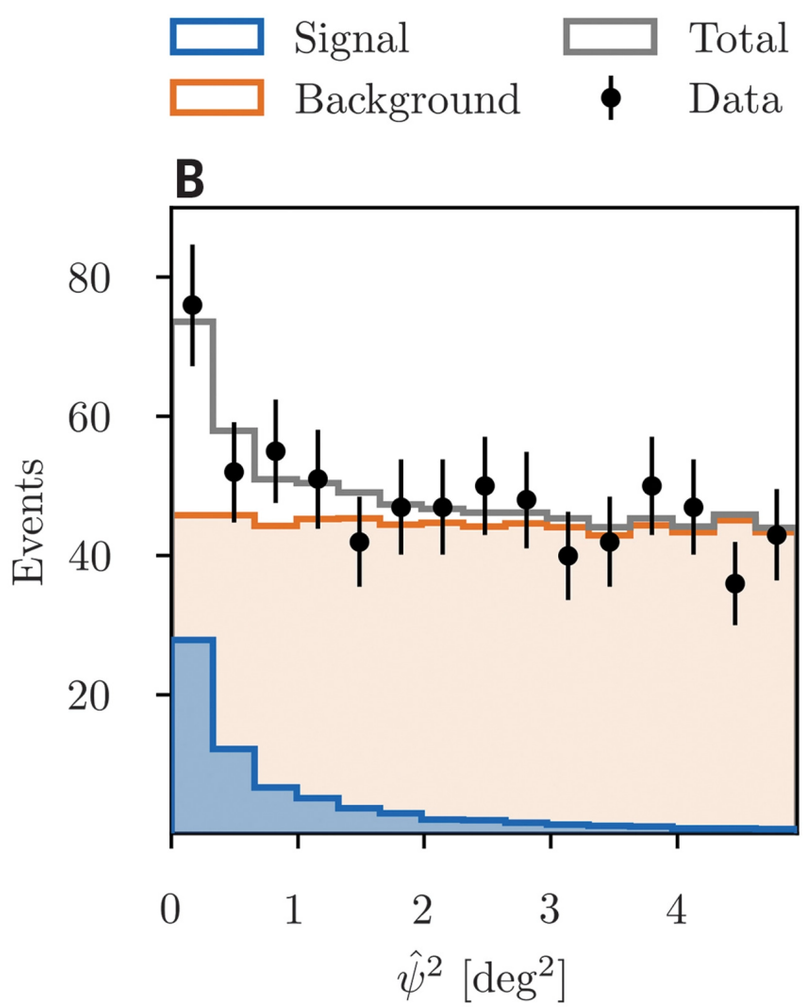
Neutrino experiments: IceCube experiment in Antarctica

Some highlights from IceCube experiment

First evidence of high-energy neutrino emission from an active galactic nuclei (AGN)







IceCube found 79^{+22}_{-20} more events than expected from the atmospheric and diffuse astrophysical background.

IceCube Collaboration: R. Abbasi et al., *Science* 378, 6619 (2022)

Neutrino experiments: KM3NeT telescope in the Mediterranean sea

Artist impression

Incoming neutrino

Muon

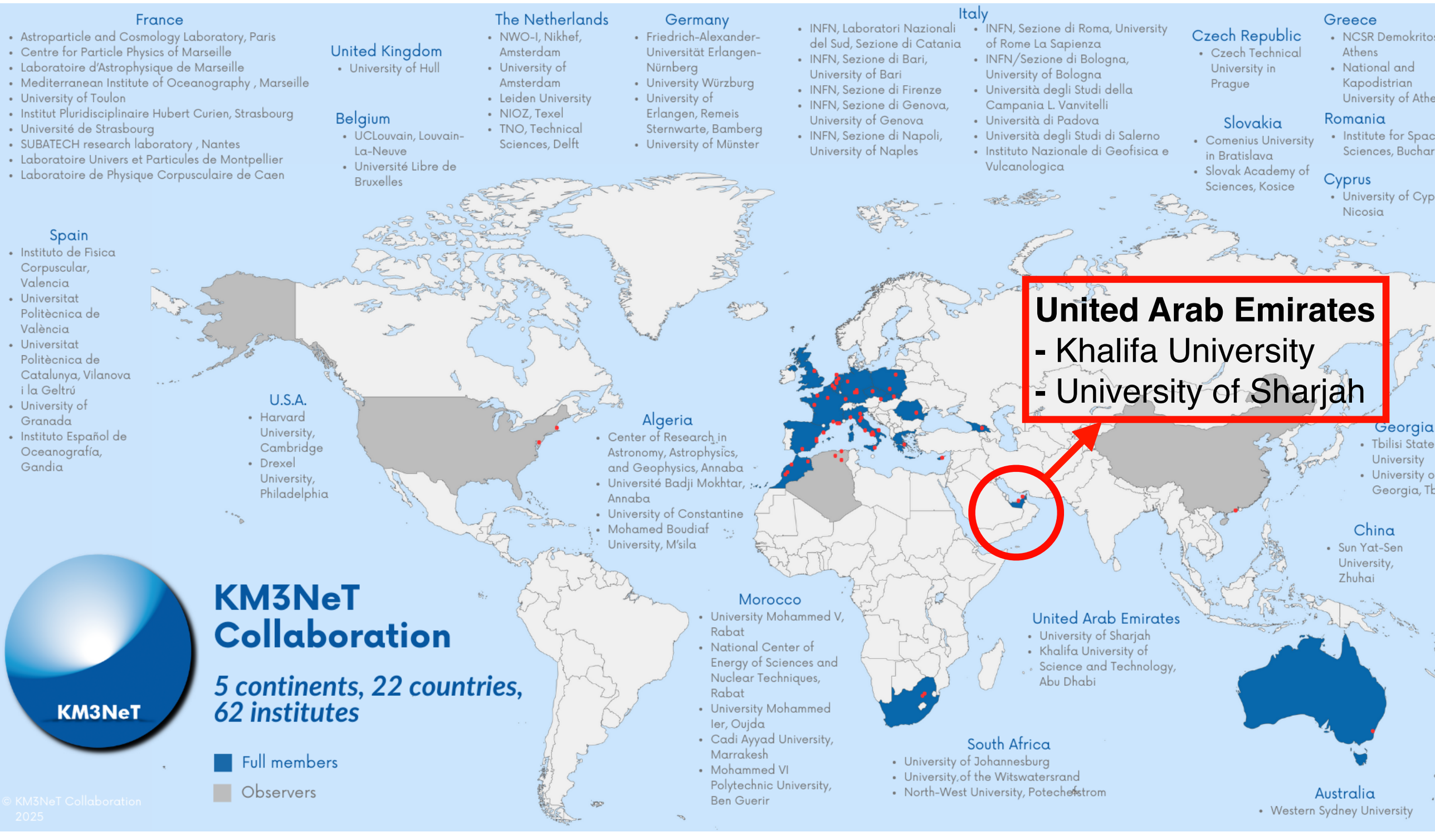
1 km

Light sensor

KM3NeT Collaboration: ~ 250 Members from 16 countries

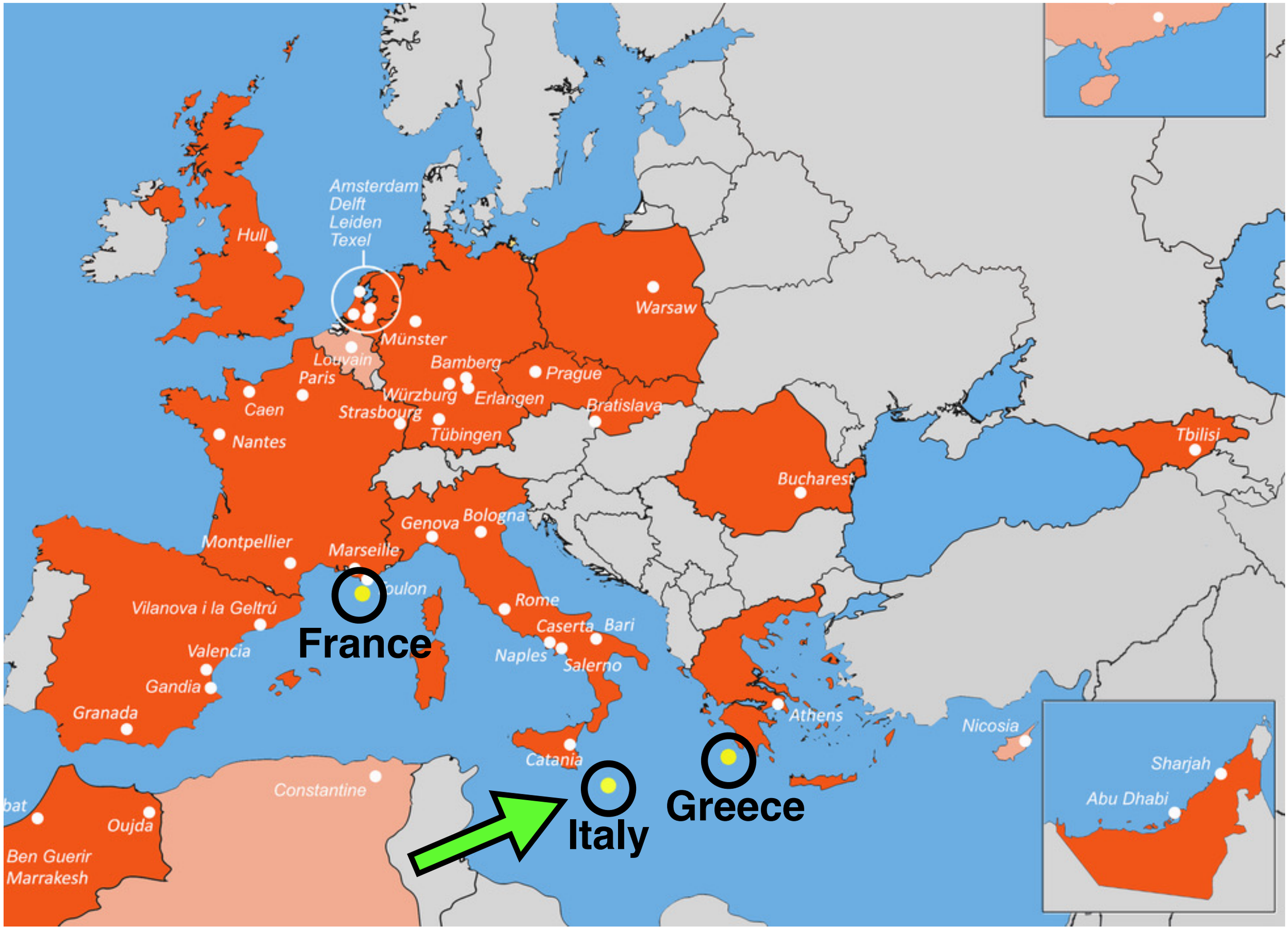
Picture credit: KM3NeT (https://www.km3net.org)

KM³ Neutrino Telescope (KM3NeT) in the Mediterranean Sea
Collaboration map: Over 360 members, 62 institutes, 22 countries



https://www.km3net.org/

Installation sites of the KM3NeT experiment



https://www.km3net.org/

Neutrino experiments: KM3NeT telescope in the Mediterranean sea

About KM3NeT

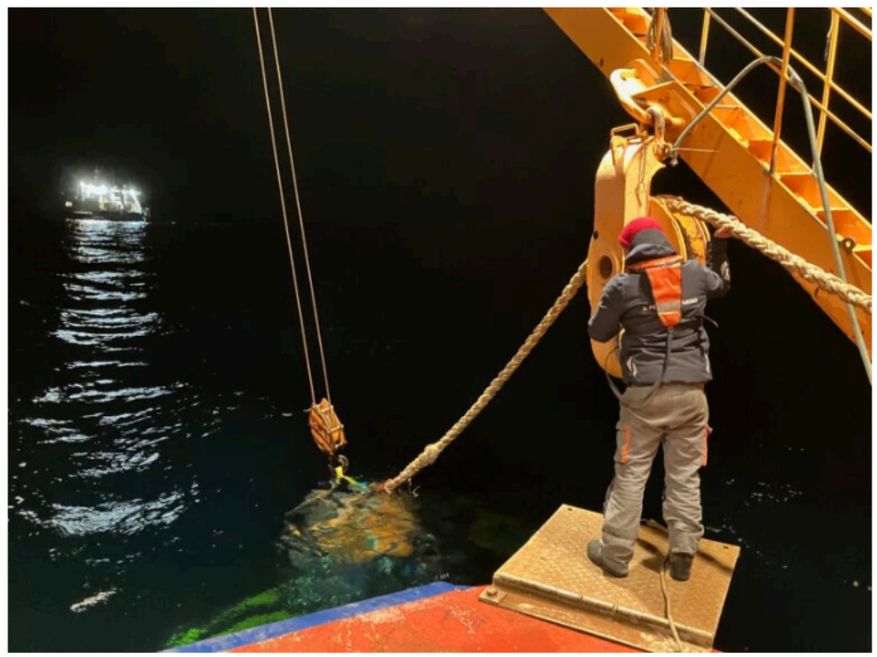
KM3NeT is a next-generation neutrino telescope currently under development in the Mediterranean sea.

KM3NeT has two kind of instruments:

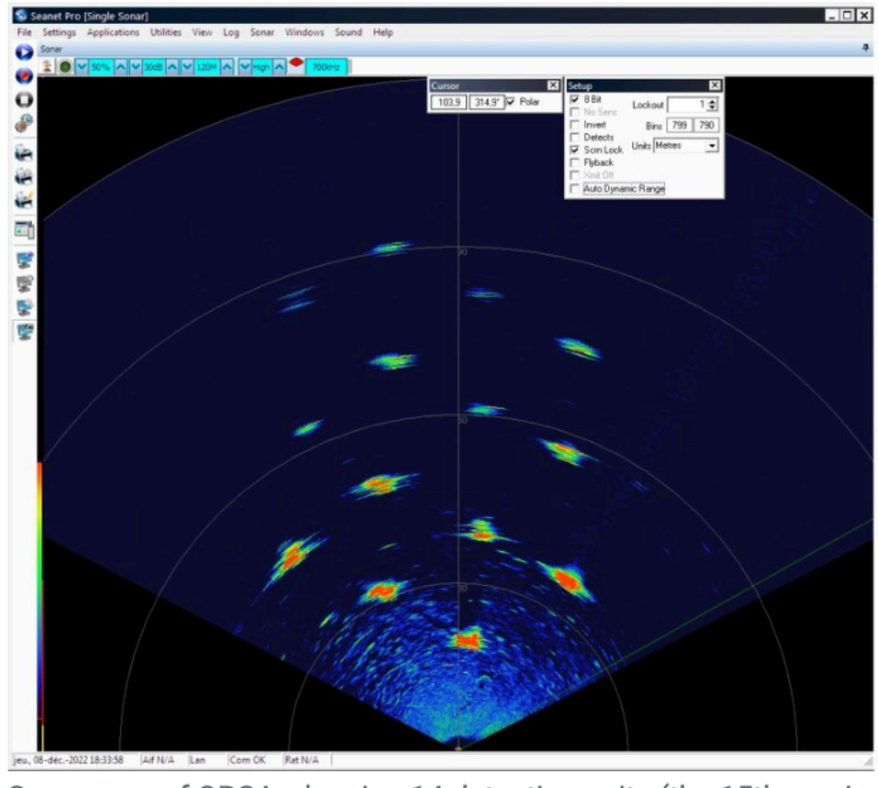
ARCA: To study Astrophysical neutrinos
- Discovery and observation of high-energy neutrino sources in the Universe.

ORCA: To study atmospheric neutrinos
- Determination of the relative masses of the neutrinos, referred to as the neutrino mass hierarchy.

KM3NeT in the Northern Hemisphere will complement neutrino measurements by IceCube in the Southern Hemisphere.



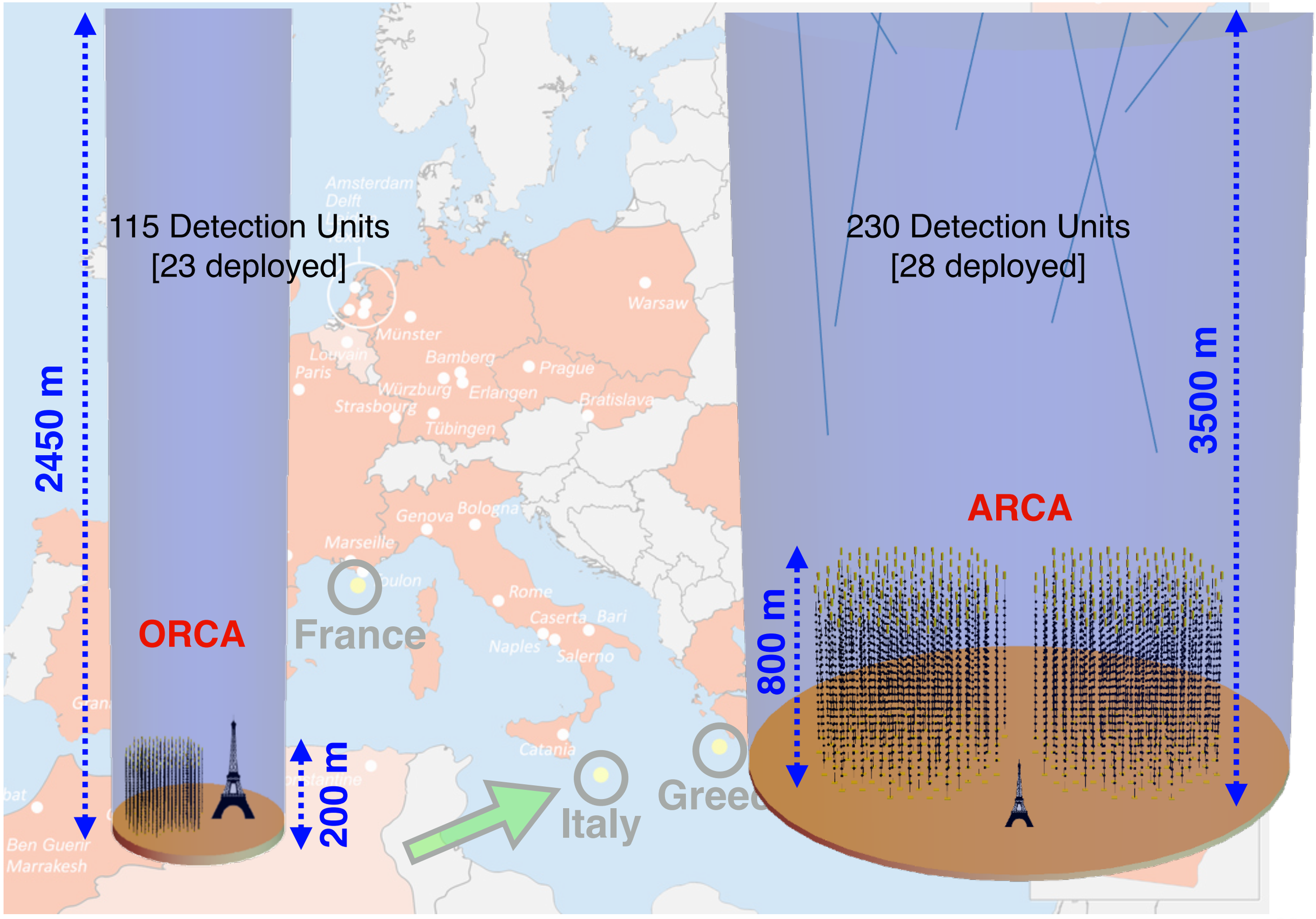
One of the ORCA detection units starting its journey from the Castor to the sea floor, 2500 m below. In the background is the Janus II.



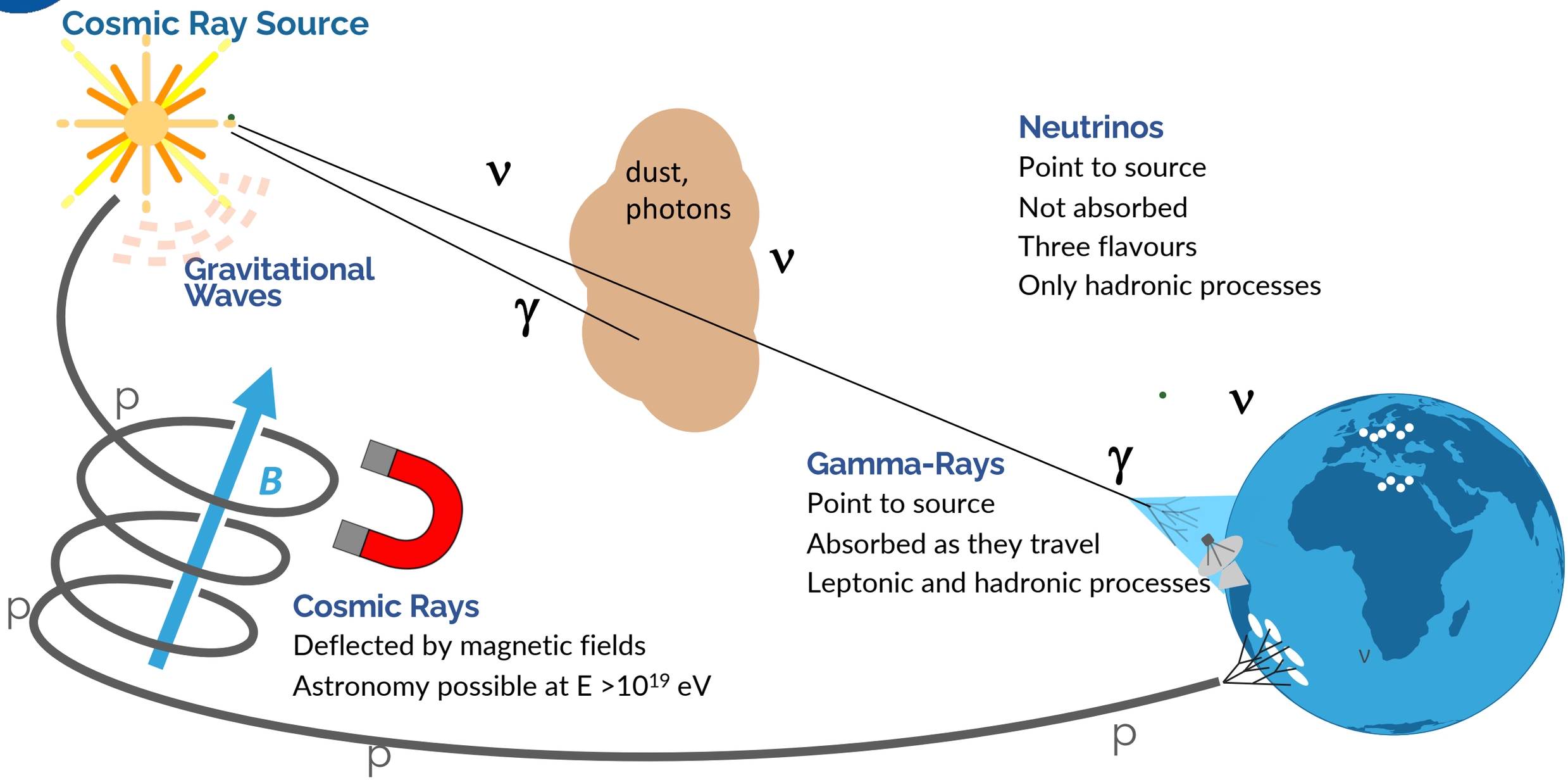
Sonar map of ORCA, showing 14 detection units (the 15th one is beyond the boundary of the map). The 4 in the bottom part of the picture are the new ones installed.

Picture credit: KM3NeT (https://www.km3net.org)

The KM3NeT neutrino experiment in the Mediterranean Sea



Astronomy/Astrophysics with neutrinos



Neutrinos

Point to source

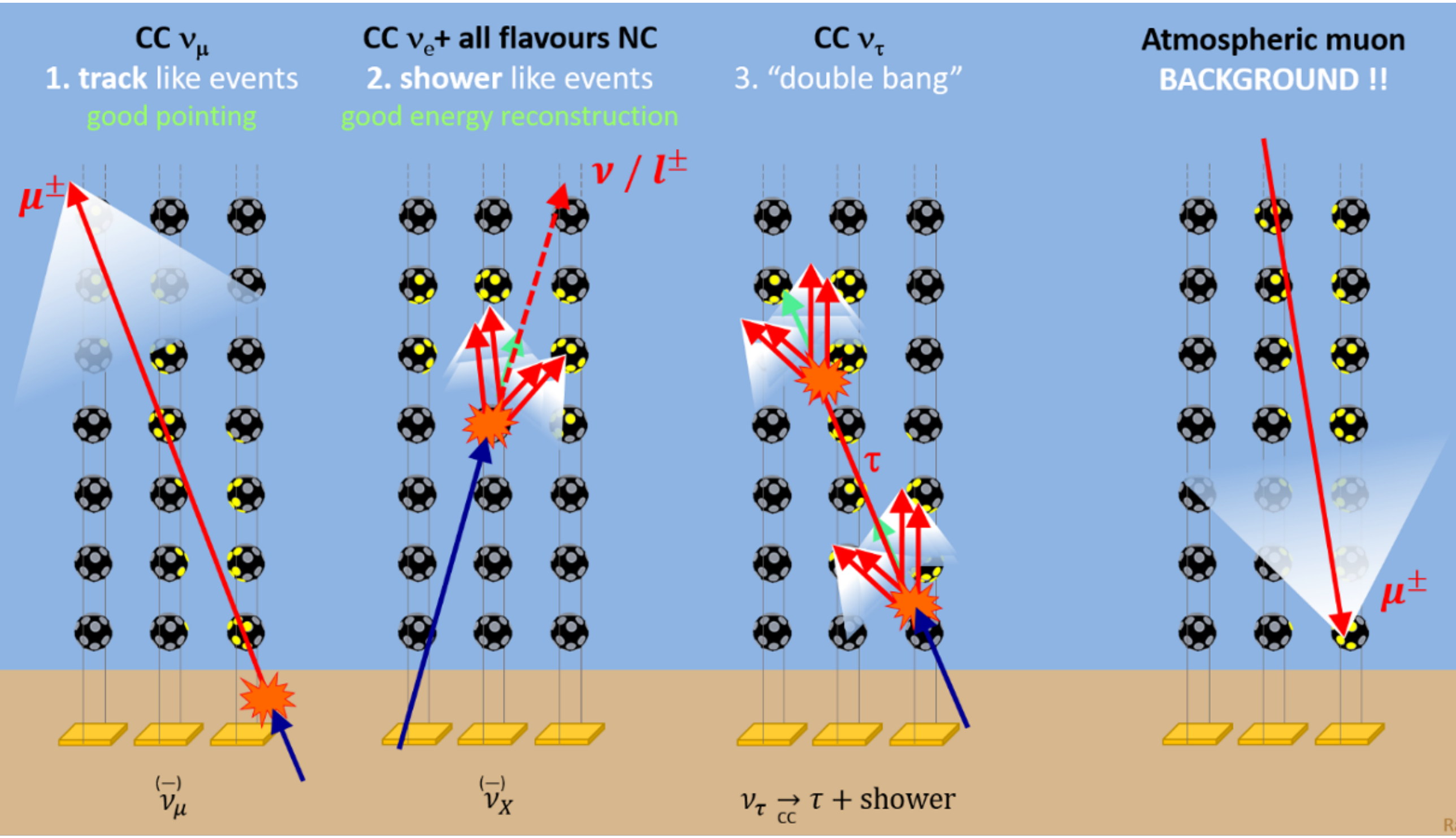
Not absorbed

Three flavours

Only hadronic processes

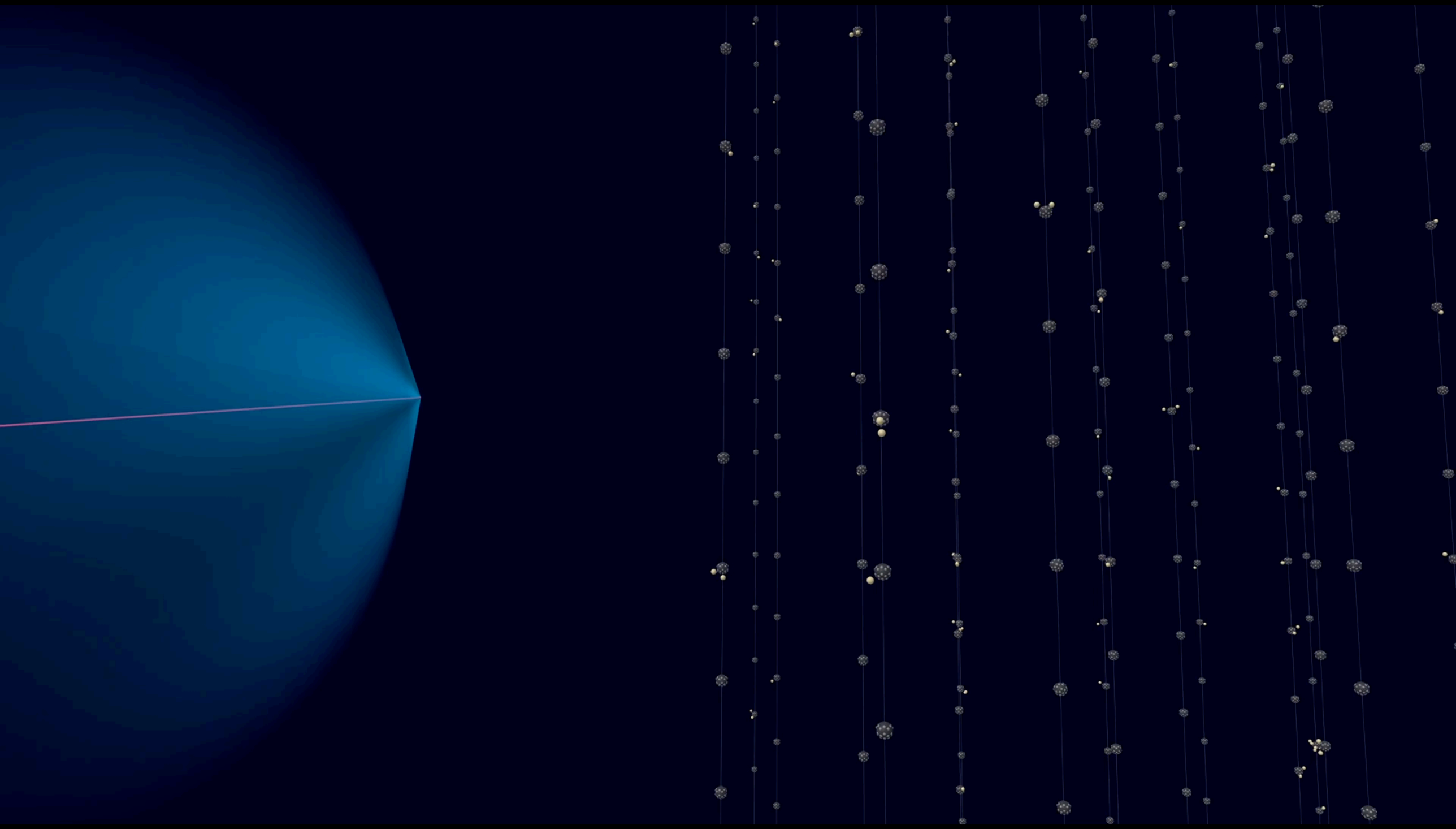
Credit: Paschal Coyle, KM3NeT Collaboration

Different flavor/type of neutrinos produce different signals



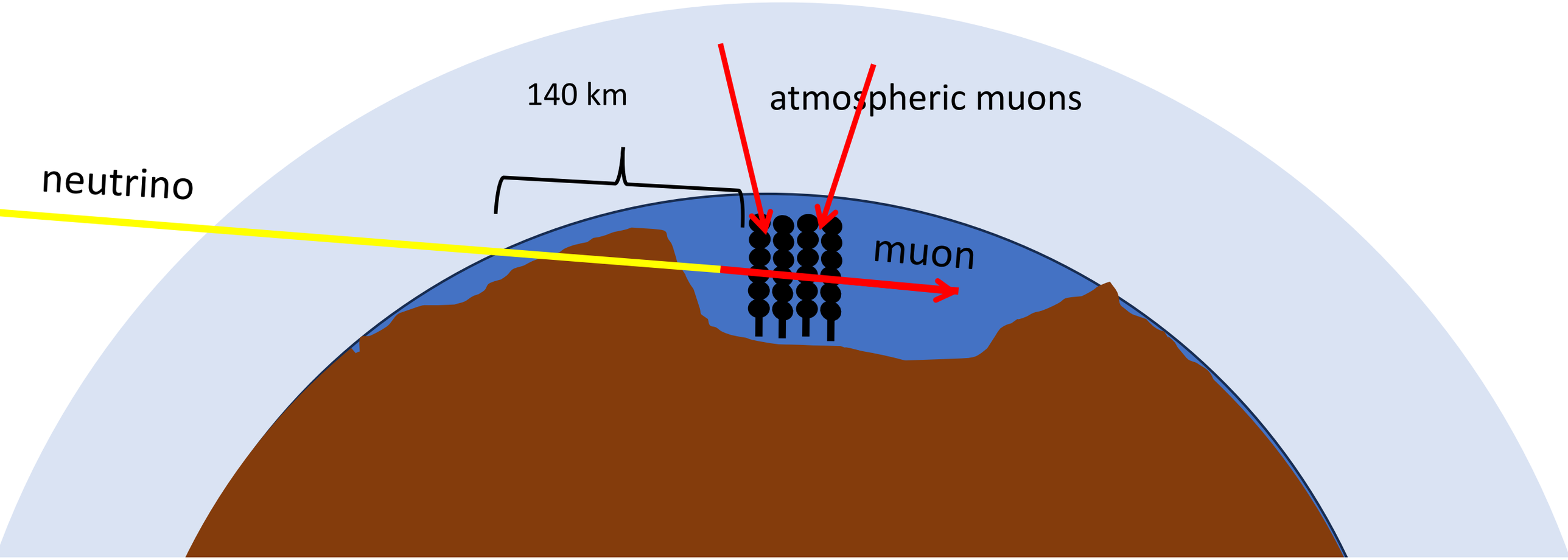
Feb 13, 2023, 01:16:47 UTC, an extraordinary event was detected!

1 in 110 million events!!



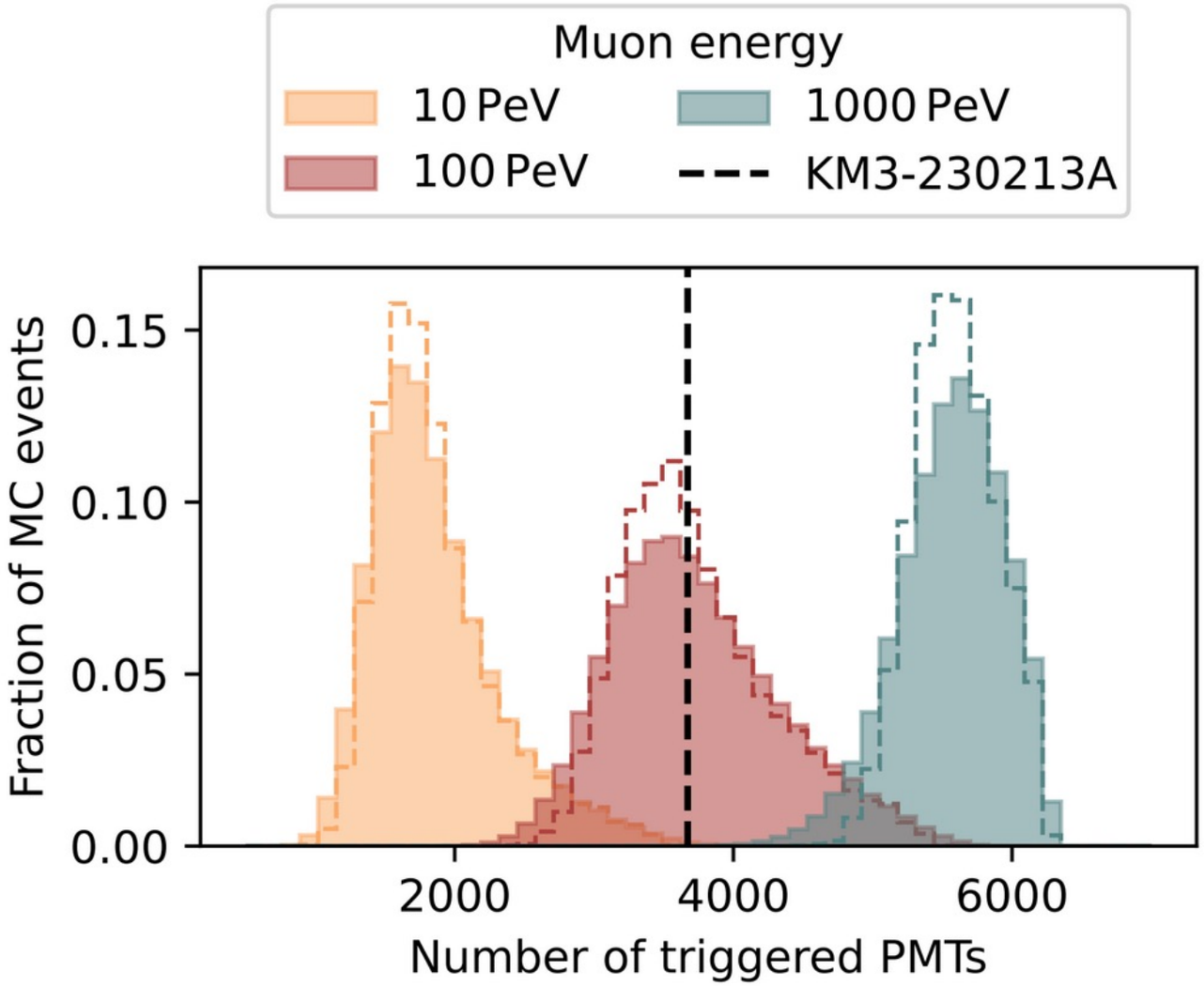
Credit: KM3NeT Collaboration

The event is a horizontal muon caused by a neutrino



Credit: Aart Heijboer, KM3NeT Collaboration

KM3-230213A is the highest energy neutrino ever measured (>10^{17} eV)!



- Energy is measured from the amount of light:

$$E_\mu = 120^{+110}_{-60}\ \mathrm{PeV}$$

(10000 times the energy of the LHC)

- The neutrino Energy is higher

$$E_\nu = 220^{+570}_{-100}\ \mathrm{PeV}$$

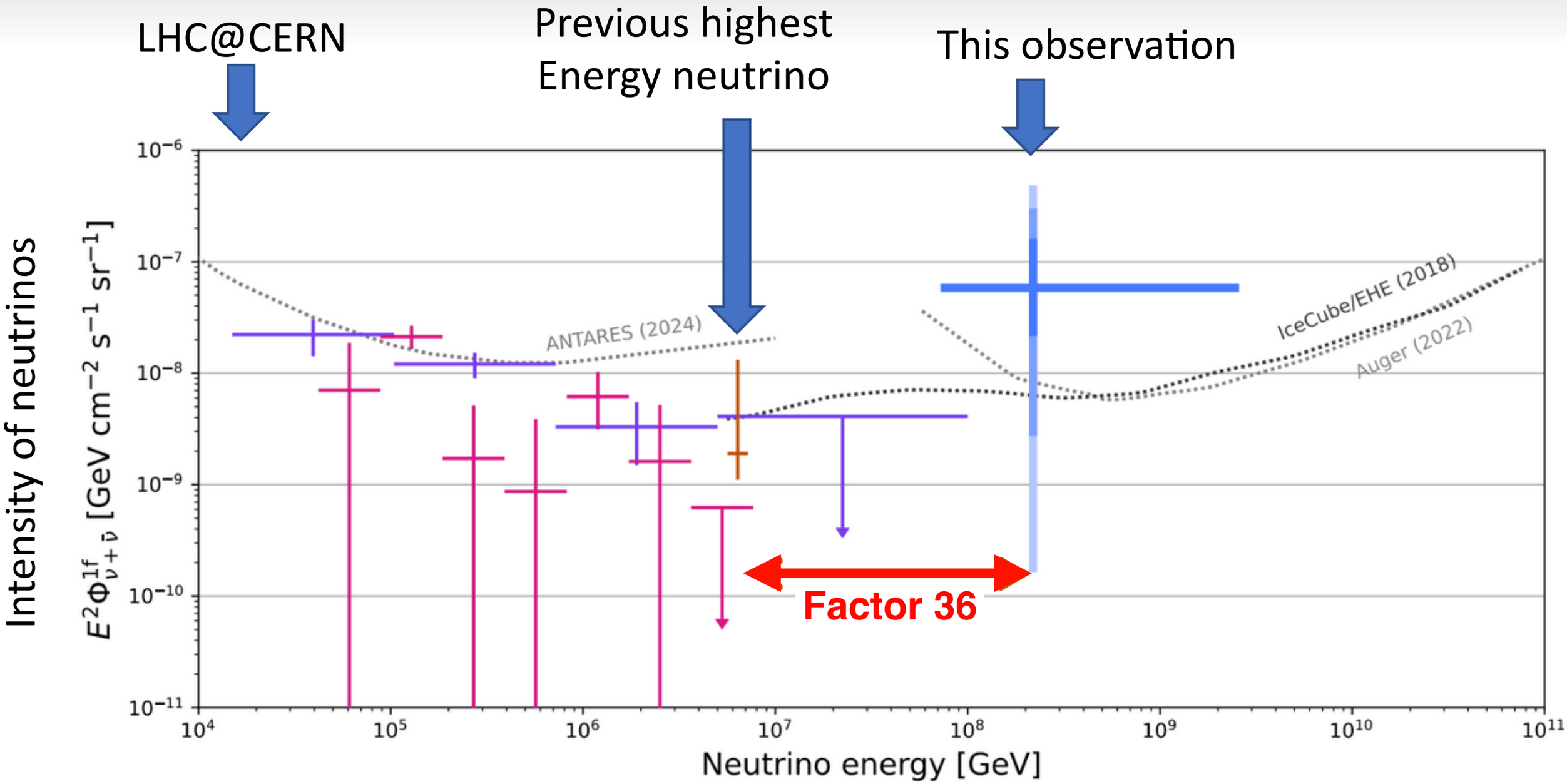
(assuming an E^{-2} source spectrum)

- Approximate celestial Origin: RA = 94.3◦, Dec= −7.8◦ with 1.5 ◦ error circle

Credit: Aart Heijboer, KM3NeT Collaboration

A milestone in Neutrino Astronomy

[More than 30 times higher than the previous highest energy !]

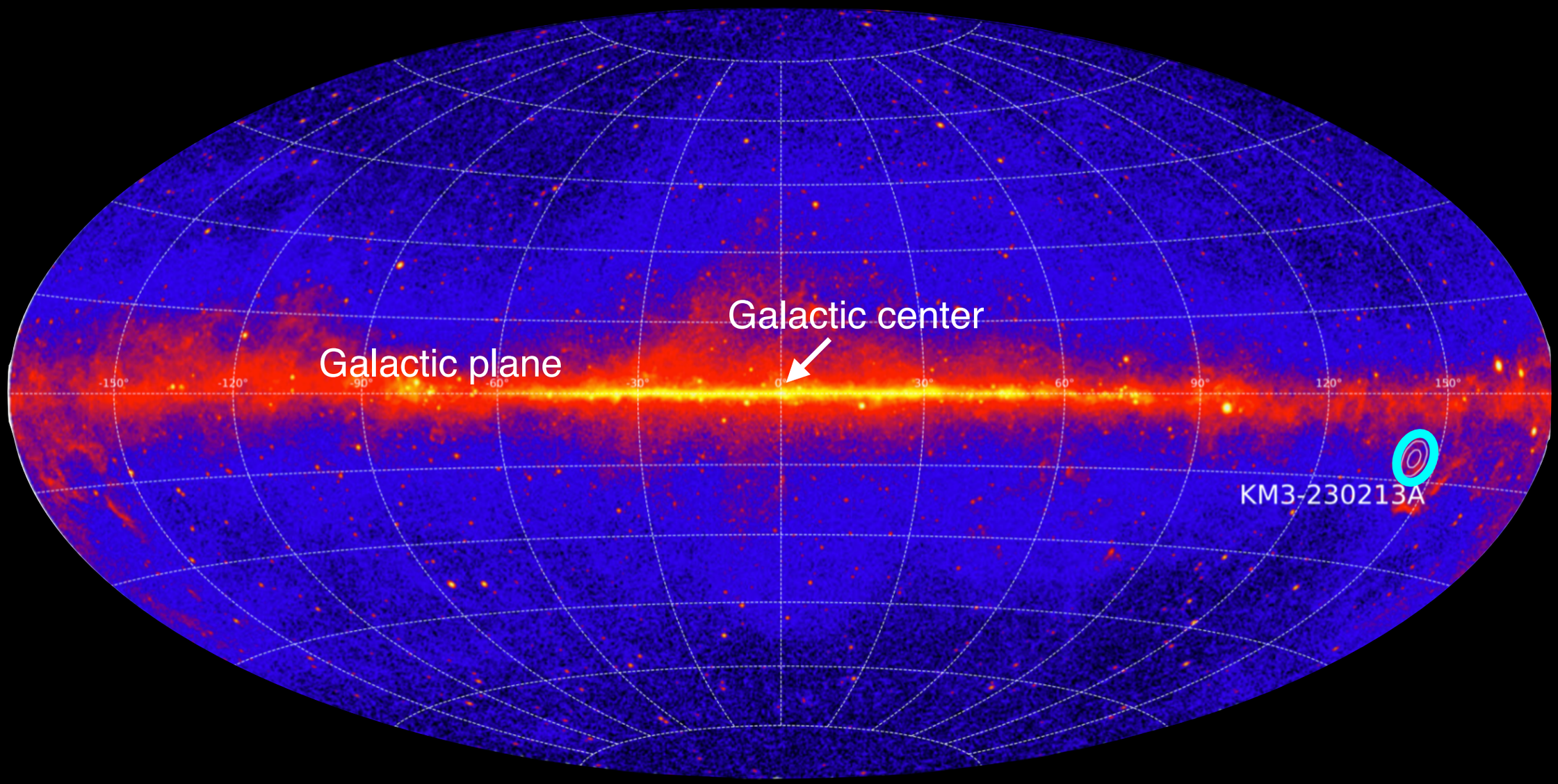


KHALIFA UNIVERSITY

Credit: Aart Heijboer, KM3NeT Collaboration

Direction of the event in Galactic coordinates

Diffuse gamma-ray map
from Fermi Telescope

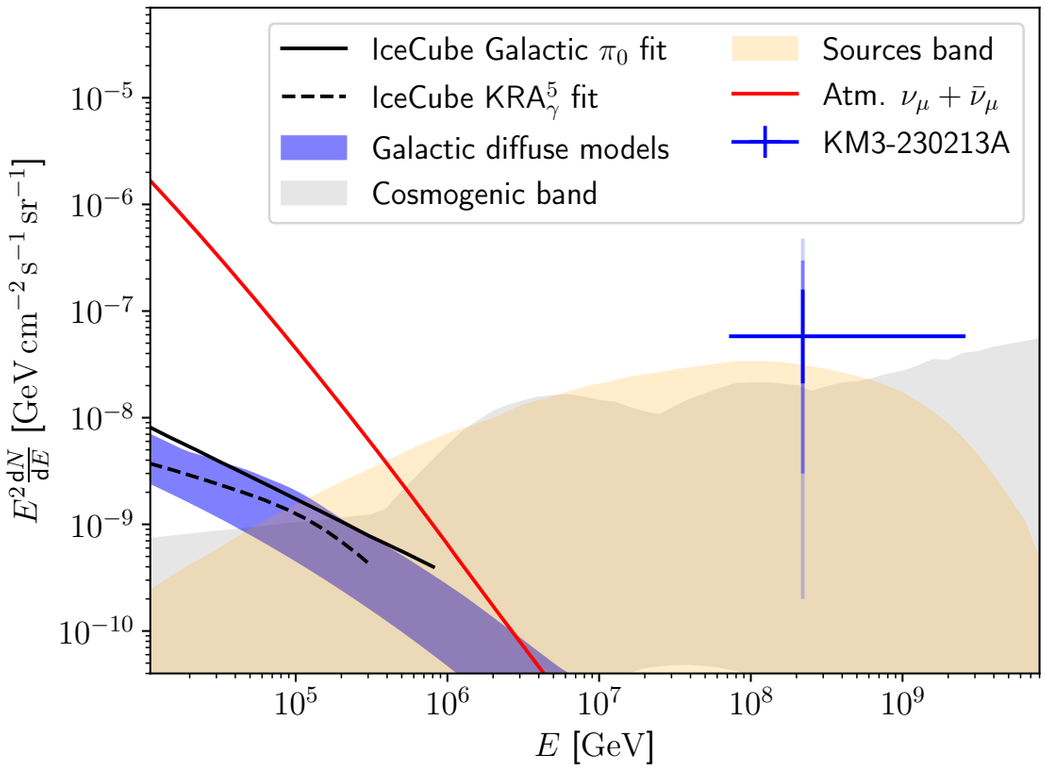


Adapted from Rosa Coniglione, KM3NeT Collaboration

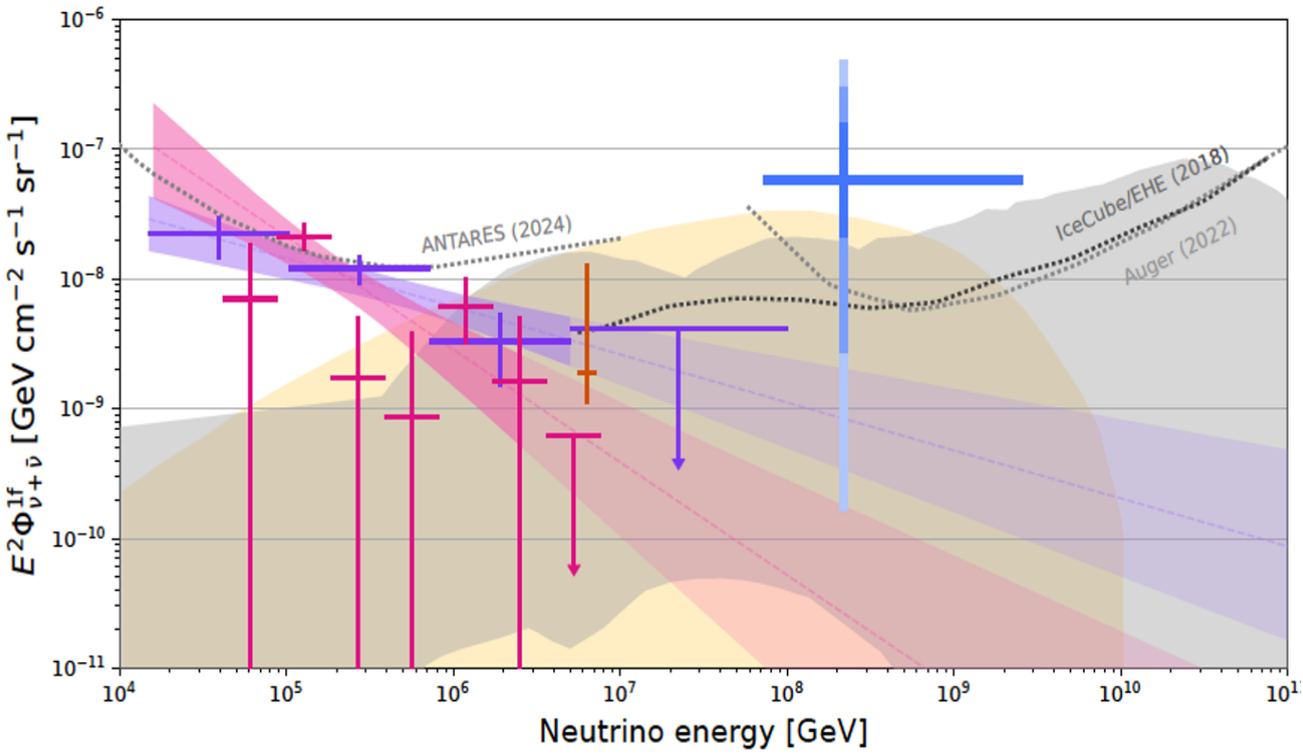
Main contribution of the KU group in the discovery paper

- Participated in the astrophysical interpretations of the neutrino event.
- Were part of the Galactic diffuse and cosmogenic origin tasks force.

Galactic diffuse emission



Cosmogenic origin



Credit: KM3NeT Collaboration

End of the course